



15'-60' Grain Bins

Wide-Corr® Centurion® Grain Bin Installation and Storage Instructions

This manual applies to models:

1513 – 1514
1813 – 1814
2113 – 2114
2413 – 2416
2713 – 2716
3013 – 3016
3613 – 3616
3913 – 3916
4213 – 4216
4513 – 4516
4813 – 4816
5111 – 5116
5410 – 5416
6008 – 6016



Read this manual before using product. Failure to follow instructions and safety precautions can result in serious injury, death, or property damage. Keep manual for future reference.

Part Number: 198893 R29

Revised: August 2021

Original Instructions

New in this Manual

The following changes have been made in this revision of the manual:

Description	Section
Added Boxed Upright Cap Plate	Boxed Upright Cap Plate on page 67
Updated Wall Sheet and Bottom Angle Assembly	Section 5.13 – Wall Sheet and Bottom Angle Assembly on page 54
Updated Bin Hardware	Table 41 on page 109

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1. Introduction

Before assembling, please read this manual. Familiarize yourself with the process and the necessary precautions for efficient and safe assembly of this Westeel 15'-60' Grain Bins.

Everyone present at the assembly site is required to be familiar with all safety precautions.

Keep this manual available for frequent reference and review it with new personnel. Call your local distributor or dealer if you need assistance or additional information.

2. Safety

2.1. Safety Alert Symbol and Signal Words



This safety alert symbol indicates important safety messages in this manual. When you see this symbol, be alert to the possibility of injury or death, carefully read the message that follows, and inform others.

Signal Words: Note the use of the signal words **DANGER**, **WARNING**, **CAUTION**, and **NOTICE** with the safety messages. The appropriate signal word for each message has been selected using the definitions below as a guideline.



Indicates an imminently hazardous situation that, if not avoided, will result in serious injury or death.



Indicates a hazardous situation that, if not avoided, could result in serious injury or death.



Indicates a hazardous situation that, if not avoided, may result in minor or moderate injury.



Indicates a potentially hazardous situation that, if not avoided, may result in property damage.

2.2. General Safety

Read and understand all safety instructions, safety decals, and manuals and follow them when assembling the equipment.

- Only experienced personnel who are familiar with this type of assembly and installation should perform this work. Untrained assemblers/installers expose themselves and bystanders to possible serious injury or death.
- Do not modify the grain bin in any way or deviate from the instructions in this manual without written permission from the manufacturer. Unauthorized modification or methods may impair the function and/or safety. Any unauthorized modification will void the warranty.
- Follow a health and safety program for your worksite. Contact your local occupational health and safety organization for information.
- Contact your local representative or Westeel if you need assistance or additional information.
- Always follow applicable local codes and regulations.



2.3. Personal Protective Equipment

The following Personal Protective Equipment (PPE) should be worn when installing the equipment.

Safety Glasses

- Wear safety glasses at all times to protect eyes from debris.



Coveralls

- Wear coveralls to protect skin.



Hard Hat

- Wear a hard hat to help protect your head.



Steel-Toe Boots

- Wear steel-toe boots to protect feet from falling debris.



Work Gloves

- Wear work gloves to protect your hands from sharp and rough edges.



2.4. Safety Decals

- Keep safety decals clean and legible at all times.
- Replace safety decals that are missing or have become illegible. See decal location figures that follow.
- Replaced parts must display the same decal(s) as the original part.
- Replacement safety decals are available **free of charge** from your distributor, dealer, or factory as applicable.

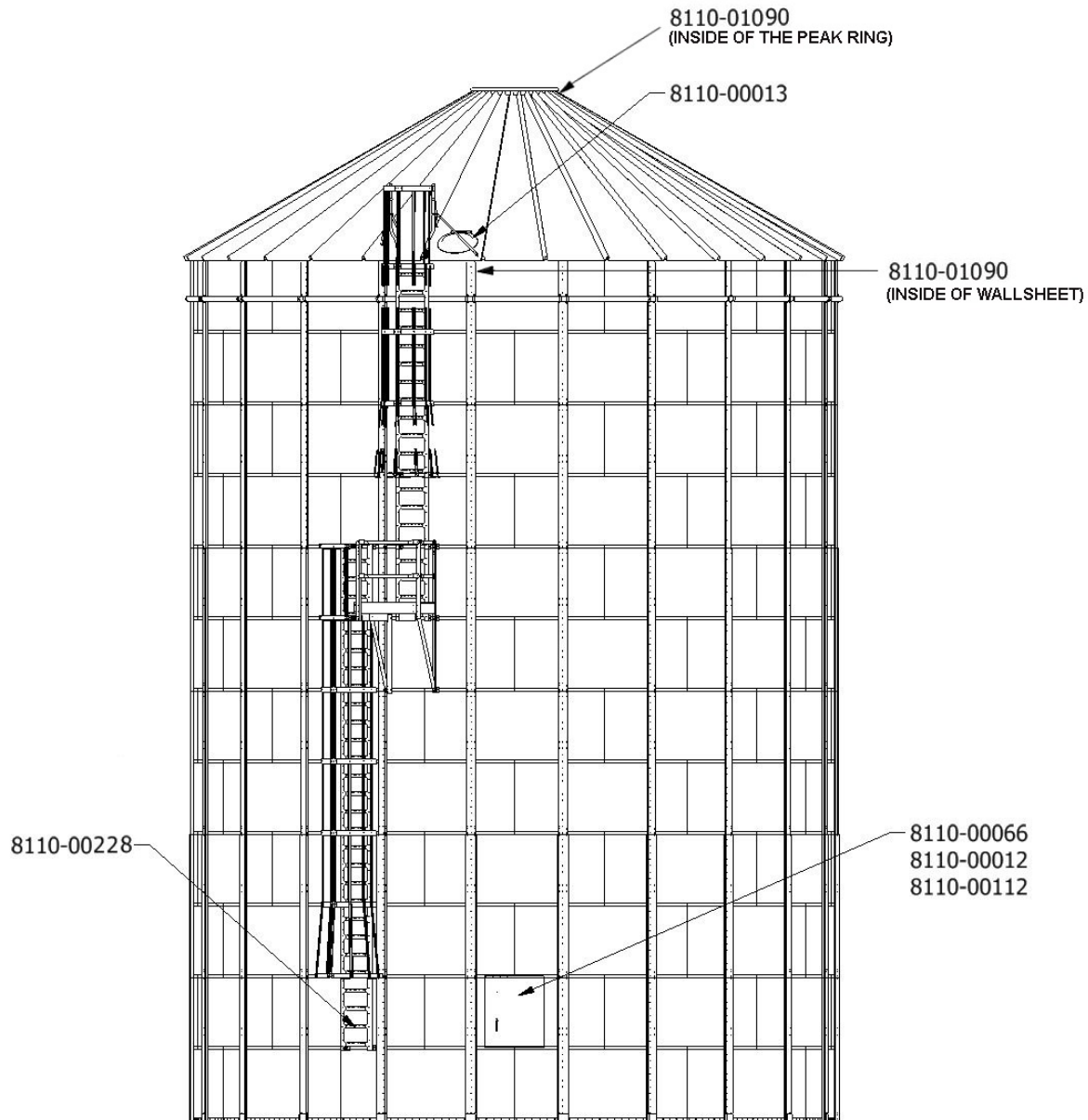
2.5. Decal Installation/Replacement

1. Decal area must be clean and dry, with a temperature above 50°F (10°C).
2. Decide on the exact position before you remove the backing paper.
3. Align the decal over the specified area and carefully press the small portion with the exposed sticky backing in place.
4. Slowly peel back the remaining paper and carefully smooth the remaining portion of the decal in place.
5. Small air pockets can be pierced with a pin and smoothed out using the decal backing paper.

2.6. Safety Decal Locations and Details

Replicas of the safety decals that are attached to the grain bin and their messages are shown in the figure(s) that follow. Safe operation and use of the grain bin requires that you familiarize yourself with the various safety decals and the areas or particular functions that the decals apply to, as well as the safety precautions that must be taken to avoid serious injury, death, or damage.

Figure 1. Safety Decals



 WARNING	
	
SAFETY INSTRUCTIONS • Read operator's manual and all safety decals before assembling, using, or servicing bin. • Close/latch all access doors when not in use. • Do not alter or modify bin structure. • Replace any damaged components only with factory made components. • This bin should only be used to store free flowing, granular material, unless specifically designed and marked otherwise. • When filling, use top filler cap and direct material to center of bin. • Do not over-fill bin. Material should not be in contact with or place pressure on roof sheets.	

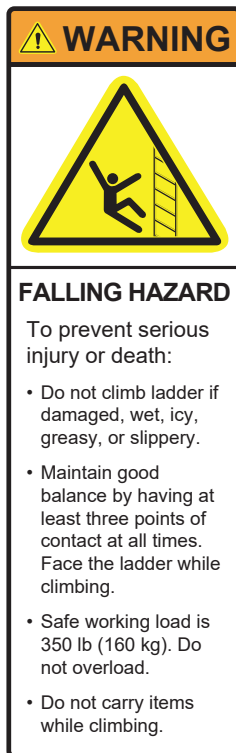
Part Number: 8110-00012

 WARNING	
	
ENTRAPMENT HAZARD Never enter the bin when loading or unloading grain. If you must enter the bin: 1. Shut off and lock out all power. 2. Use a lifeline, safety harness, and have an observer outside before entering the bin. 3. Wear proper breathing equipment or a respirator. 4. Avoid the center of the bin. Failure to heed these warnings could result in serious injury or death.	

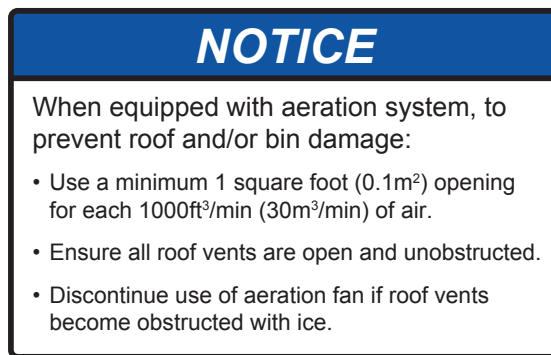
Part Number: 8110-00013

 WARNING		
		
Rotating flighting could kill or dismember.	Flowing material could trap and suffocate.	Crusted material could collapse and suffocate.
Keep clear of all augers. DO NOT ENTER this bin! If you must enter the bin: 1. Shut off and lock out all power. 2. Use a safety harness and safety line. 3. Station another person outside the bin. 4. Avoid the center of the bin. 5. Wear proper breathing equipment or respirator. Failure to heed these warnings could result in serious injury or death.		

Part Number: 8110-00112



Part Number: 8110-00228



Part Number: 8110-00066



Part Number: 8110-01090

3. Before You Begin

3.1. Bin Design and Capacity

Standard Westeel Grain Bins are designed for:

1. Non-corrosive free-flowing materials up to 55 lbs/ft³ (880 kg/m³) average compacted bulk density.
2. Maximum horizontal wind pressure based on 94 mph (151 km/h) as per NBCC 2015 and 105 mph (169 km/h) as per ASCE 7-16.
3. Zero seismic activity.

Note

Seismic resistance in grain bins varies with height and diameter. Many standard designs have significant seismic capabilities. Designs can be reviewed and/or modified to reflect local seismic requirements.

4. Roof loading capabilities vary with diameter, peak load and snow load.

- a. Peak Loads – standard peak loads follow. **Upgrades are available.**

Table 1. Peak Loads for Various Roofs

Size	Type of Roof	Load (lbs)	Load (kg)
15' to 24'	non-structural	4000 lbs	1814 kg
27' to 48'	non-structural	5000 lbs	2268 kg
51' & 54'	non-structural	8000 lbs	3629 kg
48' to 108'	structural	20,000 lbs	9072 kg

- b. Roof Snow Loads (RSL) – at the above stated standard peak loads, standard RSLs vary with diameter and range from 16 psf (78 kg/m²) to 45 psf (220 kg/m²). **Upgrades are available.**

Note

The correlation between ground snow load (GSL) and roof snow load (RSL) for grain bin designs vary with jurisdictions. In the US $GSL = 2 \times RSL$. In Europe $GSL = 1.25 \times RSL$. In Canada the correlation between GSL and RSL varies and is site specific.

- c. For maximum roof snow load capacities for various sizes and types of roofs, refer to the Roof Design Capacities sections that follow.

3.1.1 Roof Design Capacities for Non-Structural Roofs

Table 2. Maximum Roof Snow Load at STANDARD Peak Load

Bin Series	Std Peak Load	Standard Roof		Plus Upgrade 1		Plus Upgrade 2	
	lbs (kN)	psf	kPa	psf	kPa	psf	kPa
15	4000 (17.8)	45	2.15	n/a		n/a	
16		45	2.15				
18		45	2.15				
21		30	1.44	49	2.35		
24		21	1.01	40	1.92		
27	5000 (22.2)	24	1.15	39	1.87	40	1.92
30		20	0.96	32	1.53		
33		23	1.10	33	1.58	44	2.11
36		24	1.15	30	1.44	38	1.82
39		22	1.05	27	1.29	36	1.72
42		19	0.91	24	1.15	34	1.63
45		16	0.77	23	1.10	32	1.53
48		21	1.01	26	1.24	33	1.58
51	8000 (35.6)	20	0.96	28	1.34	n/a	
54		17	0.81	27	1.29		

Table 3. Maximum Roof Snow Load at UPGRADED Peak Load

Bin Series	Upgraded Peak Load	Standard Roof		Plus Upgrade 1		Plus Upgrade 2	
	lbs (kN)	psf	kPa	psf	kPa	psf	kPa
15	8000 (35.6)	29	1.39	n/a		n/a	
16		29	1.39				
18		29	1.39				
21		24	1.15	40	1.92		
24		17	0.81	27	1.29		
27	10000 (44.5)	19	0.91	28	1.34	33	1.58
30		15	0.72	24	1.15		
33		18	0.86	24	1.15	36	1.72
36		18	0.86	23	1.10	31	1.48
39		16	0.77	21	1.01	30	1.44
42		14	0.67	19	0.91	27	1.29
45		13	0.62	18	0.86	25	1.20
48*		16	0.77	21	1.01	26	1.24
51*	12000 (53.4)	14	0.67	21	1.01	n/a	
54*		13	0.62	20	0.96		

Note

- Standard roofs are adequate for many applications but additional capacity is available when optional upgrade packages are used.
- Upgrade packages include roof stiffening rings and/or rib supports.
- For peak load between standard and upgrade limits, a straight line interpolation can be used to determine maximum roof snow load.
- *Structural roofs for 48' – 54' with rafters are available to support peak ring loads greater than loads on [Table 3](#).
- Higher level upgrade kits include all components from lower level kit; only one upgrade kit needs to be ordered for any given bin.

3.1.2 Roof Design Capacities for Structural Roofs

Table 4. Maximum Roof Snow Load at STANDARD Peak Loads

Bin Series	Std Peak Load	Standard Roof	
	lbs (kN)	psf	kPa
48	20000 (89.0)	39	1.87
51		39	1.87
54		39	1.87
60		39	1.87
66		38	1.82
72		38	1.82
75		37	1.77
78		37	1.77
84		37	1.77
90		37	1.77
96		37	1.77
102		32	1.53
105		32	1.53
108		32	1.53

Table 5. Maximum Roof Snow Load at UPGRADED Peak Loads

Bin Series	Upgraded Peak Load	Standard Roof	
	lbs (kN)	psf	kPa
48	60000 (266.9)	38	1.82
51		38	1.82
54		38	1.82
60		38	1.82
66		37	1.77
72		37	1.77
75		36	1.72
78		36	1.72
84		36	1.72
90		34	1.63
96		34	1.63
102		31	1.48
105		31	1.48
108		31	1.48

Note

Standard capacities are provided. Additional capacity is available with optional upgrades.

3.1.3 Roof Snow Load vs. Ground Snow Load

The Roof Design Capacity tables reflect roof snow load (RSL) values. These are design values. Often, comparisons are made to ground snow values (GSL). These are not the same. The conversion from GSL to RSL varies between jurisdictions and is governed by building codes:

- In the United States, for grain bins, $GSL = RSL \times 2$
- In Europe, for grain bins, $GSL = RSL \times 1.25$

- In Canada, for grain bins, the GSL/RSL conversion varies with every location across the country. However, for comparison purposes, the US conversion can be used as an approximation.

Therefore, when comparing against competitive GSL values in the US, double the above values. When comparing against competitive GSL values in Canada, double the above values for a reasonably close approximation.

3.2. Guidelines for Supporting Catwalks and other External Loads on Westeel

Frequently catwalk and related equipment loads are supported on grain bins. Such connections are commonly made into the grain bin stiffeners and across the peak. A grain bin is a thin shell structure primarily designed to withstand the internal uniformly distributed loads inherent with the stored bulk material inside of the bin. Special considerations must be given to the manner in which external loads are supported. Westeel has developed products which are compatible with these requirements and considerations. If a third party solution is provided, the provider assumes full responsibility of the structure, its load distribution, and the manner in which it is connected to the grain bin. The following guidelines must form part of the third party design considerations.

Connection to Stiffeners

1. The available catwalk support stiffeners in Westeel stiffened bins are for 10,000 lb incremental catwalk loads and 20,000 lb incremental catwalk loads per upgraded stiffener. The actual loads subjected to a single stiffener by the mating catwalk support shall not exceed these maximum capacities.
2. Westeel recommends that the vertical load transfer between the catwalk supports and the stiffener occur over a minimum distance of 66" for 10,000 lb loads and 120" for 20,000 lb loads. Adequate connection strength must be provided.
3. The catwalk support stiffener in Westeel bins are designed to provide vertical load support only. Any lateral loads subjected to the grain bin must be negligible.
4. There is a restriction of 2 upgraded catwalk support stiffeners per bin location. Therefore, the maximum supported load at the grain bin eave is 20,000 lbs (for two 10,000 lb upgrades) and 40,000 lbs (for two 20,000 lb upgrades). This can be repeated on the opposing side of the bin at a second location. Deviation from this must be approved by Westeel Engineering.

Connection to Peak Rings

1. The allowable vertical peak load to any Westeel bin roof is restricted to its published rated capacity. The load must be centered and evenly distributed into the peak ring. Any off-centre load and/or improper load distribution may cause roof failure.
2. A Westeel structural roof requires the peak support loads to be transferred directly into the compression ring/roof rafter system. This is accomplished with peak load support brackets that are included with the structural roof. They must be installed as shown in the structural roof manual, connecting the peak support structure to the compression ring. They are required even if a non-Westeel peak support structure is used. A non-Westeel peak support structure needs to be designed to be able to connect with the brackets. The required bolt pattern is shown in the structured roof manual.
3. A Westeel non-structural roof that is supporting a catwalk requires six clips to be installed in order to attach the flat cap to the peak ring. These clips are available from Westeel.

3.3. Foundation Design and Loads

The foundations for the stiffened bin models are based on 4000 lbs. per sq. ft. (192 kPa) soil bearing capacity. All foundation designs use 3000 lbs. per sq. in. (21 MPa) ultimate compressive strength (after 28 days) for concrete and 43,500 lbs. per sq. in. (300 MPa) re-bar. The foundation designs included in this manual are suggestions only, and will vary according to local soil conditions. Westeel will not assume any liability for results arising from their use.

Important

Foundation should be uniform and level. Level should not vary by more than ¼" over a span of four feet under the bottom ring angle. Any variance from level must be shimmed under upright base assembly. If being utilized to support a full floor aeration system, this levelness requirement should extend across the complete floor area.

3.4. Site and Assembly

Unless otherwise specifically provided in writing, Westeel does not take responsibility for any defects or damages to any property, or injury to any persons, arising from or related to any site or assembly considerations, including but not limited to:

- Bin location and bin siting
- Soil conditions and corresponding foundation requirements
(Note that the examples provided in manuals are for specifically stated soil conditions.)
- Bin assembly (Westeel recommends the use of qualified bin installers. Contact Westeel for information on installers in your area.)
- Field modifications or equipment additions that affect the bin structure
- Interconnections with neighboring structures
- Compliance with all applicable safety standards, including but not limited to fall restraint systems (ladders or other systems). Local safety authorities should be contacted as standards vary between jurisdictions.

3.5. Methods of Installation

The recommendations for assembling and installing Westeel grain bins must be closely followed to achieve the full strength of the bin and to achieve adequate weather sealing. The product warranty is void if:

1. Wall sheets and/or uprights not specified for a given tier are used.
2. Foundations are found to be inadequate or out-of-level.
3. Anchor bolts (cast-in-place, drill-in, chemical type or other) are found to be inadequate.
4. Off-center loading or unloading is used. (This does not apply to the use of approved side unloading systems.)
5. Materials stored are not free-flowing or have a compacted bulk density greater than 55 lbs/ft³ (880 kg/m³).

If using bin jacks during assembly, always lift on an upright. Choose a hoist with a adequate capacity for the expected empty bin deadload. Make sure the rated capacity of the hoist is not exceeded.

3.6. Cutting Openings in Wide-Corr® Grain Bins

This section provides instructions for cutting openings to accommodate fan transitions, unloading augers and roof vents.

General Rules for Cutting openings

1. Never cut any uprights, roof ribs, or wall sheet bolted vertical seams to create an opening.
2. Openings shall be located so equipment being installed won't interfere with any bin components/accessories.
3. Openings shall be minimized as much as possible for structural integrity of grain bins.
4. Corners in openings shall be cut with minimum radius of 1/8" to reduce stress concentration.
5. Openings shall be sealed all the way around for all weather conditions.
6. Instructions shall be followed closely to avoid damage to bin structure.
7. Except cutting openings described below, any other modification to Westeel bins shall be approved by a professional engineer.

Openings for Fan Transitions of Aeration Floors

1. Consult aeration floor installation instructions for information on Planning floor layout.
2. Openings shall be centered to a wall sheet in horizontal direction.
3. Opening shall be cut as tight as it can be for the transition to go through and shall have no more than 1/4" gap on any side to the section of a fan transition going through a bin wall.
4. Opening height for fan transition shall be limited to 12.5" inches from bottom edge of a bottom wall sheet.
5. Opening width shall not exceed 46.5" for stiffened bins and 72.5" for unstiffened bins.
6. Vertical support shall be required to support load above opening.
7. Bottom angles may be cut flush to the sides of an opening to form part of opening.

Openings for Unloading Augers of Wide-Corr® Bins with Full Floor Aeration

1. Consult aeration floor installation instructions for information on Planning floor layout.
2. Openings shall be centered to a wall sheet in horizontal direction.
3. Openings shall be cut as tight as it can be for unloading auger to go through and shall have no more than 1/4" gap to auger flange section on any side.
4. Opening height for any auger shall be limited to 12.5" from the bottom edge of a bottom wall sheet.
5. Vertical flange of a bottom angle may be cut flush to sides of an opening to form part of opening.

Openings for Roof Vents in Roof Sheets

1. Openings shall be centered between roof ribs and have 2.5" minimum distance between edge of opening and base of a roof rib.
2. Openings can be square, rectangular, or round.
3. Openings shall be the same size as the inlet opening of a vent being installed.
4. Any side of a square/rectangular opening shall have a maximum length of 18" and a circular opening shall have a maximum diameter of 24".

3.7. Critical Assembly Requirements

To ensure a successful, safe and reliable outcome you must comply with the following assembly techniques and practices:

1. Comply with all local code and jurisdictional requirements applicable to your grain bin installation.
2. Design and build foundations with the necessary strength for the loads they must support, and for local soil conditions. Westeel foundation guidelines are based on specific stated conditions and may not be applicable to local conditions.
3. Your foundation must provide uniform and level support to the structure being supported. Surface imperfections causing gapping must be remedied. This may involve, but not be limited to a) grouting under the bottom ring of a non-stiffened bin or tank, and b) shimming under the uprights of a stiffened bin or tank, or under the legs of a hopper.
4. Make sure that the proper hardware is utilized for all bolted connections. If a shortage occurs, do not substitute. Take the necessary steps to obtain the proper hardware. Make sure nuts are tightened to the required torque values as specified in the appropriate assembly manual.
5. Comply with all assembly instructions provided in the appropriate assembly manual to make sure your whole grain bin is constructed safely. **Important: Do not deviate from the wall sheet and upright layouts provided.**
6. Before anchoring your structure to its foundation, make sure the structure is round. The maximum variation from perfect roundness is 3/4" on the radius. Locate anchor bolts toward the outside of the anchor bolt holes (away from the circle) to permit the incremental expansion that can occur with the initial filling.
7. When installing roof stiffening rings, if it is necessary to shorten the stiffening ring tubes, shorten them as little as possible. Initially the nuts on the expanders should be centered and as close together as possible. When tightening, share the amount of take-up between expanders such that the nuts remain centered, and the amount of engagement between all expanders on the same ring is equalized.
8. If extending an existing bin or tank, ensure that the foundation is adequate for the increased loads it must support.
9. If installing an existing bin on a hopper, make sure the bin is designed for a hopper application, and that the foundation is capable of withstanding the substantial point loads that the hopper legs apply. If uprights are present, make sure that they are supported.
10. Make sure that an integral end-to-end connection exists between all mating uprights. Successive uprights must not overlap.
11. Vertical tolerances between uprights and wall sheets are tight. This can be affected by “jacking” techniques, which can allow the tolerance to grow or shrink depending on the technique used. The gapping between successive uprights must be monitored to ensure that upright holes align with wall sheet holes.
12. If catwalks are being installed on the structure, upright catwalk upgrades are likely required. The upgraded stiffeners must be installed in the correct locations to support the intended catwalk loads. Also, the structure must be properly oriented to ensure the eventual correct alignment between the catwalks and the supporting uprights. Finally, the connectors that tie into the uprights and support the catwalks are best installed during assembly of the structure. See the catwalk assembly manual for additional details.

3.8. Product Storage

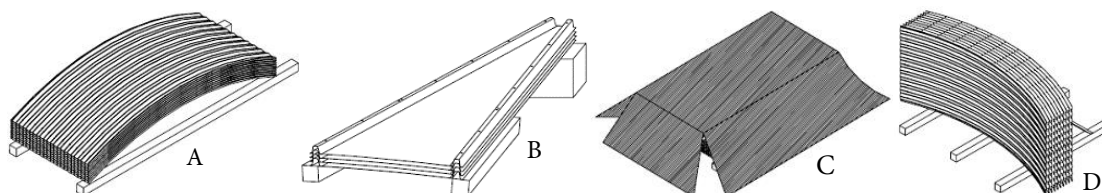
Rust on Galvanized Parts

1. White rust forms when moisture is allowed to collect on galvanized surfaces that have yet to develop the durable zinc oxide layer. This zinc oxide layer naturally occurs as the surface interacts with carbon dioxide, and is characterized over time by the dull grey appearance that weathered galvanized surfaces get.
2. Parts that are not well ventilated or well drained can collect water between surfaces and develop white rust.
3. White rust is not a structural concern if its development is stopped in the early stages. A light film or powdery residue can occur after a period of heavy rainfall or a short time of improper storage. If white rust has started to develop, separate parts and wipe off any moisture. Next, using a clean cloth, apply a thin layer of petroleum jelly or food-grade oil to the entire part.
4. If moisture is left on parts, this white rust can become more aggressive and turn into red rust. Red rust can cause degradation in the material and become a structural concern. Any parts that have red rust should be replaced immediately.

Storage Guidelines

- Keep all bundles dry before assembly of the bin.
- Start assembly as soon as possible.
- Do not lay bundles on the bare ground. Raise all bundles 6" to 8" off the ground on wood blocks or timbers. (See Detail A in [Figure 2 on page 18.](#))
- Store curved wall sheets 'hump-up'. (See Detail A in [Figure 2 on page 18.](#))
- All other bundles material should be placed so they are well sloped to promote good drainage. (See Detail B in [Figure 2 on page 18.](#))
- Roof sheets must be elevated at least 12" at the small end of the sheets. (See Detail B in [Figure 2 on page 18.](#))
- Temporary storage can be provided by erecting a simple framework supporting a waterproof tarp. (See Detail C in [Figure 2 on page 18.](#))
- All bin boxes, ladder boxes and hardware boxes should be stored inside. These are not waterproof, and will deteriorate in normal weather conditions, allowing moisture to contact the parts inside.

Figure 2. Product Storage



If Parts Become Wet

1. If parts become submerged or wet, the bundles should be opened as soon as possible, sheets or material separated and dried. Keep separated until assembly.

Brace parts properly so as to avoid damage or injury from material falling when in storage. (See Detail D in [Figure 2 on page 18.](#))

2. Any boxed parts that become wet should be dried and stored in a new box that is free of moisture.

3. In addition to wiping down wall sheets, a food-grade oil can also be applied with a clean, lint-free cloth. This will assist in preventing any further moisture from contacting the galvanizing on the steel. Due to safety concerns with installation and use, Westeel does not recommend the use of oil on other parts such as roof sheets and safety ladders.

3.9. Grain Bin Use

- Do not off-center unload a grain bin. It is imperative to unload from the center of the bin first, until as much grain as possible has been removed, and only then proceed to unload from the next closest unload gate to the center. Continue utilizing the unload gates in succession from the center towards the outside. Gate control mechanisms should be clearly marked and interconnected to prevent an external gate from being opened first.
- The only exception to center unloading is when a properly designed and installed side draw system is utilized. However, as bins tend to go out of round when employing side draws, the bin must be completely emptied before refilling.
- When unloading a bin with a mobile auger through a properly designed auger chute, the entry end of the auger should be pushed into the center of the bin before the auger is engaged. Slower rates of flow are preferable and should not exceed the capacity of an 8" auger.
- Ensure that the inner door panels of grain bin doors are completely closed and latched before filling the grain bin.
- Never enter a loaded grain bin for any reason. Grain can be a killer.

3.10. Important Notes

- Westeel does not provide a foundation design for this product, and is not liable for any damages or injuries related to inadequately designed or constructed foundations. Customers must contract professional services for all foundation design and construction work.
- In order to maintain your wall sheets in good condition separate sheets and allow air circulation between them. Store sheets in a dry place. Do not store sheets with sheet ends pointing upwards.
- To keep an even pressure on walls, the bin must always be unloaded from the center.
- Contact local power officials for minimum power line clearance.
- See [Section 3.7 – Critical Assembly Requirements on page 17](#) for mandatory siting and assembly requirements.
- Store only non-corrosive, free-flowing materials up to 55 lbs/ft³ (880 kg/m³) average compacted density in Westeel .
- Tighten all bolts to the recommended torque settings.
- Do not locate grain bins close to high buildings, which might cause snow to fall onto or build up on the roof of the grain bin. Consider future expansion and allow space for loading and unloading of the bin. Your dealer and local government agricultural consultants can help you plan your storage system for maximum efficiency.

4. Preparation

4.1. Check Shipment

Unload the parts at the assembly site and compare the packing slip to the shipment. Ensure that all items have arrived and that none are damaged.

Report damaged parts or shortages immediately to your dealer. Your dealer will order replacement parts immediately to ensure that assembly will not be held up by missing parts. All parts will be charged for and credit will be issued by party at fault. No credit will be issued if freight bills are signed as received in good condition.

4.2. List of Tools and Equipment

Use quality tools and equipment. Use them safely, and correctly, for their intended use. Tools for this application should include:

Tools

- Electric or pneumatic (air) impact tools
- Power drill and drill bits
- Sockets (multiple 9/16" and 1/2" sockets recommended)
- Large-pocket carpenter pouch
- 8" (20 cm) metal punches (for aligning bolt holes)
- Step and extension ladders, construction grade
- 6-point wrenches (Imperial, box end)
- Metal-cutting saw suitable for cutting roof rings and wind rings
- Scaffolding
- Centre-post bin stand
- Crane and/or bin jacks

Minimum Recommended Safety Equipment

- A properly-stocked first-aid kit
- Eye, foot, head, and hand protection (safety glasses, steel-toed boots, hard hat, work gloves)
- Cable, chain, or rope to tie-off bin or jacks in case of wind
- Body harness and lifeline (for use where falling hazard exists)
- Ground fault interrupt protected electrical hook-ups

4.3. Order Optional Equipment

Optional equipment such as unloading augers, aeration equipment, anchor bolts, foundation sealant, external ladders, safety cage and platforms, etc., should all be on site and checked before assembly starts. Plan your installation in advance. For details, see assembly instruction supplied with optional equipment.

5. Assembly



Before continuing, ensure you have completely read and understood this manual's Safety section, in addition to the safety information in the section(s) below.

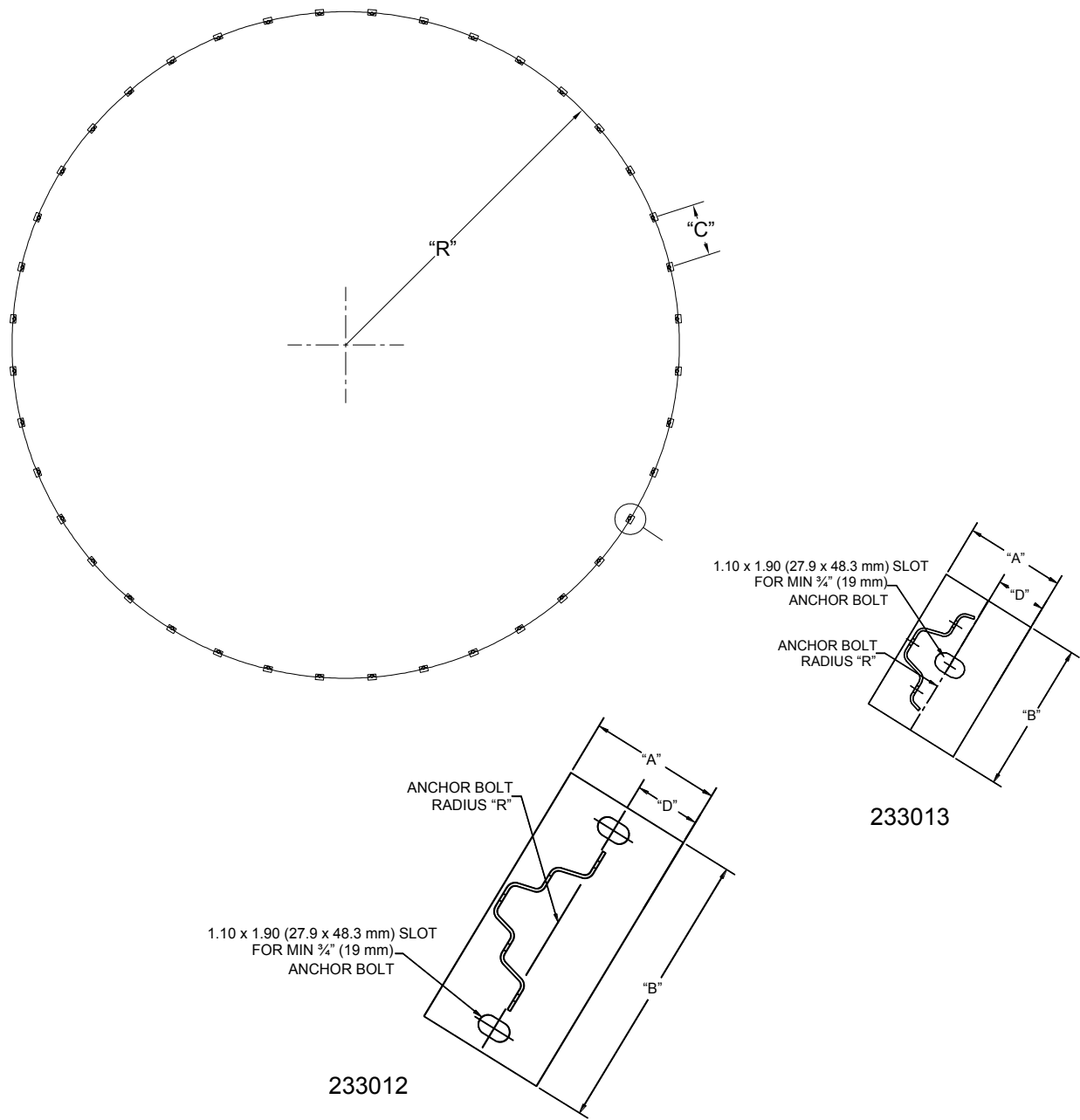
5.1. Assembly Safety

WARNING

- Do not take chances with safety. The components can be large, heavy, and hard to handle. Always use the proper tools, rated lifting equipment, and lifting points for the job.
- Carry out assembly in a large open area with a level surface.
- Always have two or more people assembling the grain bin.
- Make sure you have sufficient lighting for the work area.
- Tighten all fasteners according to their specifications. Do not replace or substitute bolts, nuts, or other hardware that is of lesser quality than the hardware supplied by the manufacturer.
- Stay away from overhead power lines and other obstructions during assembly. Contact with power lines can cause electrocution.
- Do not work in high winds.
- The equipment shall be installed in accordance with applicable local codes and regulations.

5.2. Anchor Bolt Radius

Figure 3. Anchor Bolt Radius



Base Plate			
Part Number	A	B	D
233013	6"	9"	3"
233012	7.5"	16"	3.75"

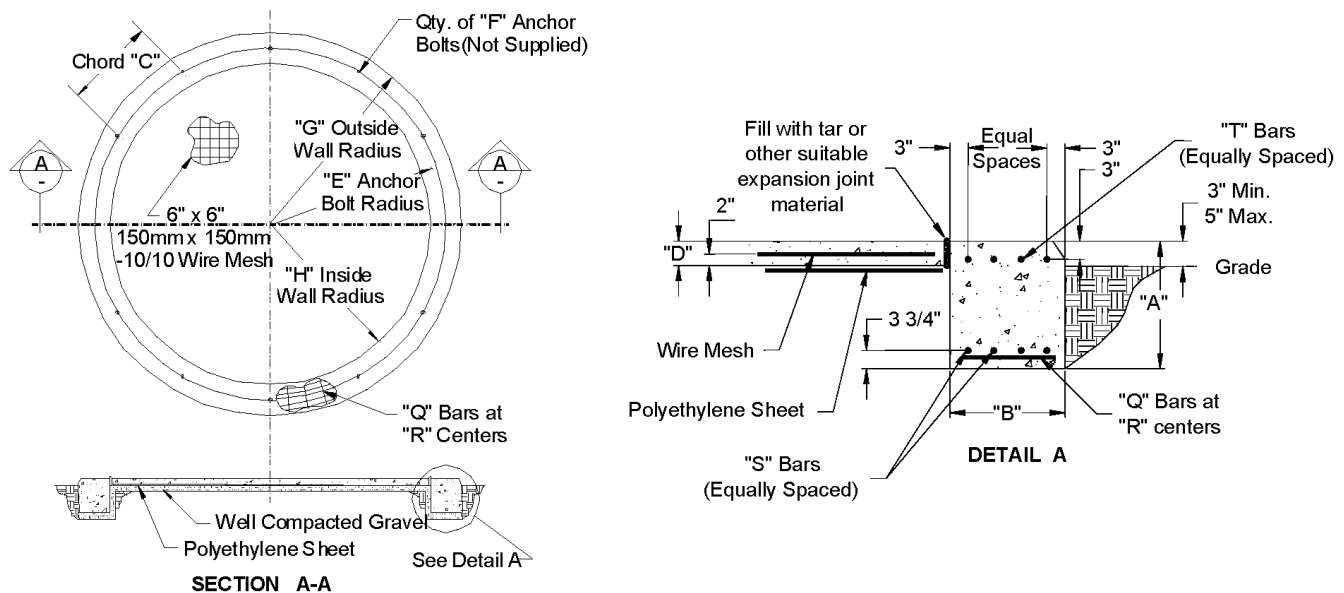
Table 6. Anchor Bolt Radius

Bin Model	Number of Tiers	Base Plate	Anchor Bolt Radius "R"		Anchor Bolt Cord "C"		No. of Anchor Bolts
			Inch	mm	Inch	mm	
15	13–14	233013	92.7	2355	57.29	1455	10
18	13–14	233013	110.6	2809	57.25	1454	12
21	13–14	233013	128.6	3266	57.23	1454	14
24	13–16	233013	146.5	3721	57.16	1452	16
27	13–16	233013	164.4	4176	57.10	1450	18
30	13–15	233013	182.3	4630	57.04	1449	20
30	16	233012	183.0	4648	57.26	1454	40
33	13–16	233013	200.2	5085	56.98	1447	22
36	13–16	233013	218.1	5540	56.94	1446	24
39	13–16	233013	236.0	5994	56.89	1445	26
42	13–16	233013	253.9	6449	56.86	1444	28
45	13–16	233013	271.8	6904	56.82	1443	30
48	13–16	233013	289.7	7358	56.79	1442	32
51	11–16	233013	307.6	7813	56.76	1442	34
54	10–16	233013	325.5	8268	56.74	1441	36
60	8–15	233013	361.4	9180	56.71	1440	40
60	16	233012	362.1	9197	56.82	1443	80

5.3. Curb Footing — Floating Slab Foundation

1. Choose a site that is well drained, and has a minimum soil-bearing capacity of 4000 lbs/ft² (192 kPa). If soil-bearing capacity is not known, consult a local engineering representative.
2. Use minimum 4" to 6" (100 mm to 150 mm) of well-compacted coarse granular material below slab and curb footing.
3. Concrete strength shall be minimum 3000 psi [21 MPa] compressive strength.

Figure 4. Curb Footing Foundation Layouts



Note

For dimensions for specific bin sizes, refer to the tables in the following sections.

4. The foundation must be level to within 1/4" over a span of 4 feet [6 mm over 1200 mm]. Any variance from level must be shimmed under an upright. (See [Section 5.19 – Commercial Bin Upright Assembly on page 61.](#))
5. Locate anchor bolts as shown in [Figure 4](#), using the dimensions provided in the following Curb Specifications tables, to ensure bin roundness.
6. Re-bar end laps are not included. Add 15" [380mm] for each lap. To estimate weight of end lap: add 0.5 lbs. for #3 and 1.0 lbs. for #4, 0.3 kg for 10M and 0.6 kg for 15M.
7. Concrete slump for reinforced wall to be 5" (130 mm) maximum and 2" (50 mm) minimum.
8. Reinforcing bars for concrete reinforcement shall conform to Grade 40, 40000 psi (280 MPa) or Grade 60, 60000 psi (420 MPa) as called for in the following: ASTM² A615, A616, or A617.
9. Welded wire mesh for concrete reinforcement shall conform to ASTM² A185, or A497.

¹ American Concrete Institute

² American Society for Testing Materials

5.3.1 Curb Footing Specifications — 15' to 27' Series

Table 7. Curb Footing Specifications — 15' to 27' (externally stiffened)

BIN MODEL		1513 1514	1813 1814	2113 2114	2413	2414 2415 2416	2713	2714 2715 2716
A		To Be Determined	2'4"	2'0"	2'0"	2'0"	2'0"	2'0"
B			2'8"	2'10"	3'0"	4'2"	3'2"	4'6"
C			4'9-1/4"	4'9-1/4"	4'9-5/32"	4'9-5/32"	4'9-1/8"	4'9-1/8"
D			6"	6"	6"	6"	6"	6"
E			9'2-5/8"	10'8-5/8"	12'2-1/2"	12'2-1/2"	13'8-7/16"	13'8-7/16"
F			12	14	16	16	18	18
G			10'7"	12'2"	13'8"	14'3"	15'3"	15'11"
H			7'11"	9'4"	10'8"	10'1"	12'1"	11'5"
Q	Metric ¹		39 - 10M	45 - 10M	54 - 10M	51 - 15M	69 - 10M	58 - 15M
	Imperial		41 - #3	58 - #3	77 - #3	51 - #5	94 - #3	58 - #5
R	Metric (mm c/c)		432	356	301	457	279	457
	Imperial (in. c/c)		17	14	12	18	11	18
S	Metric ¹		6 - 10M	6 - 10M	6 - 10M	4 - 15M	6 - 10M	5 - 15M
	Imperial		5 - #4	5 - #4	5 - #4	4 - #5	5 - #4	5 - #5
T	Metric ¹		6 - 10M	5 - 10M	6 - 10M	4 - 15M	6 - 10M	4 - 15M
	Imperial		4 - #4	4 - #4	4 - #4	4 - #5	5 - #4	4 - #5
Reinforcing Rod	M (kgs.)		188	203	252	383	291	481
	I (lbs.)		383	457	532	833	668	1049
Wire Mesh (sq. ft.)			197	274	358	320	459	410
Concrete (3000 psi) Cubic Yards	Footing		13.4	14.2	17.0	23.6	20.2	28.7
	Slab		3.7	5.1	6.7	6.0	8.5	7.6

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.3.2 Curb Footing Specifications — 30' to 39' Series

Table 8. Curb Footing Specifications — 30' to 39' (externally stiffened)

BIN MODEL		3013	3014 3015	3016	3313	3314 3315 3316	3613	3614 3615 3616	3913	3914 3915 3916
A		2'0"	2'0"	2'0"	2'0"	2'0"	2'0"	2'0"	To Be Determined	
B		3'4"	4'8"	4'8"	3'6"	4'10"	3'8"	5'2"		
C		4'9"	4'9"	4'9.6"	4'9"	4'9"	4'8-15/16"	4'8-15/16"		
D		6"	6"	6"	6"	6"	6"	6"		
E		15'2-5/16"	15'2-5/16"	15'3"	16'8-3/16"	16'8-3/16"	18'2-1/8"	18'2-1/8"		
F		20	20	40	22	22	24	24		
G		16'10"	17'6"	17'6"	18'5"	19'1"	20'0"	20'9"		
H		13'6"	12'10"	12'10"	14'11"	14'3"	16'4"	15'7"		
Q	Metric ¹	82 - 10M	68 - 15M	68 - 15M	97 - 10M	84 - 15M	115 - 10M	92 - 15M		
	Imperial	115 - #3	68 - #5	68 - #5	140 - #3	84 - #5	153 - #3	92 - #5		
R	Metric (mm c/c)	254	432	432	229	381	229	381		
	Imperial (in. c/c)	10	17	17	9	15	9	15		
S	Metric ¹	7 - 10M	6 - 15M	6 - 15M	7 - 10M	5 - 15M	7 - 10M	5 - 15M		
	Imperial	5 - #4	5 - #5	5 - #5	5 - #4	5 - #5	6 - #4	5 - #5		
T	Metric ¹	6 - 10M	4 - 15M	4 - 15M	6 - 10M	4 - 15M	6 - 10M	5 - 15M		
	Imperial	4 - #5	4 - #5	4 - #5	5 - #4	4 - #5	5 - #4	5 - #5		
Reinforcing Rod	M (kgs.)	353	547	547	396	626	443	752		
	I (lbs.)	760	1191	1191	858	1363	1021	1639		
Wire Mesh (sq. ft.)		573	518	518	700	638	839	763		
Concrete (3000 psi) Cubic Yards	Footing	23.6	33.0	33.0	27.2	37.5	31.0	43.7		
	Slab	10.7	9.6	9.6	13.0	11.9	15.6	14.2		

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.3.3 Curb Footing Specifications — 42' to 51' Series

Table 9. Curb Footing Specifications — 42' to 51' (externally stiffened)

BIN MODEL		4213	4214 4215 4216	4513	4514 4515 4516	4813	4814 4815 4816	5111 5112 5113	5114 5115 5116
A		2'0"	2'0"	2'0"	2'0"	2'0"	2'0"	To Be Determined	
B		3'10"	5'6"	3'6"	4'10"	4'0"	5'8"		
C		4'8-13/16"	4'8-13/16"	4'8-13/16"	4'8-13/16"	4'8-3/4"	4'8-3/4"		
D		6"	6"	6"	6"	6"	6"		
E		21'1-15/ 16"	21'1-15/ 16"	22'7-13/ 16"	22'7-13/ 16"	24'1-11/ 16"	24'1-11/ 16"		
F		28	28	30	30	32	32		
G		23'1"	23'11"	18'5"	19'1"	26'2"	27'0"		
H		19'3"	18'5"	14'11"	14'3"	22'2"	21'4"		
Q	Metric ¹	145 - 10M	123 - 15M	97 - 10M	84 - 15M	183 - 10M	152 - 15M		
	Imperial	114 - #4	123 - #5	140 - #3	84 - #5	141 - #4	152 - #5		
R	Metric (mm c/c)	356	330	229	381	330	305		
	Imperial (in. c/c)	14	13	9	15	13	12		
S	Metric ¹	7 - 10M	5 - 15M	7 - 10M	5 - 15M	8 - 10M	6 - 15M		
	Imperial	6 - #4	5 - #5	5 - #4	5 - #5	6 - #4	6 - #5		
T	Metric ¹	7 - 10M	5 - 15M	6 - 10M	4 - 15M	7 - 10M	5 - 15M		
	Imperial	6 - #4	5 - #5	5 - #4	4 - #5	6 - #4	5 - #5		
Reinforcing Rod	M (kgs.)	562	931	396	626	699	1175		
	I (lbs.)	1320	2029	858	1363	1547	2562		
Wire Mesh (sq. ft.)		1165	1066	700	638	1544	1430		
Concrete (3000 psi) Cubic Yards	Footing	37.8	54.2	27.2	37.5	45.0	63.8		
	Slab	21.6	19.8	13.0	11.9	28.6	26.5		

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.3.4 Curb Footing Specifications — 54' to 60' Series

Table 10. Curb Footing Specifications — 54' to 60' (externally stiffened)

BIN MODEL	5410		5413	5414		6008		6011		6013	6014	
	5411	5412		5415	5416	6009	6010	6012	6015		6016	
A		To Be Determined										
B												
C												
D												
E												
F												
G												
H												
Q	Metric ¹											
	Imperial											
R	Metric (mm c/c)											
	Imperial (in. c/c)											
S	Metric ¹											
	Imperial											
T	Metric ¹											
	Imperial											
Reinforcing Rod	M (kgs.)											
	I (lbs.)											
Wire Mesh (sq. ft.)												
Concrete (3000 psi) Cubic Yards	Footing											
	Slab											

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

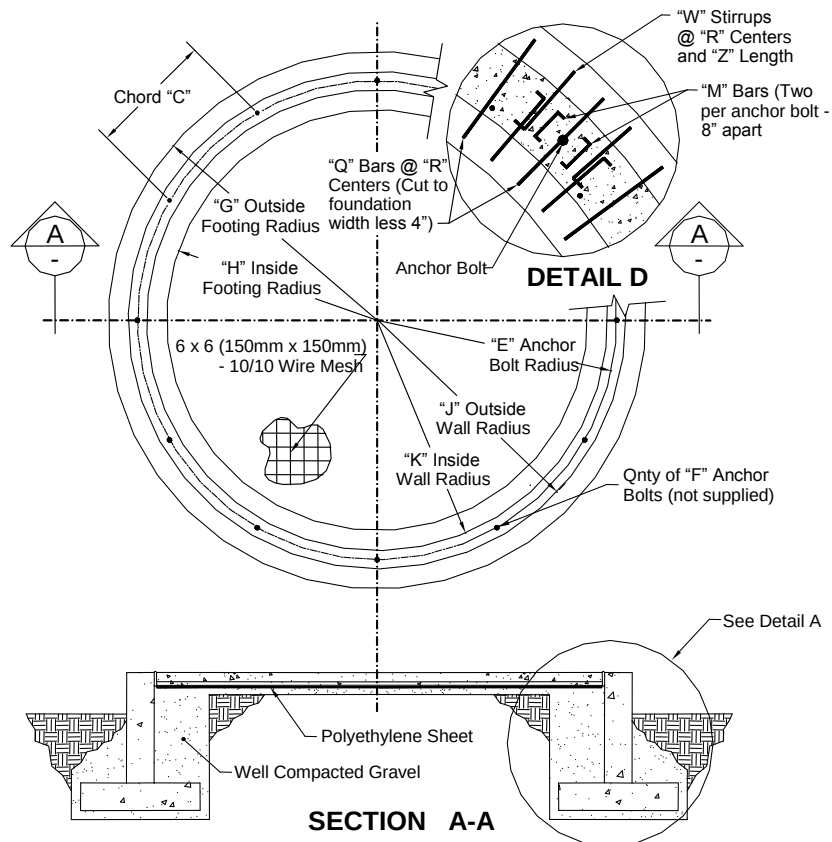
Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.4. T-Footing — Floating Slab foundation

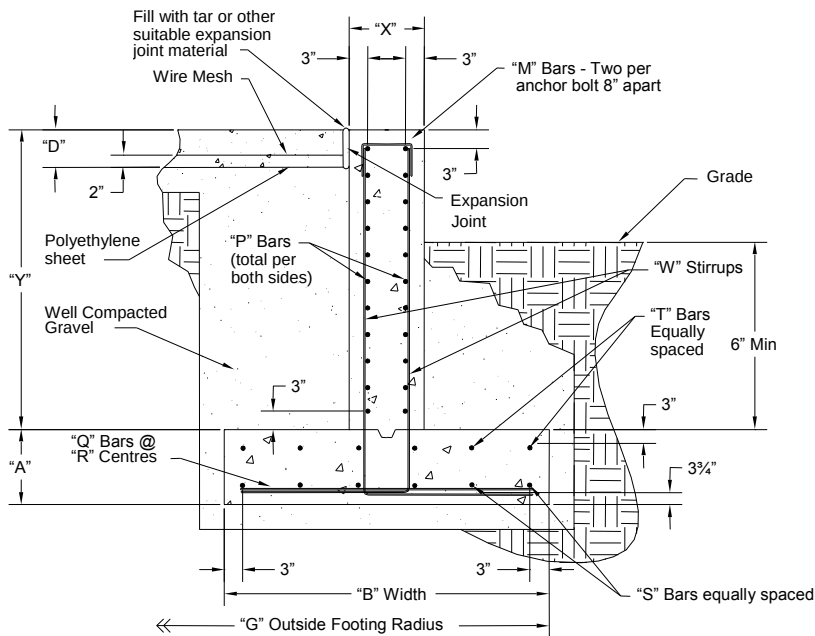
1. Choose a site that is well drained, and has a minimum soil-bearing capacity of 4000 lbs. per sq. ft. (192 kPa). If soil-bearing capacity is not known, consult a local engineering representative.
2. Use minimum 4" to 6" (100 - 150mm) of well-compacted coarse granular material below slab and curb footing.

Figure 5. T-Footing Foundation Layout

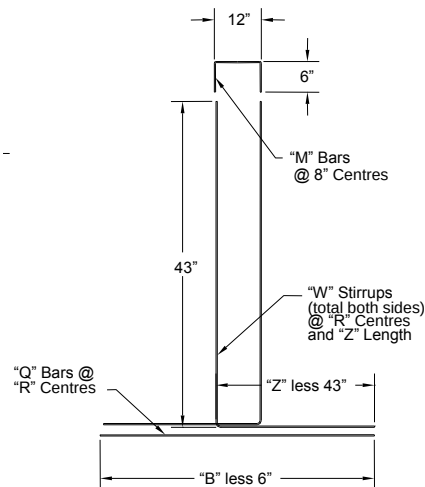


3. Concrete strength shall be minimum 3000 lbs/sq inch [21 MPa] compressive strength.
4. The foundation must be level to within 1/4" over a span of 4 feet [6 mm over 1200 mm]. Any variance from level must be shimmed under an upright. (See [Section 5.19 – Commercial Bin Upright Assembly on page 61.](#))

The bin wall must have a uniform support to carry the wall loads. Weather-sealing the bottom is also made easier if the foundation is level. Camber the outside edge of the concrete to ensure that water drains off.

Figure 6. T-Footing Foundation Layout - Detail A**DETAIL A**

5. Locate anchor bolts as shown in [Figure 5 on page 29](#) and specified in the following T-Footing Specification tables, to ensure bin roundness
6. Re-bar end laps are not included. Add 15" [380mm] for each lap. To estimate weight of end lap: add 0.5 lbs. for #3 and 1.0 lbs. for #4, 0.3 kg for 10M and 0.6 kg for 15M.

Figure 7. T-Footing Foundation Layout - Detail C (Re-bar)**DETAIL C (RE-BAR)**

(See Detail D for Bar Placement)

7. Concrete slump for reinforced wall to be 5" (130 mm) maximum and 2" (50 mm) minimum.

5.4.1 T-Footing Specifications – 15' to 27' Series

Table 11. T-Footing Specifications — 15' to 27' (externally stiffened)

BIN MODEL		1513 1514	1813	1814	2113	2114	2413	2414 2415 2416	2713	2714 2715 2716	
A		To Be Determined	2'0"	To Be Determined	2'0"	To Be Determined	2'0"	2'0"	2'0"	2'0"	
B			2'10"		3'2"		3'4"	4'6"	3'6"	4'10"	
C			4'9-1/4"		4'9-3/16"		4'9-1/8"	4'9-1/16"	4'9-1/16"	4'9-1/16"	
D			6"		6"		6"	6"	6"	6"	
E			9'2-9/16"		10'8-1/2"		12'2-3/8"	12'2-3/8"	13'8-5/16"	13'8-5/16"	
F			12		14		16	16	18	18	
G			10'7"		12'3"		13'10"	14'5"	15'5"	16'1"	
H			7'9"		9'1"		10'6"	9'11"	11'11"	11'3"	
J			9'11"		11'5"		12'11"	12'11"	14'5"	14'5"	
K			8'5"		9'11"		11'5"	11'5"	12'11"	12'11"	
X			1'6"		1'6"		1'6"	1'6"	1'6"	1'6"	
Y			2'0"		2'0"		2'0"	2'0"	2'0"	2'0"	
M	Metric ¹		24 - 10M		28 - 10M		32 - 10M	32 - 10M	36 - 10M	36 - 10M	
	Imperial		24 - #3		28 - #3		32 - #3	32 - #3	36 - #3	36 - #3	
P	Metric ¹		6 - 10M		6 - 10M		6 - 10M	6 - 10M	6 - 10M	6 - 10M	
	Imperial		8 - #3		8 - #3		8 - #3	8 - #3	8 - #3	8 - #3	
Q	Metric ¹		0		0		0	0	51 - 10M	0	65 - 10M
	Imperial		0		0		0	0	54 - #3	0	65 - #3
R	Metric (mm c/c)		457		457		457	432	432	406	
	Imperial (in. c/c)		18		18		18	17	17	16	
S	Metric ¹		4 - 10M		4 - 10M		5 - 10M	6 - 10M	5 - 10M	7 - 10M	
	Imperial	6 - #3	6 - #3	6 - #3	9 - #3	7 - #3	9 - #3				
T	Metric ¹	4 - 10M	4 - 10M	4 - 10M	6 - 10M	4 - 10M	6 - 10M				
	Imperial	5 - #3	6 - #3	6 - #3	8 - #3	6 - #3	9 - #3				
W	Metric ¹	78 - 10M	96 - 10M	132 - 10M	102 - 10M	160 - 10M	130 - 10M				
	Imperial	78 - #4	90 - #4	102 - #4	108 - #4	122 - #4	130 - #4				
Z		5'3"	5'5"	5'6"	6'1"	5'7"	6'3"				
Reinforcing Rod	M (kgs.)	303	364	465	542	540	670				
	I (lbs.)	704	851	974	1263	1161	1516				
Wire Mesh (sq. ft.)		223	309	410	410	524	524				
Concrete (3000 psi) Cubic Yards	Footing	12.1	15.8	18.9	25.5	22.3	60.1				
	Wall	6.4	7.5	8.5	8.5	9.6	9.6				
	Slab	4.2	5.8	7.6	7.6	9.7	9.7				

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.4.2 T-Footing Specifications – 30' to 39' Series

Table 12. T-Footing Specifications — 30' to 39' (externally stiffened)

BIN MODEL		3013	3014 3015 3016	3313	3314 3315 3316	3613	3614 3615 3616	3913	3914 3915 3916
A		2'0"	2'0"	2'0"	2'0"	2'0"	2'0"	To Be Determined	
B		3'8"	5'0"	3'10"	5'4"	4'0"	5'6"		
C		4'9"	4'9"	4'8-15/16"	4'8-15/16"	4'8-15/16"	4'8-15/16"		
D		6"	6"	6"	6"	6"	6"		
E		15'2-3/16"	15'2-3/16"	16'8-1/8"	16'8-1/8"	18'2"	18'2"		
F		20	20	22	22	24	24		
G		17'0"	17'8"	18'7"	19'4"	20'2"	20'11"		
H		13'4"	12'8"	14'9"	14'0"	16'2"	15'5"		
J		15'11"	15'11"	17'5"	17'5"	18'11"	18'11"		
K		14'5"	14'5"	15'11"	15'11"	17'5"	17'5"		
X		1'6"	1'6"	1'6"	1'6"	1'6"	1'6"		
Y		2'0"	2'0"	2'0"	2'0"	2'0"	2'0"		
M	Metric ¹	40 - 10M	40 - 10M	44 - 10M	44 - 10M	48 - 10M	48 - 10M		
	Imperial	40 - #3	40 - #3	44 - #3	44 - #3	48 - #3	48 - #3		
P	Metric ¹	6 - 10M	6 - 10M	8 - 10M	8 - 10M	8 - 10M	8 - 10M		
	Imperial	8 - #3	8 - #3	10 - #3	10 - #3	10 - #3	10 - #3		
Q	Metric ¹	64 - 10M	76 - 10M	70 - 10M	90 - 10M	76 - 10M	106 - 10M		
	Imperial	64 - #3	82 - #3	70 - #3	90 - #3	76 - #3	106 - #3		
R	Metric (mm c/c)	457	356	457	356	457	330		
	Imperial (in. c/c)	18	14	18	14	18	13		
S	Metric ¹	5 - 10M	7 - 10M	5 - 10M	7 - 10M	6 - 10M	8 - 10M		
	Imperial	6 - #3	6 - #3	9 - #3	6 - #4	9 - #4	6 - #4		
T	Metric ¹	4 - 10M	4 - 10M	6 - 10M	7 - 10M	6 - 10M	7 - 10M		
	Imperial	7 - #3	6 - #4	7 - #3	6 - #4	4 - #4	6 - #4		
W	Metric ¹	128 - 10M	152 - 10M	140 - 10M	180 - 10M	152 - 10M	212 - 10M		
	Imperial	128 - #4	164 - #4	140 - #4	180 - #4	152 - #4	212 - #4		
Z		5'8"	6'4"	4'9"	6'6"	5'10"	6'7"		
Reinforcing Rod	M (kgs.)	607	765	721	955	819	1112		
	I (lbs.)	1344	1850	1604	2142	1768	2512		
Wire Mesh (sq. ft.)		653	653	796	796	953	953		
Concrete (3000 psi) Cubic Yards	Footing	26.3	35.3	29.8	41.4	33.9	46.5		
	Wall	10.6	10.6	11.7	11.7	12.7	12.7		
	Slab	12.1	12.1	14.8	14.8	17.7	17.7		

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.4.3 T-Footing Specifications – 42' to 51' Series

Table 13. T-Footing Specifications — 42' to 51' (externally stiffened)

BIN MODEL		4213	4214 4215 4216	4513	4514 4515 4516	4813	4814 4815 4816	5113	5114 5115 5116
A		2'0"	2'0"	To Be Determined		2'0"	2'0"	To Be Determined	
B		4'2"	5'10"			4'4"	6'2"		
C		4'8-13/16"	4'8-13/16"			4'8-3/4"	4'8-3/4"		
D		6"	6"			6"	6"		
E		21'1-3/16"	21'1-3/16"			24'1-5/8"	24'1-5/8"		
F		28	28			32	32		
G		23'3"	24'1"			26'4"	27'3"		
H		19'1"	18'3"			22'0"	21'1"		
J		21'11"	21'11"			24'11"	24'11"		
K		20'5"	20'5"			23'5"	23'5"		
X		1'6"	1'6"			1'6"	1'6"		
Y		2'0"	2'0"			2'0"	2'0"		
M	Metric ¹	56 - 10M	56 - 10M			64 - 10M	64 - 10M		
	Imperial	56 - #3	56 - #3			64 - #3	64 - #3		
P	Metric ¹	8 - 10M	10 - 10M			10 - 10M	10 - 10M		
	Imperial	8 - #4	8 - #4			8 - #4	10 - #4		
Q	Metric ¹	89 - 10M	133 - 10M			102 - 10M	166 - 10M		
	Imperial	89 - #3	133 - #3			102 - #3	166 - #3		
R	Metric (mm c/c)	457	305			457	279		
	Imperial (in. c/c)	18	12			18	11		
S	Metric ¹	6 - 10M	7 - 10M			6 - 10M	8 - 10M		
	Imperial	5 - #4	6 - #4			5 - #4	7 - #4		
T	Metric ¹	5 - 10M	8 - 10M			6 - 10M	8 - 10M		
	Imperial	4 - #3	6 - #4			4 - #4	6 - #4		
W	Metric ¹	178 - 10M	266 - 10M			204 - 10M	332 - 10M		
	Imperial	178 - #4	266 - #4			204 - #4	332 - #4		
Z		5'11"	6'9"			6'0"	6'11"		
Reinforcing Rod	M (kgs.)	962	1454			1216	1750		
	I (lbs.)	2396	3285			2738	4269		
Wire Mesh (sq. ft.)		1310	1310			1723	1723		
Concrete (3000 psi) Cubic Yards	Footing	41.1	57.5			48.8	69.4		
	Wall	14.8	14.8			16.9	16.9		
	Slab	24.3	24.3			31.9	31.9		

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.4.4 T-Footing Specifications – 54' to 60' Series

Table 14. T-Footing Specifications — 54' to 60' (externally stiffened)

BIN MODEL		5410 5411	5412	5413	5414 5415 5416	6008 6009 6010	6011 6012 6013	6014 6015 6016
A		To Be Determined						
B								
C								
D								
E								
F								
G								
H								
J								
K								
X								
Y								
M	Metric ¹							
	Imperial							
P	Metric ¹							
	Imperial							
Q	Metric ¹							
	Imperial							
R	Metric (mm c/c)							
	Imperial (in. c/c)							
S	Metric ¹							
	Imperial							
T	Metric ¹							
	Imperial							
W	Metric ¹							
	Imperial							
Z								
Reinforcing Rod	M (kgs.)							
	I (lbs.)							
Wire Mesh (sq. ft.)								
Concrete (3000 psi) Cubic Yards	Footing							
	Wall							
	Slab							

¹For Canadian Metric to European Standard EN 10080/BS 4449 conversion, see [Section 5.4.5 – Canadian Metric to European Conversion Table on page 35](#).

Note

1. The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.06" x 1.80" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.
2. One 1/2" x 3" (minimum embedment) anchor bolt needs to be installed into every slot into the bottom angle (6 per angle) to ensure that the bin is fixed round at the bottom.

5.4.5 Canadian Metric to European Conversion Table

Table 15. Canadian Metric to European Standard EN 10080/BS 4449 Conversion

Canadian Metric	European Metric	Nominal Diameter (mm)
10 M	12,0	12 mm
15 M	16,0	16 mm
20 M	20,0	20 mm
25 M	25,0	25 mm

5.5. Typical Non-Structural Roof Installation

The following is a step-by-step procedure for assembling a non-structural roof system.

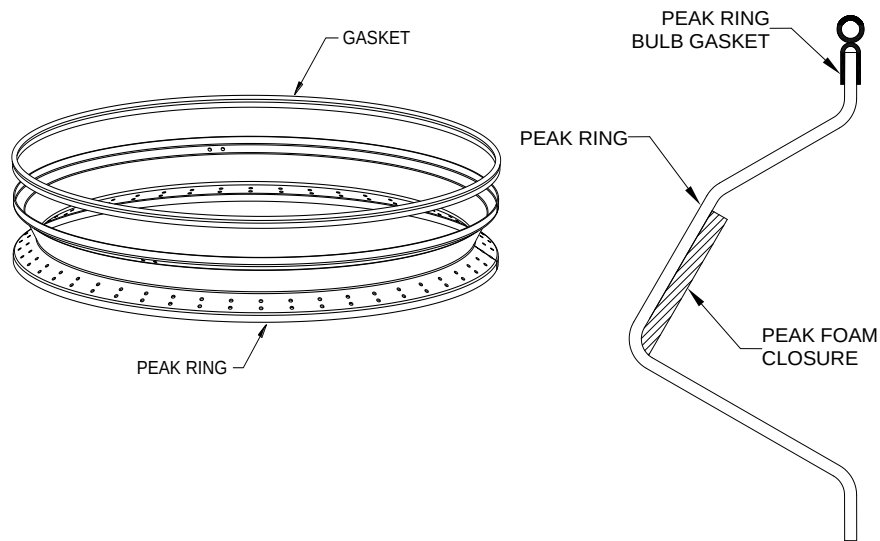
Preparation

1. Inspect the concrete foundation to insure that the foundation meets all the requirements of the installation.
2. Plan the assembly:
 - a. Determine the desired bin orientation.
 - b. Determine the locations of bin features and accessories (Westeel logo, fall restraint brackets, Grain Gauge, unloading devices, outside ladder, spiral stairs).

These considerations affect the location of the inspection hatch roof panel and the placement of the roof ladder or roof stairs.

3. Prepare the peak ring:
 - a. Install the supplied bulb gasket around the top of the peak ring. Trim to fit.
 - b. Install the foam closure gasket around the center section of the peak ring. Trim to fit.

Refer to [Figure 8 on page 36](#).

Figure 8. Gasket and Foam Closure Assembly to Peak Ring

4. Install the center post making sure the post is vertical, braced and anchored properly for safe installation.
5. Lay out the bin circumference (for the bottom tier of wall sheets) on the foundation:
 - a. Anchor a string to the exact center of the concrete foundation.
 - b. Determine the required string length using [Table 16 on page 36](#).

Note

The radius values given in the chart are 3/4" smaller than the wall sheet radius at the bottom. This ensures that the scribed circle can be seen during assembly. A perfectly placed ring of sheets should be 3/4" on the outside of this scribed circle.

- c. Scribe the bin circumference onto the foundation.

Important

Follow these steps carefully. It is imperative that the bin be as round as possible.

Table 16. Scribe Radius and Peak Ring Height (1 and 2 tier)

Nominal Bin Dia.	Scribe Radius		Top of Peak Ring Height "H" with 1 tier of wall sheets		Top of Peak Ring Height "H" with 2 tiers of wall sheets	
	(ft)	(ft in)	(ft in)	(m)	(ft in)	(m)
15		7'4-3/4"	7'9-3/8"	2.372	11'5-3/8"	3.489
18		8'10-11/16"	8'7-3/4"	2.635	12'3-3/4"	3.753
21		10'4-9/16"	9'6-1/16"	2.897	3'2-1/16"	4.015
24		11'10-1/2"	10'4-7/16"	3.160	14'0-7/16"	4.279
27		13'4-3/8"	11'2-3/4"	3.423	14'10-3/4"	4.540
30		14'10-5/16"	11'7-5/8"	3.546	15'3-5/8"	4.664
33		16'4-3/16"	12'5-15/16"	3.808	16'1-15/16"	4.910
36		17'10-1/8"	13'4-1/4"	4.071	17'0-1/4"	5.188
39		19'4"	14'2-5/8"	4.334	17'10-5/8"	5.452

Table 16 Scribe Radius and Peak Ring Height (1 and 2 tier) (continued)

Nominal Bin Dia.	Scribe Radius		Top of Peak Ring Height "H" with 1 tier of wall sheets		Top of Peak Ring Height "H" with 2 tiers of wall sheets	
(ft)	(ft in)	(m)	(ft in)	(m)	(ft in)	(m)
42	20'9-15/16"	6.348	15'0-15/16"	4.596	18'8-15/16"	5.714
45	22'3-13/16"	6.803	15'11-5/16"	4.859	19'7-5/16"	5.977
48	23'9-3/4"	7.258	16'9-5/8"	5.121	20'5-5/8"	6.239
51	25'3-5/8"	7.712	17'5-5/8"	5.325	21'1-5/8"	6.442
54	26'9-9/16"	8.167	18'4"	5.588	22'0"	6.706

Assemble the Top Tier of Wall Sheets

1. Assemble a single tier of wall sheets if single-tier uprights are included in the bin package.
2. Assemble two tiers if two-tier uprights are included in the bin package.
3. Refer to the Appendix for information on proper hardware usage.
4. After the first ring of wall sheets has been assembled, check the position and roundness of the ring:
 - a. Verify that the bin is round, with **no more than 0.75" variation** on the radius, when measured from the center of the bin.
 - b. Verify that the wall sheets form a smooth circle with no flat spots or cauliflower shaped curves.

Note

Correcting for roundness becomes much more difficult the longer you wait.

5. When setting jacks, make sure they are also set round and that they are anchored to the concrete.

Install the Top Ring Angle Sections

Important

Read and fully understand the following instructions before attaching the top ring angles to the top of the wall sheets. There are timing considerations for the inspection hatch, outside ladder, inside ladder, and other components.

1. Attach the top ring angle to the inside top of the wall sheets.
 - a. Do not align the top ring angle joints with wall sheet joints.
 - b. Make sure that the top ring angle joints are at least two or three wall sheet holes away from the Grain Gauge cutout.
 - c. If building a stiffened bin, do not install the top angle bolt that lines up with the stiffener holes at this time. It needs to be installed with the stiffeners.
2. There are six pairs of vertically aligned circular and square holes in the top ring angle. (See [Figure 9 on page 38](#))
 - a. These are used to attach the center holes at the bottom of the roof sheets to the top ring angle. This locks in the correct centering location of the roof sheets, which is important for locating roof accessories such as the inspection hatch, roof ladder, roof stairs, etc.
 - b. Whichever pair of vertically aligned circular or square holes you choose to attach the first roof sheet center hole to, the next pair of vertically aligned circular or square holes over, to the left or right, will be used as a Bird Stop location. (See [Figure 9 on page 38](#))
 - c. The next pair of vertically aligned circular or square holes over from the Bird Stop location, will be another roof sheet center hole location.
 - d. This pattern will repeat all the way around the bin.

Note

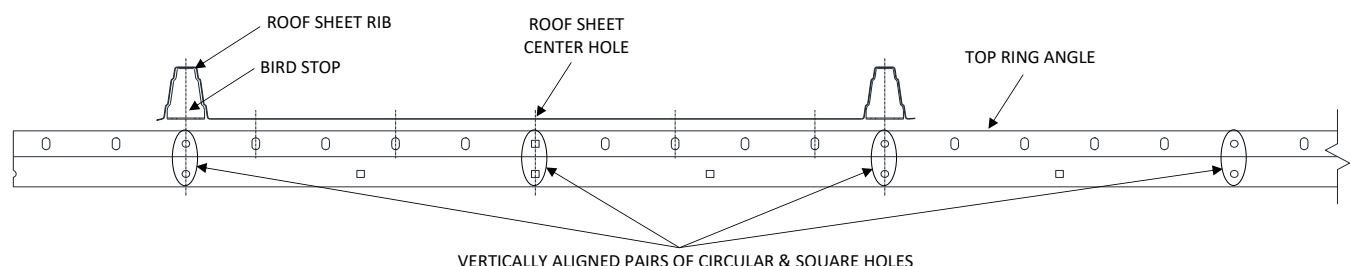
To make timing easier of other parts, and accessories on the bin, best practice is to mark all important locations on the top angle using a marker.

3. Because the inspection hatch is in the center of the roof sheet, the inspection hatch will be centered on one set of the vertically aligned circular or square holes.
 - a. If this bin is equipped with an inside ladder, center the inspection hatch above the inside ladder.
 - b. Do not align the inside ladder with stiffeners.
 - c. Do not attach inside ladder brackets at stiffener locations or vertical seams.
4. The slots in the top ring angle align with the other non-center holes in the roof sheet.

Note

If using bin jacks through the top ring angle, jack bolts will need to be longer. Knowing jack placement before attaching the top ring angles will eliminate the need for replacing bolts.

Figure 9. Top Ring Angle Timing Details

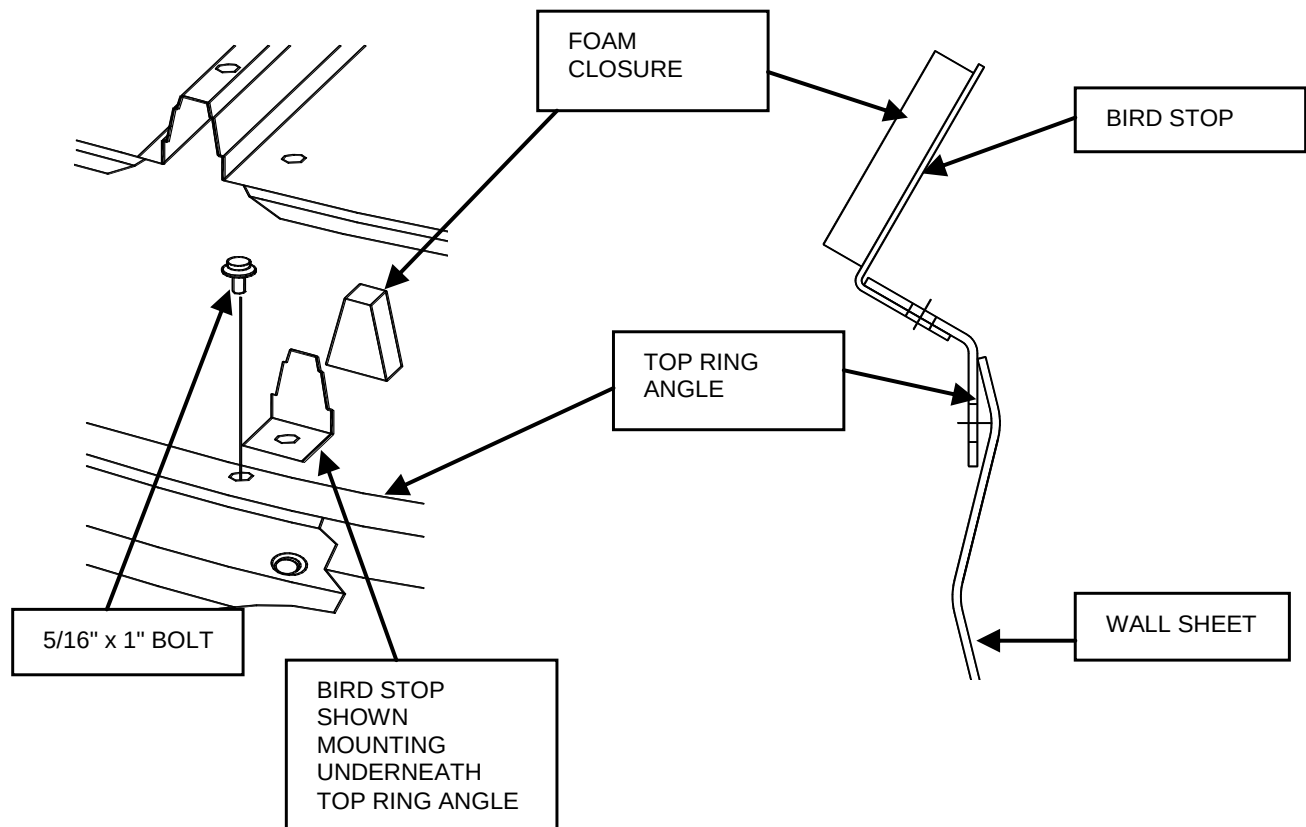


Install Bird Stops

Bird stops consist of a metal bird stop bracket, an adhesive backed foam closure and a nut and bolt.

1. Install bird stops at the bottom end of all roof panel ribs:
 - a. Best practice is to install bird stops before attaching the roof panels. (Easier access to bolts and aids with timing)
 - b. Install bird stops at locations that are five holes to the left or right of the roof panel center mounting holes in the top ring angle.
 - c. Best practice is to install the bird stop with the flange placed under the top ring angle instead of on top of it.

Figure 10. Bird Stop Installation



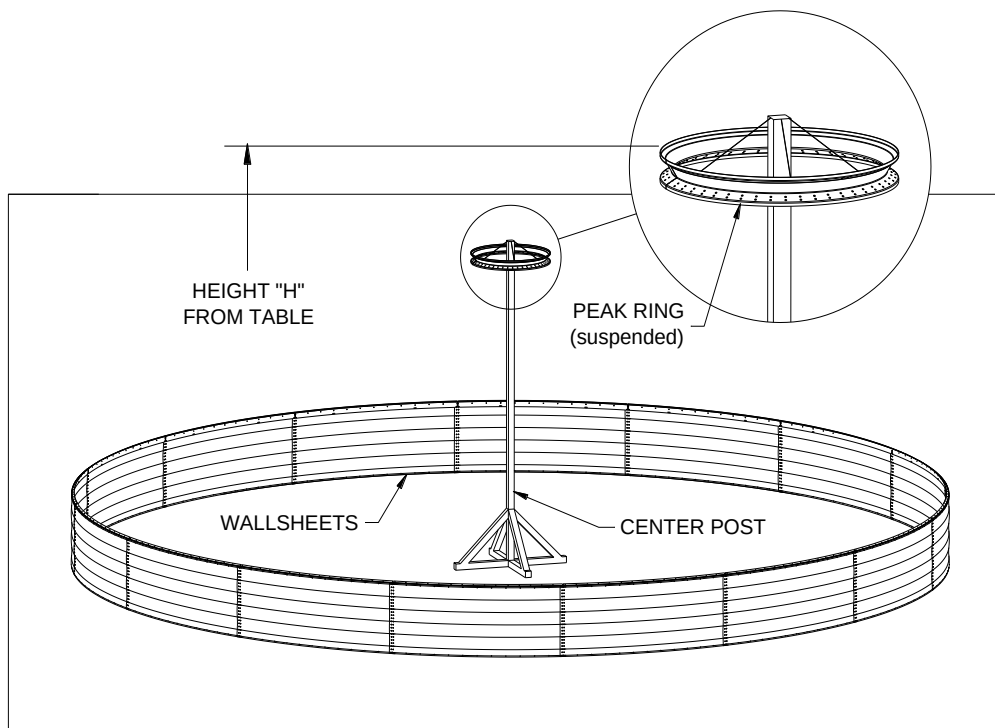
Tip

Mounting the bird stop under the top ring angle prevents it from turning when tightening the nut.

Install the Peak Ring

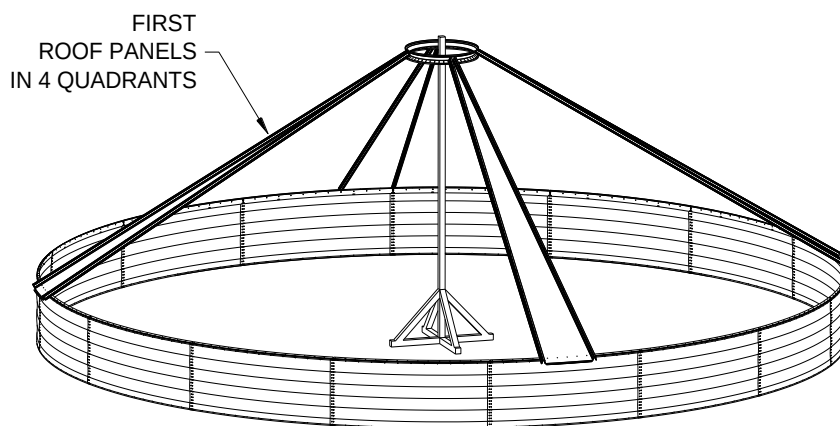
1. Determine the correct peak ring height (H) for the bin size from [Table 16 on page 36](#).
2. Attach the peak ring assembly to the top of the center post at the correct height for the bin being assembled.

Figure 11. Peak Ring Installation



Install the Roof Sheets

1. Attach roof sheets with the narrow end to the peak ring and the wide end to the top ring angle.
 2. Initially, attach four roof panels at the quarter points of the bin. (See [Figure 12 on page 41](#).)
- This will stabilize and support the peak ring during the rest of the installation.

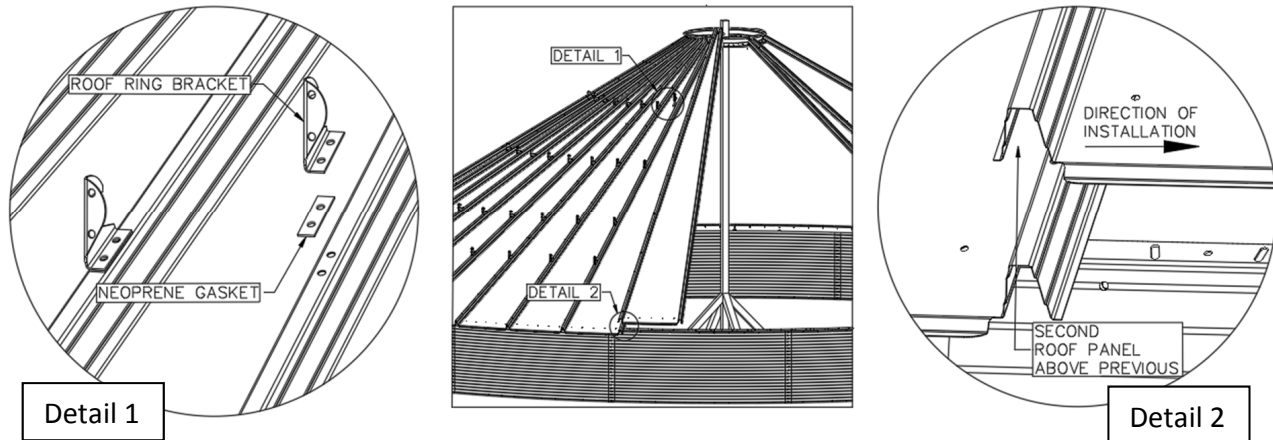
Figure 12. Roof Panel Installation**Tip**

The narrow end of the roof panels gets pushed into the foam closure. Once this occurs there is little room for adjustment at the top end as the roof panel is embedded in the foam. Sometimes the roof panels get “flattened” slightly from bundling, shipping and handling. It is relatively easy to push the ribs together slightly but this should occur before the panel is seated in the foam. Monitor the alignment of mating roof panels with the underlying holes in the peak ring and make adjustments, if necessary, before anchoring the roof panel into the foam.

Important

Be careful when attaching the bottom of the roof panels to the top ring angle. The center round holes at the bottom of the roof panels must align with either the vertically-aligned round or square holes in the top ring angle. This locks in the correct centering location for the roof sheet. The other non-centre holes in the bottom of the roof sheet align with the round slots in the top ring angle.

3. Make sure that the gap between the roof panel and the peak-ring is sealed by the foam closure.
4. Install the remaining roof panels, working in a counter clockwise direction:
 - a. Attach the center roof panel hole first.
 - b. Use two bolts at each roof panel to peak ring connection.
 - c. Fill in every bolt hole in roof panel ribs with rubber washered bolts to the outside and nuts on the underside.
 - d. Make sure the left roof rib overlaps the right rib of the preceding panel. (See Detail 2 in [Figure 13 on page 42.](#))

Figure 13. Roof Rib Orientation & Roof Ring Bracket Assembly**Important**

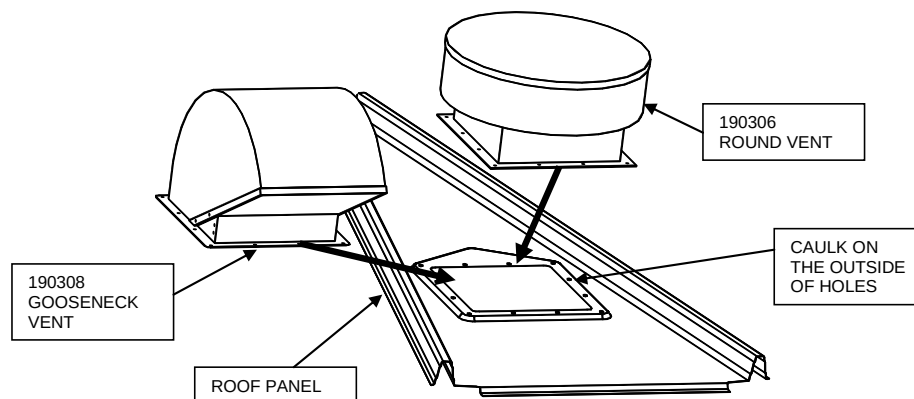
As assembly proceeds, additional support is advised to keep the peak ring level. Alternatively sequentially add roof panels in the different quadrants such that the weight of the panels on the peak ring remains uniformly distributed. Leave all roof bolts loose until the roof is completely assembled, especially those at the peak ring and top ring angle locations.

- e. If the number and diameter of optional roof stiffening rings is known, install the roof ring brackets (with a neoprene gasket under each) in the double rib hole locations as you add roof panels. (See Detail 1 [Figure 13 on page 42.](#))
5. Install vent roof panels where required, as the roof is being assembled.

Distribute vent roof panels evenly around the roof. Ensure that they do not interfere with other roof elements such as roof stairs or rungs, temperature cables, etc.

Note

Westeel supplied roof vents come in two styles: Gooseneck and Round. Both have pre-formed bolt holes for mounting to the roof panel. The vent roof panels have a raised mount section, mounting holes and a pre-cut ventilation opening. No on-site cutting is required. A recommended practice is to assemble the vents to the roof panels at ground level before installing. Place a strip of caulking all the way around the weather side of the connection, position the vent, and bolt into place.

Figure 14. Roof Vent Assembly

6. Install inspection hatch roof panel where required. The inspection hatch can be pre-assembled if desired. (See [Section 5.9 – Inspection Hatch Details on page 52.](#))

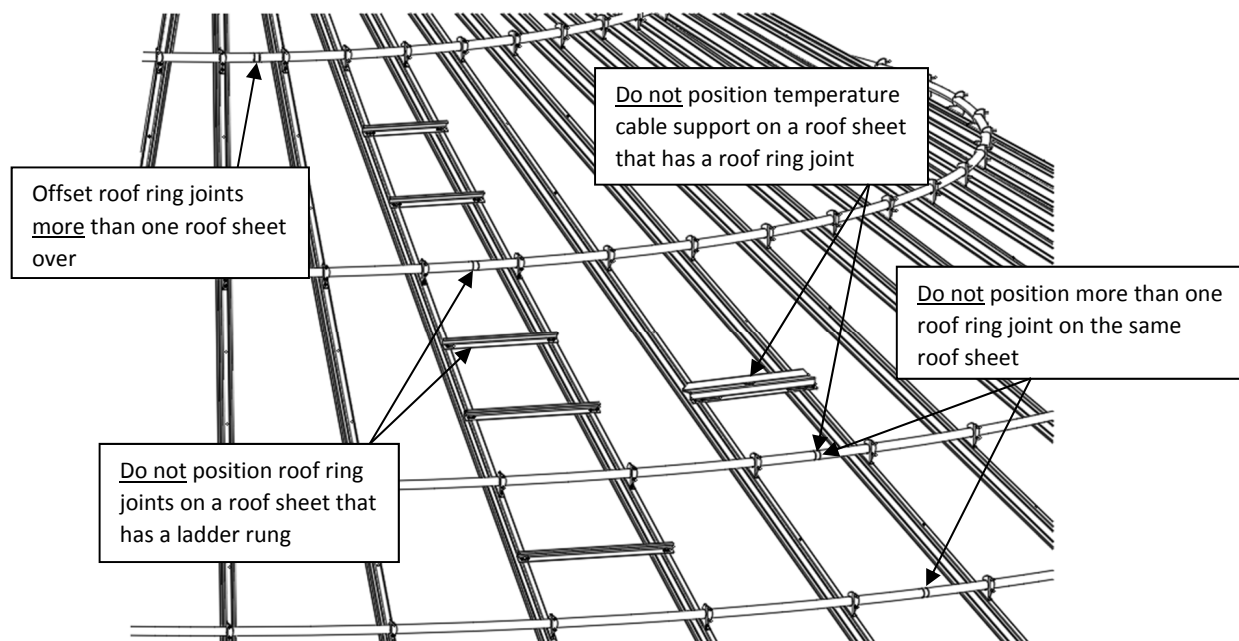
7. Install the roof ladder on the roof sheet to the left of the inspection hatch. (See [Section 5.6 – Roof Ladder Details on page 48.](#))

Enough roof ladder rungs are supplied to bridge across every pair of holes on a single roof panel. Where roof stiffening ring brackets are placed, the ladder rung can be skipped. The roof ring will serve as a rung in this location. Roof ladder rungs are installed with the higher vertical flange facing the peak ring.

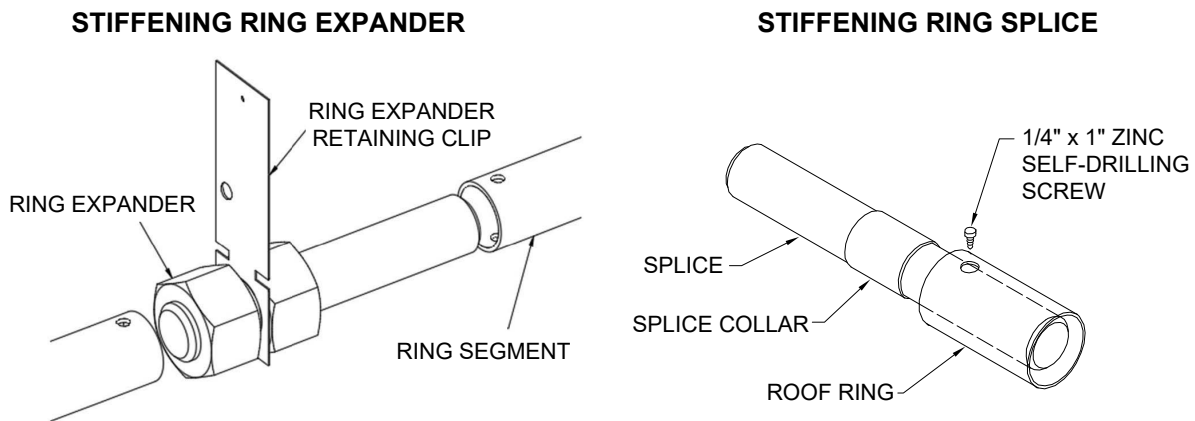
Install Roof Stiffening Rings

1. Add roof stiffening rings (if required):
 - a. See [Table 17 on page 46](#) for standard roof stiffening ring locations.
 - b. On roofs with multiple stiffening rings, stagger the ring joints to avoid having more than one joint on same roof sheet. (See [Figure 15 on page 43.](#))

Figure 15. Roof Stiffening Rings Installation



- c. Join roof stiffening rings together by inserting a ring splice into the facing ends and pushing everything together tightly.
 - d. Secure the splice to the roof ring with a self-drilling screw.

Figure 16. Stiffening Ring Connection

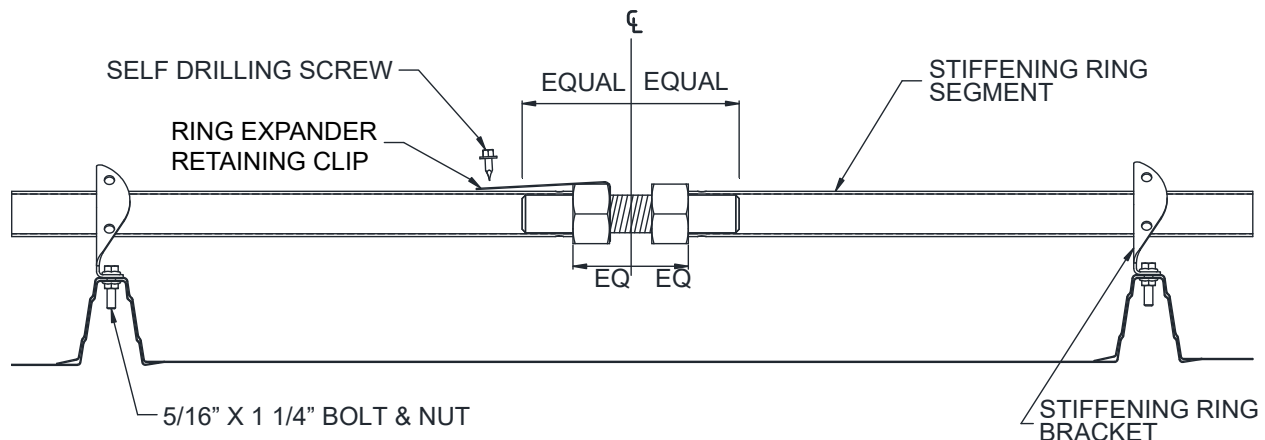
- e. Field cut the last stiffening ring segment so there is a 2½" gap between the mating tubes.
- f. To make the final connection, insert a ring expander between the final ring sections.
- g. Remove one nut from the ring expander, slide the ring expander retaining clip onto the threaded portion against other nut. Thread removed nut back on. (See [Figure 16 on page 44](#)).
- h. With nuts close to one end, insert the long end of the ring expander into one tube and, by flexing both tubes, make the connection to the mating tube. (See [Figure 16 on page 44](#).)
- i. Thread both nuts toward the center.

The green paint on the threaded portion indicates the center.

- j. Slide stiffening ring tubes into the brackets.

Note

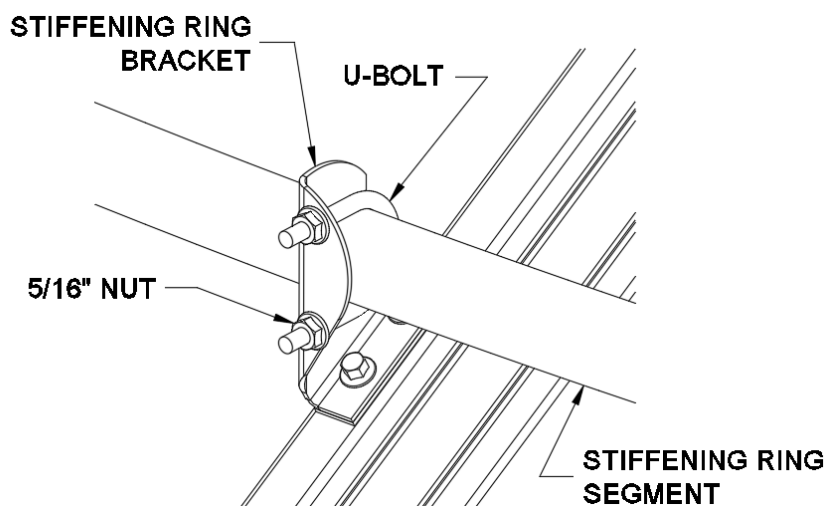
Before expanding, or tightening the roof, all nuts on the ring expanders should be tight together and centered.

Figure 17. Stiffening Ring Connection (side view)

2. Tighten the roof hardware.
3. Expand the nuts on the stiffening ring expanders until the slack has been taken up and the roof is snug.
 - a. Do not overtighten and crown the roof.
 - b. Nuts must be centered on the threaded rod.

- c. Use the painted marking as a guide.
- d. On rings with multiple expanders, the distance between the nuts on all of them should be equal. (See [Figure 17 on page 44.](#))
4. Slide retaining clip against one nut, bend clip over nut and flat against stiffening ring. Secure the retaining clip to the stiffening ring with a self drilling screw. (See [Figure 17 on page 44.](#))
5. Install u-bolts into stiffening ring brackets, tighten the u-bolts until stiffening ring is pulled tight against the brackets. (See [Figure 18 on page 45.](#))

Figure 18. Stiffening Ring Bracket Assembly



6. Once all the roof panels have been installed, make sure all nuts have been tightened.

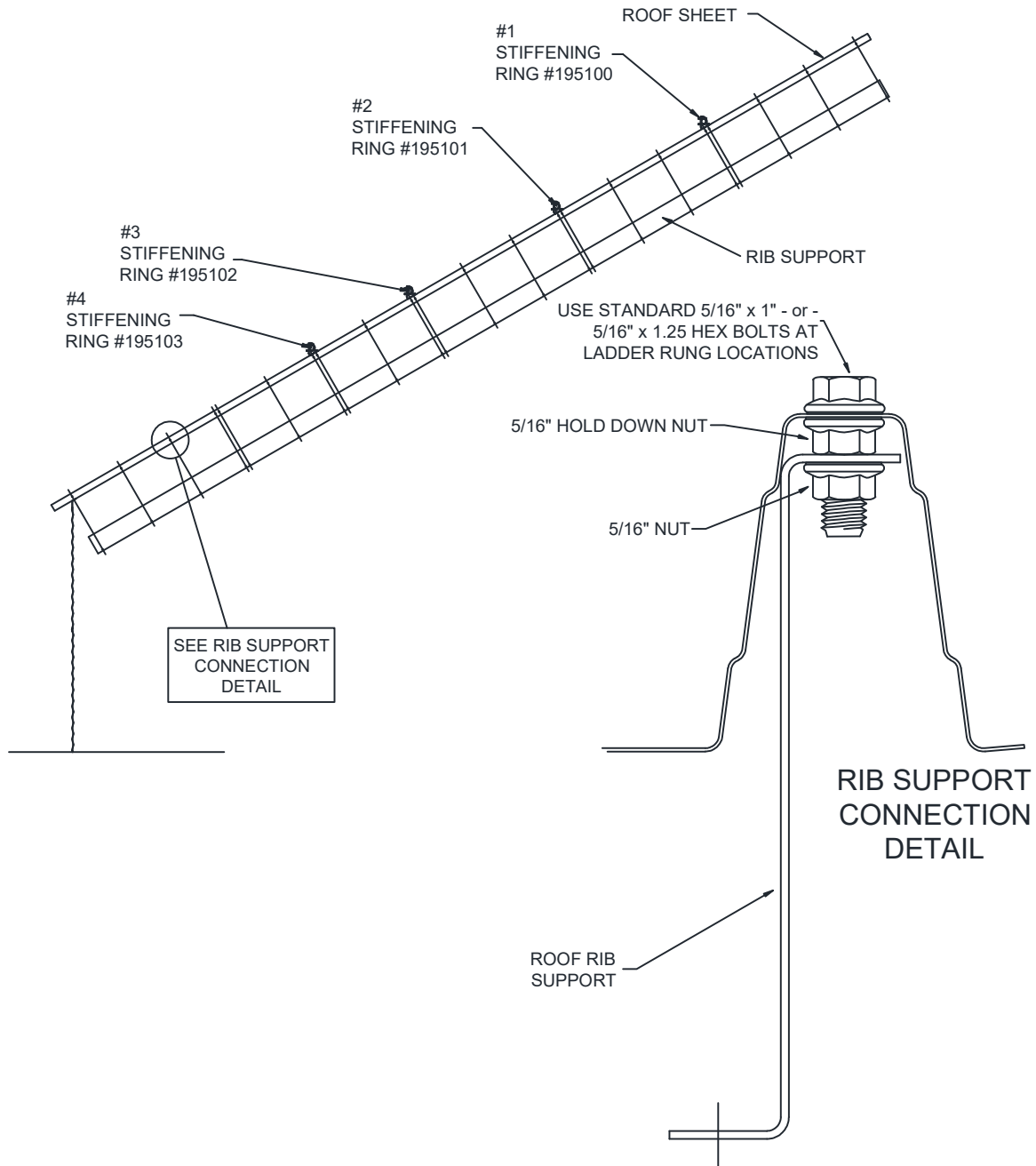
Table 17. Roof Reinforcing Matrix

Component →	Roof Stiffening Ring Tubes				Rib Supports
Location & Colour →	1st - Yellow	2nd - Black	3rd - Red	4th - Green	Under ribs
Chord length (in) →	103.8	167.4	197.6	200.4	Varies
Qty of Expanders →	2	2	3	4	n/a
Bin Series	Component Part Numbers (and Quantity)				
15	n/a	n/a	n/a	n/a	n/a
16					
18					
21					212752 (21)
24					212753 (24)
27	195100 (5)	195101 (5)	195102 (6)	n/a	n/a
30	195100 (5)				212755 (30)
33	195100 (5)				212756 (33)
36	195100 (5)				212757 (36)
39	195100 (5)				212758 (39)
42	195100 (5)	195101 (5)	195102 (6)	195103 (7)	212759 (42)
45	195100 (5)	195101 (5)	195102 (6)		212760 (45)
48	195100 (5)	195101 (5)	195102 (6)		212761 (48)
51	n/a	195101 (5)	195102 (6)		212762 (51)
54		195101 (5)	195102 (6)	195103 (7)	212763 (54)
Notes:	Roofs are supplied as standard or with optional upgrades for higher load capacity				
	<– standard components for all roofs				
	<– optional components supplied with all roof upgrade levels 1 & 2				
	<– optional components supplied with roofs upgraded to level 2				
	Structural roofs with rafter system is available for 48' bins and larger. These raftered roofs do not require stiffening rings or rib supports				

Install Roof Rib Supports

Rib supports are an upgrade that provide additional load capacity when required. Rib supports vary in length, depending on roof size and are designed to fit under the roof panel ribs. They run along the length of the rib from the eave (at the bottom) to near the peak ring (at the top).

1. Install one rib support at each roof rib location:
 - a. Fit the rib support onto the shanks of the existing bolts used to join mating roof ribs.
 - b. Add a second nut to secure the rib supports to the ribs. (See Rib Support Connection Detail in [Figure 19 on page 47.](#))

Figure 19. Roof Rib Support Assembly

(Drawing represents a non-specific example only)

Install Associated Components

1. Assemble bin entry anchor system, roof cap, roof cap opener, ladders and associated components (if applicable).
 - See [Section 5.6 – Roof Ladder Details on page 48](#)
 - See [Section 5.7 – Flat Roof Cap Assembly on page 49](#)
 - See [Section 5.8 – Bin Entry Anchor System: Non-Structured Roof on page 50](#)

5.6. Roof Ladder Details

1. Locate the roof panel containing the roof ladder components to the left or right of the inspection hatch, and in line with the outside ladder.
2. Recommended (for convenience): Attach the roof ladder and a section of the outside ladder early, when the roof section is at ground level.
3. Start at the bottom of the roof with the longest ladder rung supplied and move up the roof using progressively shorter ladder rungs.
4. Bolt ladder rungs to the roof panel ribs using the pre-drilled holes in the ribs.
5. Use 5/16" x 1-1/4" hex bolts and hex nuts (bolts above and nuts underneath).

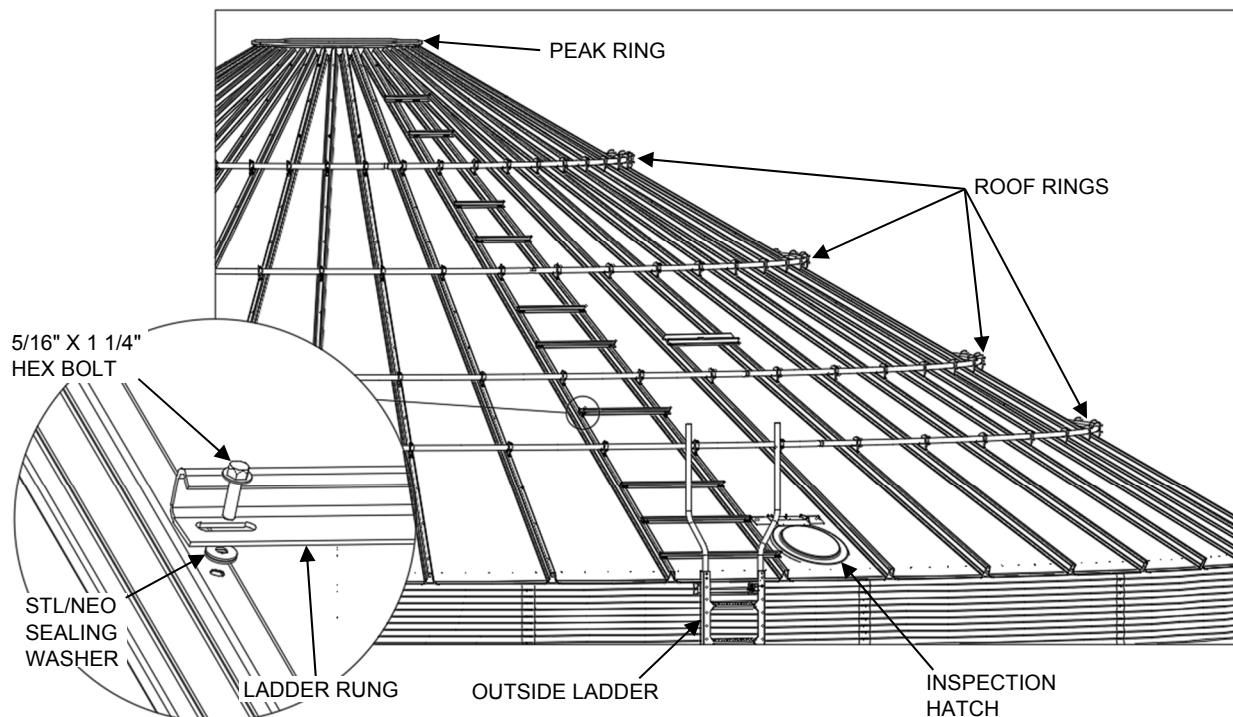
Note

Make sure that a STL/NEO sealing washer is installed between the ladder rung and the roof sheet.

Note

The ladder rung is oriented with the vertical portion facing up towards the peak ring. When a ladder rung is located at a double hole pattern designed for a roof-ring element, bolt through the upper holes and fill the other holes with a 1" hex bolt. No ladder rung is used at a roof-ring location. The ring itself will serve as a step. This ladder rung can be discarded or saved for another job.

Figure 20. Roof Ladder Details



5.7. Flat Roof Cap Assembly

Figure 21. Flat Roof Cap Assembly Detail

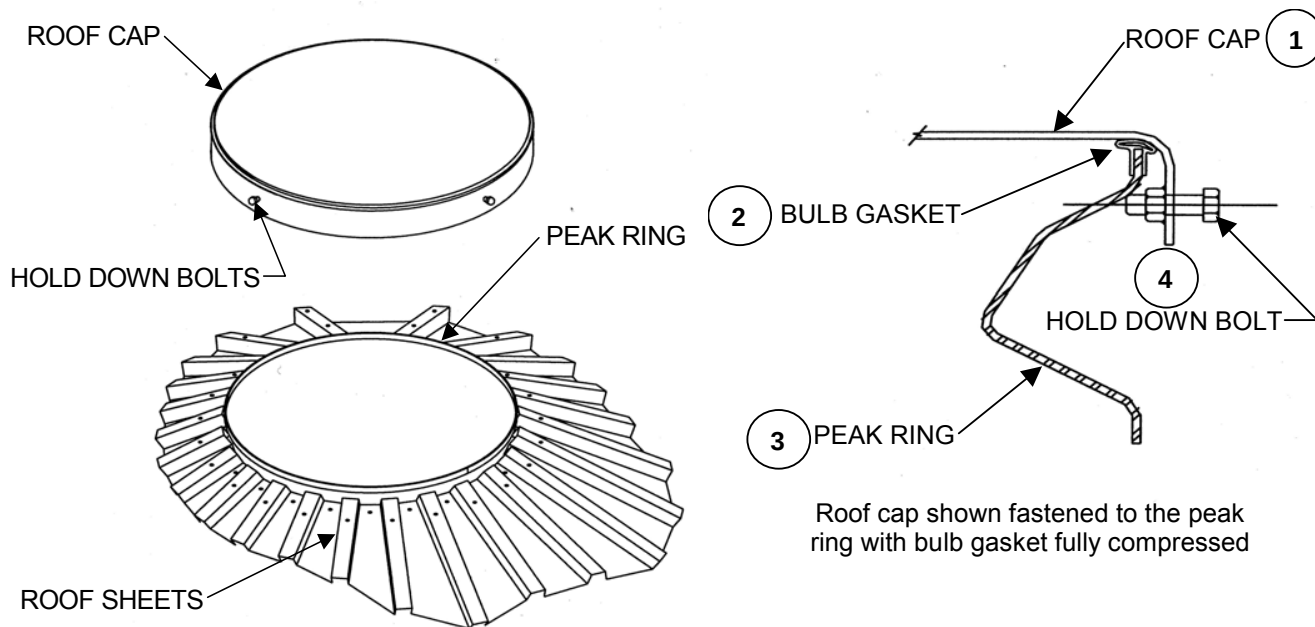


Table 18. Flat Roof Cap Part Numbers

Item	Description	Part No.	Used On
1	34" HEAVY DUTY FLAT CAP (for use with 33" peak ring)	195090	up to 27' Bin
	53.5" HEAVY DUTY FLAT CAP (for use with 52" peak ring)	195087	30' to 48' Bin
	61.5" HEAVY DUTY FLAT CAP (for use with 60" peak ring)	195091	51' & 54' Bin
2	PEAK RING BULB GASKET 105" LONG	195149	up to 27' Bin
	PEAK RING BULB GASKET 168" LONG	195150	30' to 48' Bin
	PEAK RING BULB GASKET 105" LONG	2 x 195149	51' & 54' Bin
3	3/8" x 1-1/2" HEX FLANGE BOLT (supplied with the lid)	193797	All

1. Fasten the bulb gasket onto the top rim of the peak ring. Trim to fit.
2. Place the roof cap on the peak ring with two of the hold down bolts, making sure they are clear of the roof ladder.
3. Locate bolts between the roof ribs.
4. Tighten the hold-down bolt opposite the roof ladder until approximately 3/8" of the bolt is protruding past the welded nut.
5. Tighten the two bolts near the roof ladder until the roof cap pulls down firmly and cannot be moved.
6. Tighten all other roof cap bolts similarly.
7. Ensure that the roof cap is fully secured around the peak ring.
8. For a non-structural roof that is supporting a catwalk, install six flat cap clips (213437) as shown in the Westeel catwalk manual 213440. These clips are provided in the Westeel catwalk peak support modules.

5.8. Bin Entry Anchor System: Non-Structured Roof

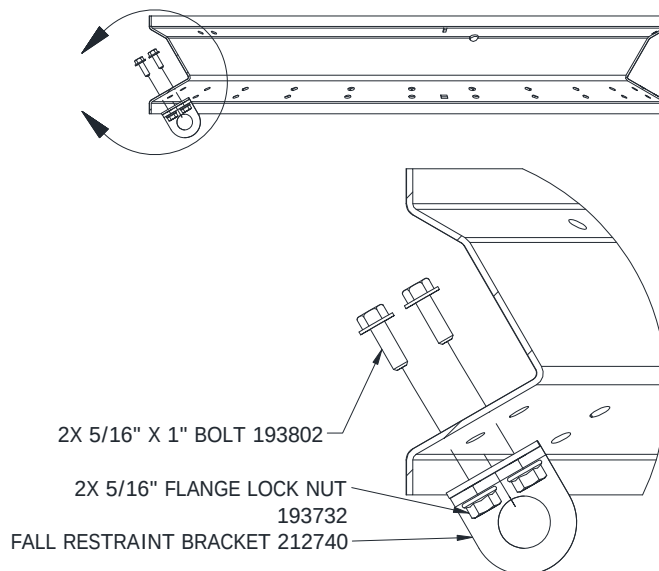
Important

The fall restraint bracket is rated for a maximum load of 2,000 lbs. The bin owner and user are responsible for correctly installing, using, and operating the Bin Entry Anchor System. The rope, pulley, and harness are not supplied by Westeel.

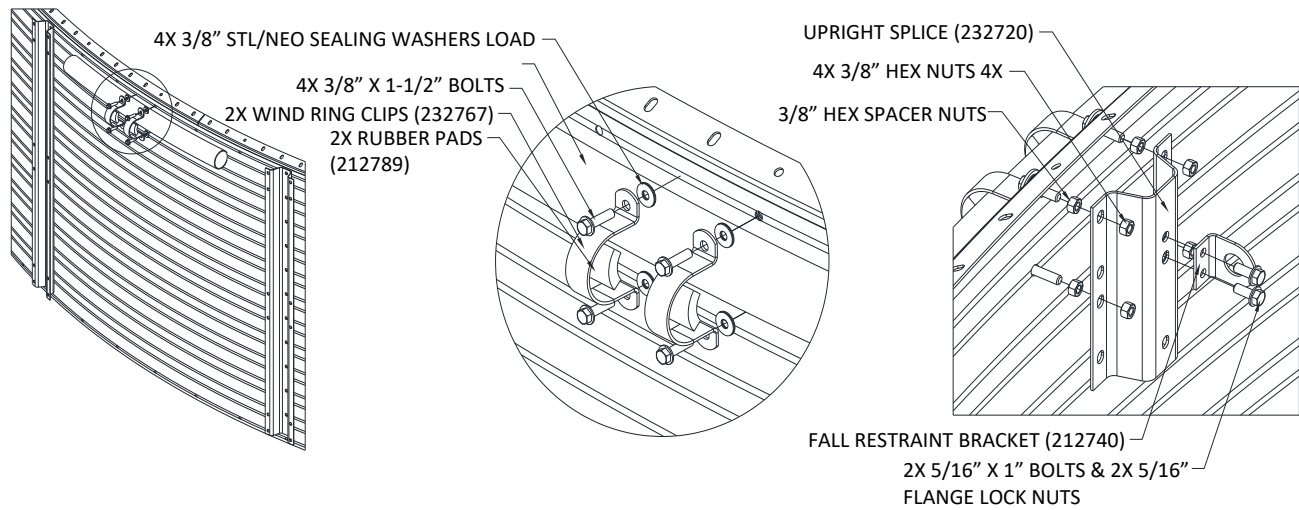
WARNING Failure to install correctly as instructed below may result in serious injury or death.

1. Install fall restraint bracket under peak ring as shown. Make sure to bolt the fall restraint bracket to the roof sheet with the inspection hatch cut out. (See [Figure 22 on page 50](#).)

Figure 22. Installing the fall restraint bracket



2. Bolt the second fall restraint bracket to the upright splice. Then bolt the upright splice along the top wall sheet horizontal seam with the wind ring clips, sandwiching the load spreader tube and rubber pads as shown. (See [Figure 23 on page 51](#).)
 - a. Field drill holes as needed on the wall sheet and top angle for the wind ring clip connections. Washers must be sandwiched between the wind ring clips and the wall sheet for sealing. (See [Figure 23 on page 51](#).)
 - b. The load spreader tube and upright splice can be installed off center on the wallsheet (between the uprights on stiffened bins), for better reachability from the inspection hatch. Do not install the load spreader tube over a vertical wall sheet seam. (See [Figure 23 on page 51](#).)

Figure 23. Installing the load spreader tube

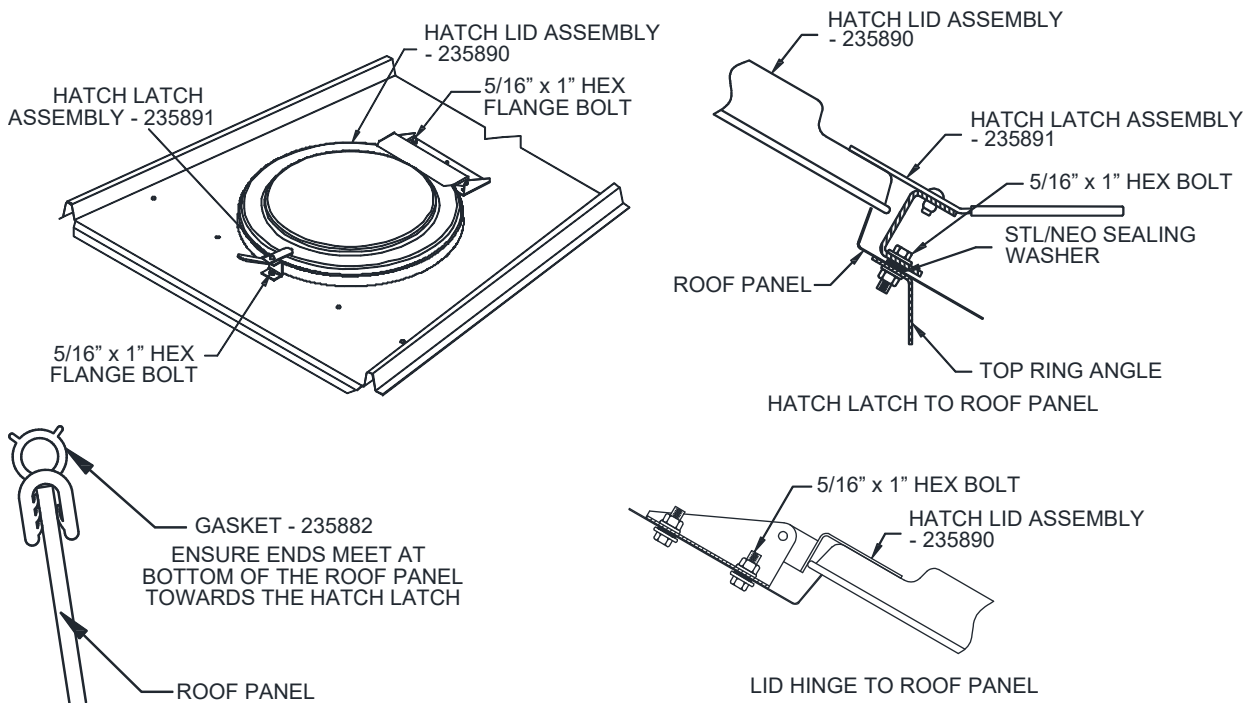
- Place the Fall Restraint Anchor Point Decals (PN 8110-01090) on the bin, see [Section 2.5 – Decal Installation/Replacement on page 7](#) and [Section 2.6 – Safety Decal Locations and Details on page 8](#) for installation instructions and placement.

5.9. Inspection Hatch Details

Installation of inspection hatch

1. Place the inspection hatch gasket (235882) around the lip of the inspection hatch opening. Trim the gasket to fit if necessary.
2. Bolt on the hatch lid assembly (235890) with 5/16" x 1" bolts provided for the roof. For best sealing results, the bolt heads should be on the underside of the roof panel, with the sealing washers pressed against the roof panel.
3. Bolt on the latch assembly (235891) as shown below. The latch is positioned on the center hole of the roof panel and bolts through the top ring angle as shown.

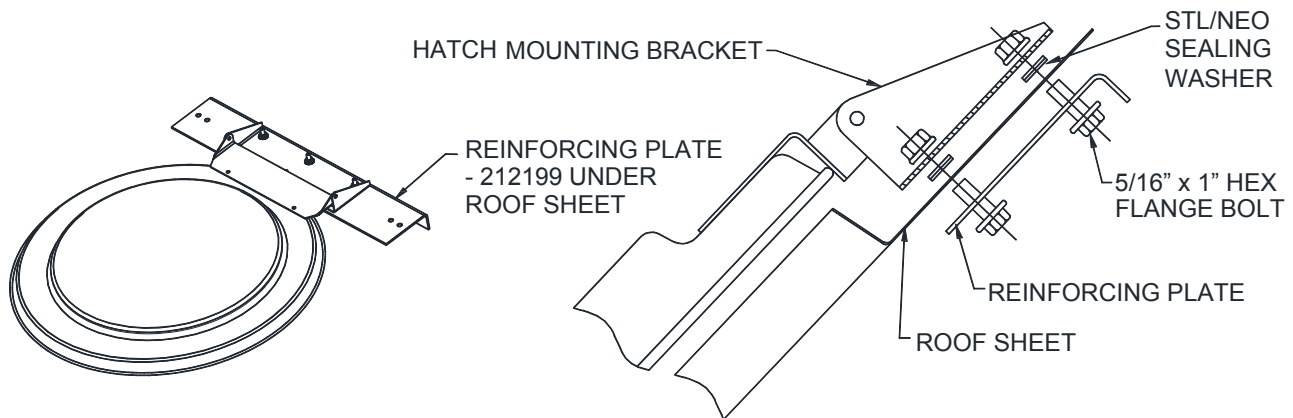
Figure 24. Inspection Hatch Details



5.10. Inspection Hatch Reinforcing Plate

1. For high wind applications, place the inspection hatch reinforcing plate under the roof sheet and secure with 5/16" x 1" hex flange bolts and nuts.
2. The flange on the plate must be located away from the hatch opening to minimize the possibility of interference or injury with inspector.
3. The four outermost mounting holes are used for extra stiffening when the roof sheet width permits. Field drill the roof sheet as needed.

Figure 25. Inspection Hatch Reinforcing Plate Detail



5.11. Typical Structural Roof Installation

For structural roof system installation instructions, please refer to **Structural Roof Manual (212453)**.

5.12. Bin Roundness

It is imperative that the bin be as round as possible. The following steps describe how to ensure the bin is round.

1. Verify that the foundation meets all the requirements of the installation.
2. Scribe the bin circumference onto the foundation as follows:
 - a. Anchor a string to the exact center of the concrete foundation.
 - b. Consult the following table to find the scribe radius for the size of the bin being assembled.
 - c. Using the required string length, scribe the bin circumference onto the foundation.

The radius values given in the chart are 3/4-inch smaller than the wall sheet radius at the bottom. This ensures that the scribed circle can be seen during assembly. A perfectly placed ring of sheets should be 3/4 inch on the outside of this scribed circle.
3. After the first ring of wall sheets has been assembled, check the position and roundness of the ring:
 - a. Verify that the maximum amount that the bin is out of round is no more than 0.75" on the radius, when measured from the center of the bin.
 - b. Verify that the wall sheets form a smooth circle with no flat spots or cauliflower shaped curves.

- c. Before anchoring the bin to the foundation, re-check to ensure that the bin is round and within tolerance.

Correcting for roundness becomes much more difficult the longer you wait.

4. Locate anchor bolts towards the outside of the anchor bolt slots (away from bin) to permit the incremental expansion that can occur with the initial filling of the bin.
5. When setting jacks, make sure they are also set round and that they are anchored to the concrete.

Table 19. Scribe Radius – 15' to 60' Bins

Nominal Bin Diameter (ft)	Scribe Radius (ft in)
15	7'4-3/4"
18	8'10-11/16"
21	10'4-9/16"
24	11'10-1/2"
27	13'4-3/8"
30	14'10-5/16"
33	16'4-3/16"
36	17'10-1/8"

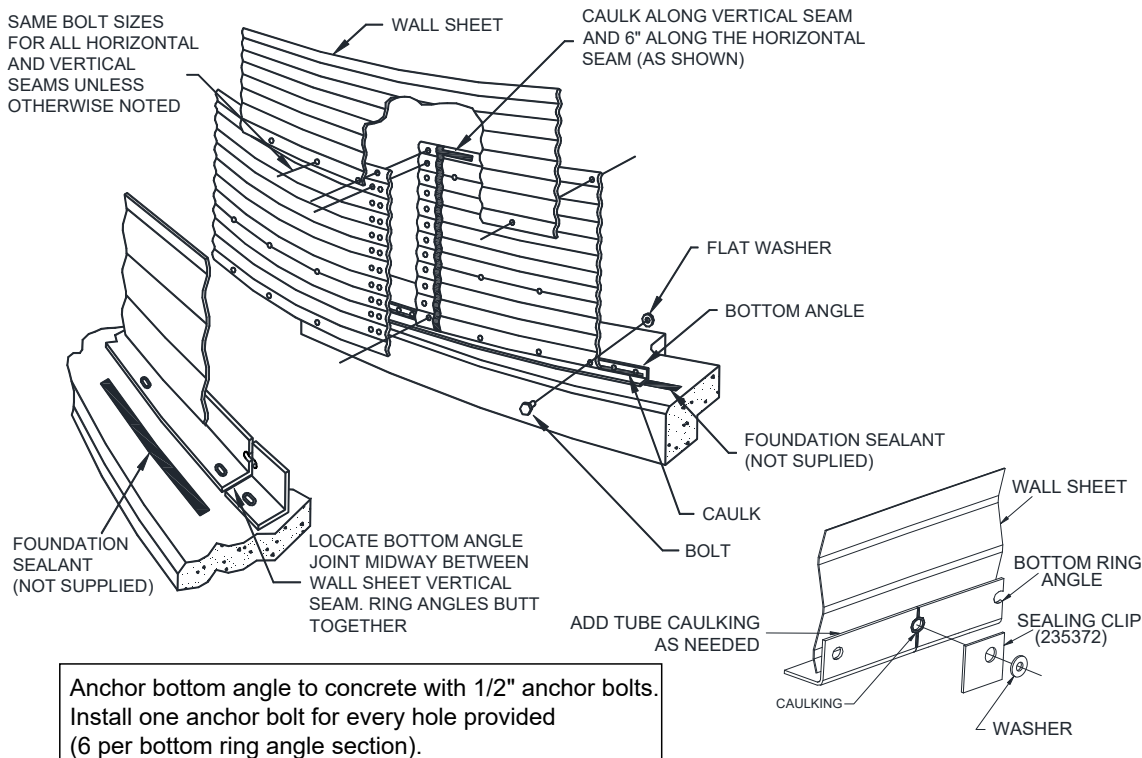
Nominal Bin Diameter (ft)	Scribe Radius (ft in)
39	19'4"
42	20'9-15/16"
45	22'3-13/16"
48	23'9-3/4"
51	25'3-5/8"
54	26'9-9/16"
60	29'9-5/16"

5.13. Wall Sheet and Bottom Angle Assembly

Note

For wall sheet hardware specification, refer to [Table 41 on page 109](#).

Figure 26. Wall Sheet and Bottom Angle Assembly Detail



5.14. Centurion Wall Sheet Part Number Matrix

Table 20. Wall Sheet Part Number Table

Corrugated Wall Sheets					Punched Wall Sheets			
Thickness nom (min)	Gauge	Label Colour	Weight lbs	Length (overall)	Flat	Regular	Bottom	
.040 (.036)	20	Yellow	58.3	116.5"	194654	194660 Stencil		
					194657	194663 Stencil		
					194679	194730		
.050 (.045)	18	Orange	72.8		194655	194661 Stencil		
					194658	194664 Stencil		
					194680	194731	194771	
.057 (.052)	17	Red	83.0		194681	194732	194772	
.066 (.061)	15	Pink	97.7		194682	194733	194773	
.076 (.070)	14	Lime	112.2		194656	194662		
					194659	194665		
					194683	194734	194774	
.096 (.088)	13	Green	141.1		194684	194735	194775	
.116 (.107)	12	Blue	171.4		194685	194736	194776	
.126 (.117)	11	Purple	189.0	117.0"	194606	194737	194777	
.139 (.130)	10	Black	209.4		194607	194738	194778	
Short Sheets Installed Beside the Door								
.057 (.052)	17	Red	66.3	93.0"		194780		
			26.2	36.8"			194783	
.076 (.070)	14	Lime	89.6	93.0"		194781		
.096 (.088)	13	Green	44.6	36.8"			194784	
.116 (.107)	12	Blue	136.8	93.0"		194782		
			54.1	36.8"			194785	
Bottom wall sheets are punched for full floor aeration flashing. Use bin bolts provided to plug unused holes if a full floor aeration system is not being used.								

5.15. Wall Sheet Caulking Detail

Figure 27. Wall Sheet Caulking Detail (inside view) — Imperial

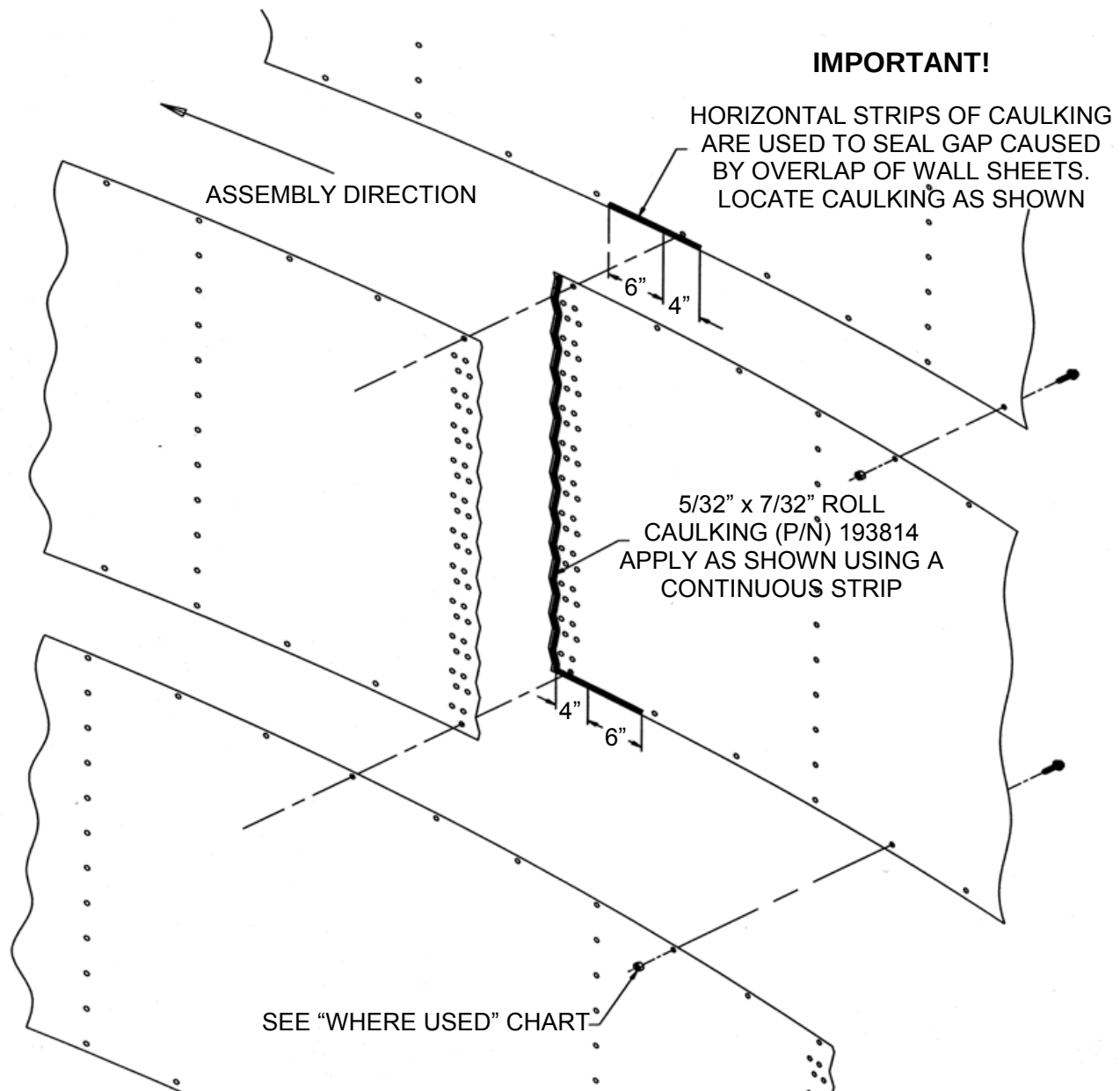
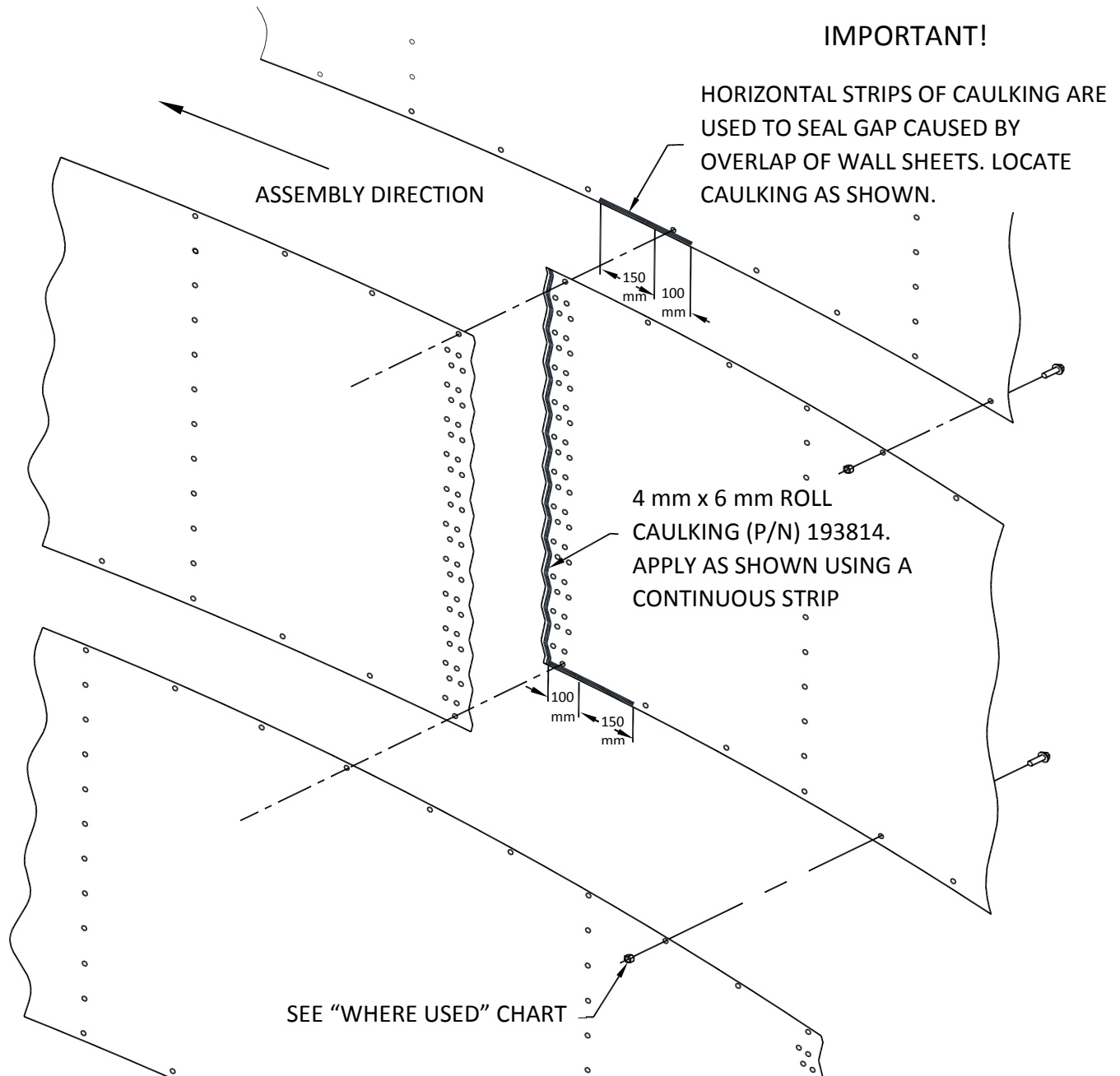
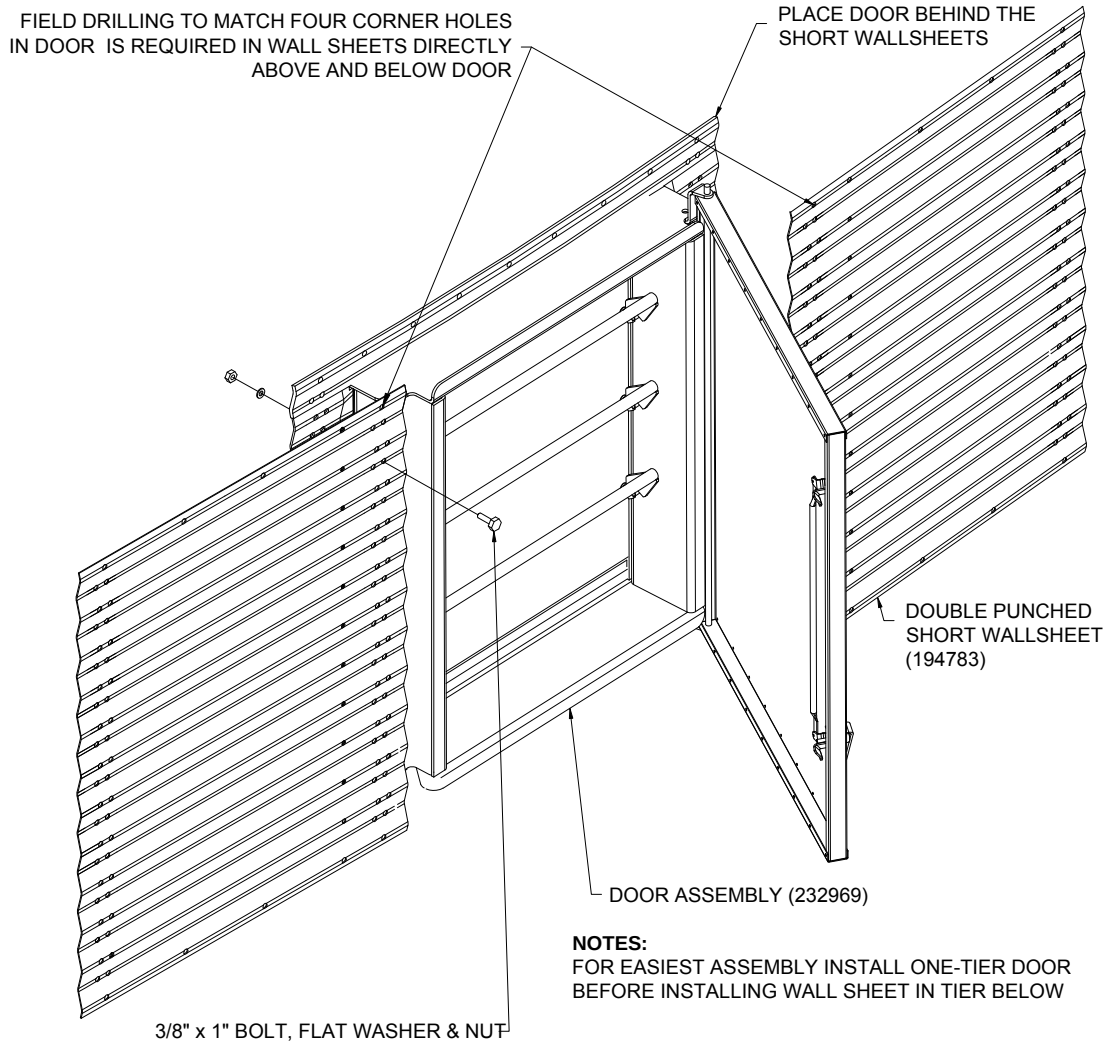


Figure 28. Wall Sheet Caulking Detail (inside view) — Metric

5.16. One-Tier Light Duty Door (15' – 27') Installation

Figure 29. One-Tier Light Duty Door (15' – 27') Detail

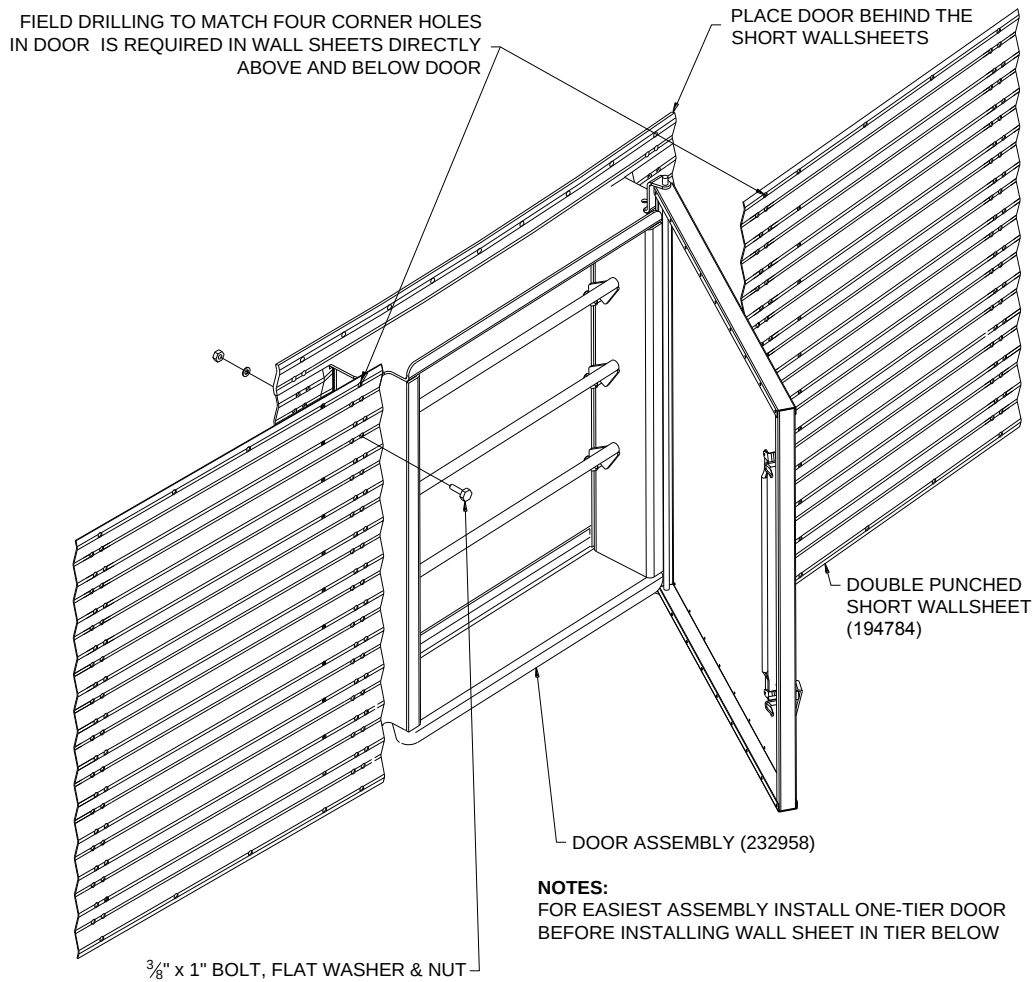


Important

The inner door board must be closed and latches completely engaged before filling. Failure and collapse of the bin could result if the bin is filled without properly closing the inner door board.

5.17. One-Tier Light Duty Door Installation

Figure 30. One-Tier Light Duty Door Detail

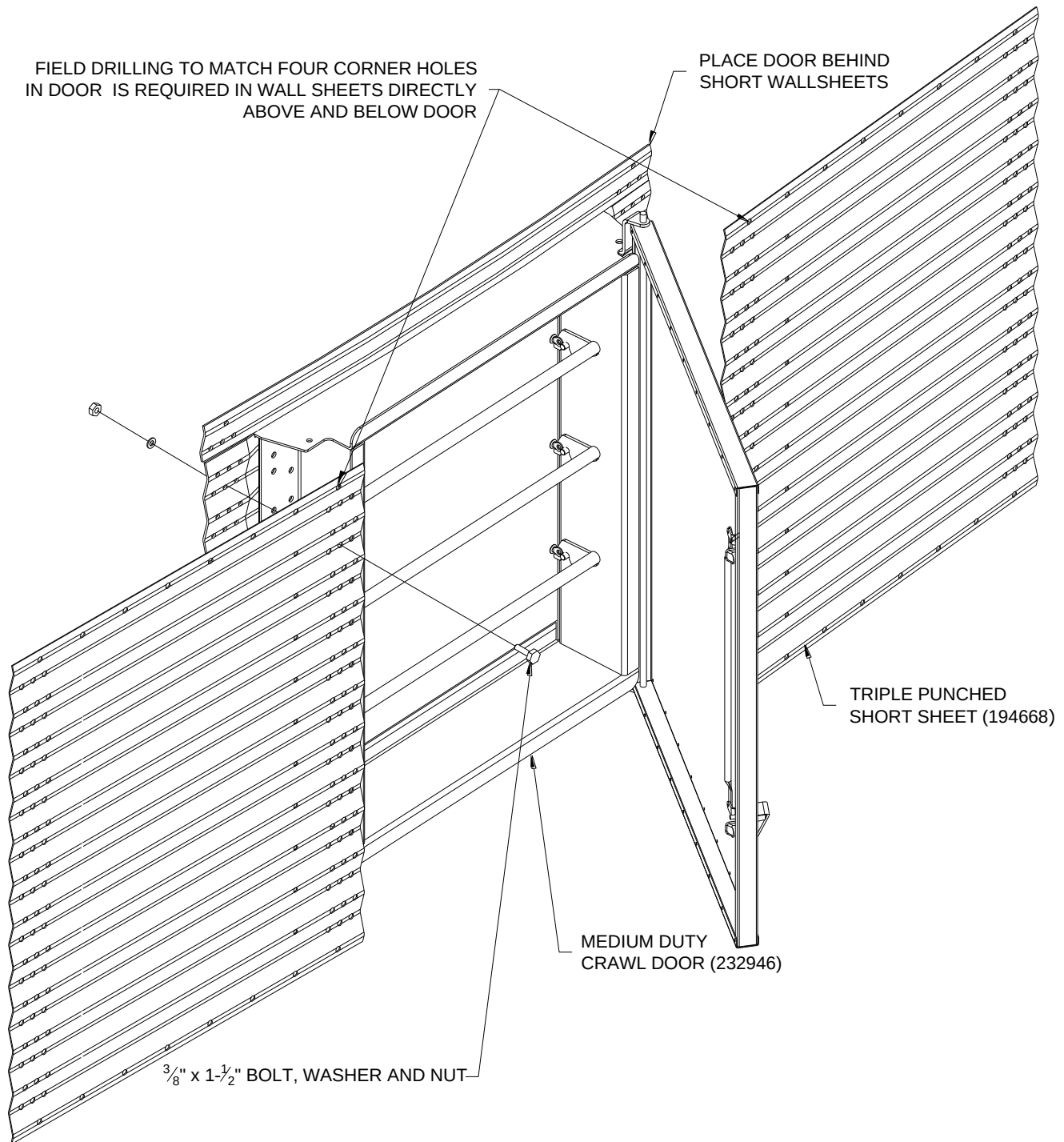


Important

Inner door board must be closed and latches completely engaged before filling. Failure and collapse of the bin could result if bin is filled without properly closing inner door board.

5.18. One-Tier Medium Duty Door Installation

Figure 31. One-Tier Medium Duty Door Detail



Important

The inner door board must be closed and latches completely engaged before filling. Failure and collapse of the bin could result if the bin is filled without properly closing the inner door board.

5.19. Commercial Bin Upright Assembly

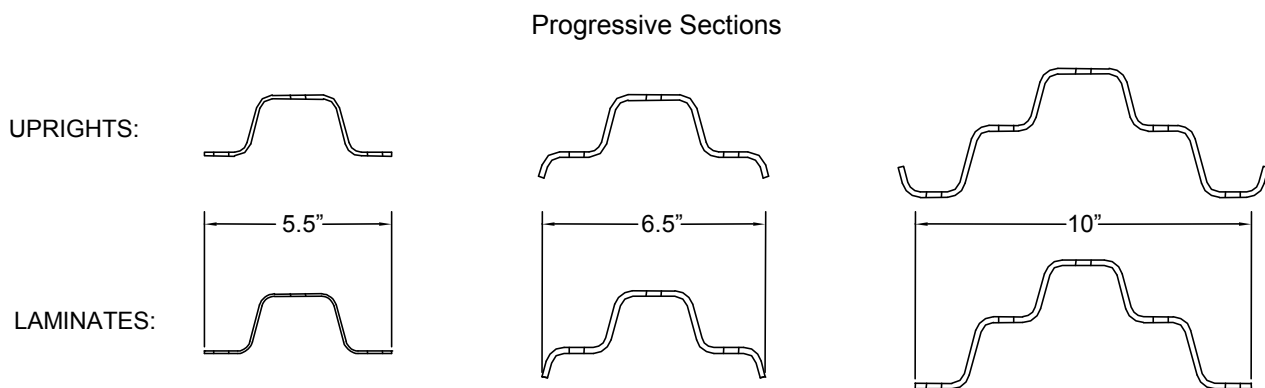
This section provides information needed to assemble uprights for commercial bins.

Introduction

The Westeel Commercial upright system consists of uprights and laminates. Single uprights, joined by splice plates, are used at the top of bins. Laminate sections are introduced when vertical load requirements dictate. Once introduced the laminates continue to be utilized for the balance of the assembly.

A unique feature of the Westeel upright system is the progressive section. Not only do the uprights and laminates increase in gauge from the top to the bottom of the bin, they also increase in section.

Figure 32. Progressive Sections



NOT ALL SECTIONS ARE USED ON ALL BINS

Both upright and laminate sections measure 88" long. In the center of each there are vertical holes spaced at 4" centers. This permits use on externally stiffened bins. There are two locations on each wall sheet for attachment of the uprights. The wall sheet holes that mate with the uprights are spaced at 4" centers. All center upright holes must be filled with bolts.

Upright/Laminate Identification

In order to properly erect the bin it is necessary to distinguish uprights from laminates, it is necessary to determine the gauge of the part, and it is necessary to determine the width of the section. The various combinations are provided in the upright/ laminate table. It is also necessary to determine the orientation of the parts as there is a distinct top and bottom. All the information that is required for assembly is contained on the label.

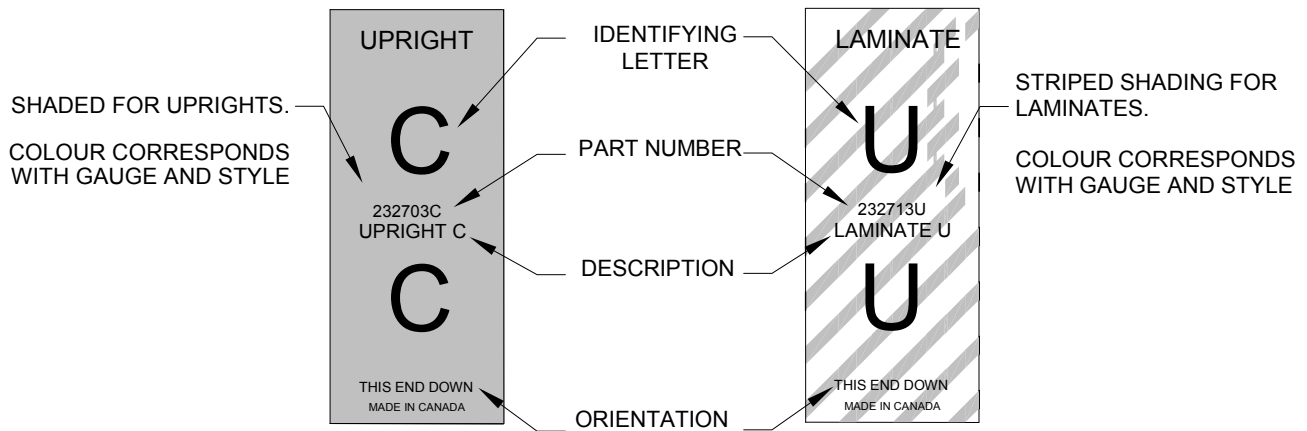
The label, is the easiest means of identification. It contains all of the necessary information. For assembly purposes, the label is placed on the bottom of both uprights and laminates.

Upright labels have solid colours and laminate labels are striped. For both uprights and laminates, the colour corresponds with gauge and style.

Each upright and laminate has a unique identifying letter. This is prominently displayed on the label, and corresponds with the respective assembly charts provided (see [Figure 33](#)).

Tip

For error free installation, make sure that the identifying letter on the label coincides with the wall sheet/upright layout for the bin being assembled, and that the labels on both uprights and laminates remain on the bottom. There is a definite top and bottom orientation for uprights and laminates. It is imperative that they are oriented correctly.

Figure 33. Upright and Laminate Labels

Short Upright

There is one short upright measuring 44" long for use in odd tier bins. The alpha character for this part is "S". There is no corresponding laminate as it is used at the top of the bin before the laminates are introduced. The short upright always goes in the top tier.

Tip

The short "S" upright is located in the top tier of odd-tiered bins.

Upright/Laminate Assembly

Use the wall sheet/upright layout provided for the bin in question, to determine the proper order of the various upright and laminate components. The identifying letter on the label is the easiest means of identification. In addition to the identifying letter, every upright and laminate is also identified by gauge and width. If for some reason the label is missing from a part, the following table contains information that will aid in the identification of the various parts.

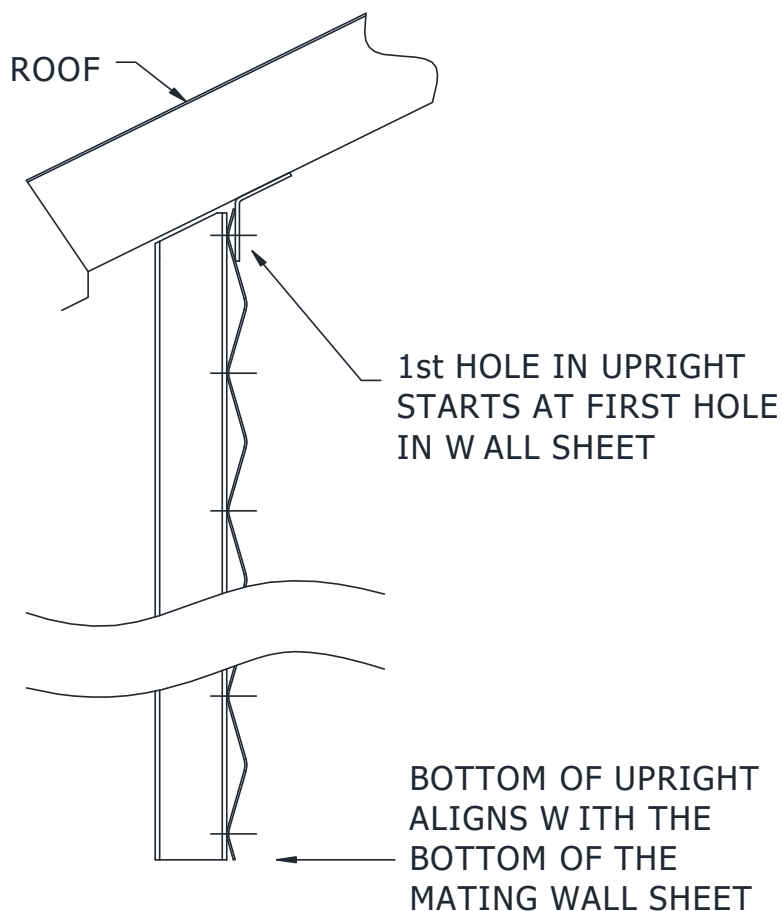
Tip

In all cases laminates nest inside uprights. The uprights are placed against the bin wall sheets and the laminates are away from the wall sheets.

It is important to get the first uprights started correctly. The top hole in the top upright bolts into the top horizontal wall sheet seam (see [Figure 34](#)).

Tip

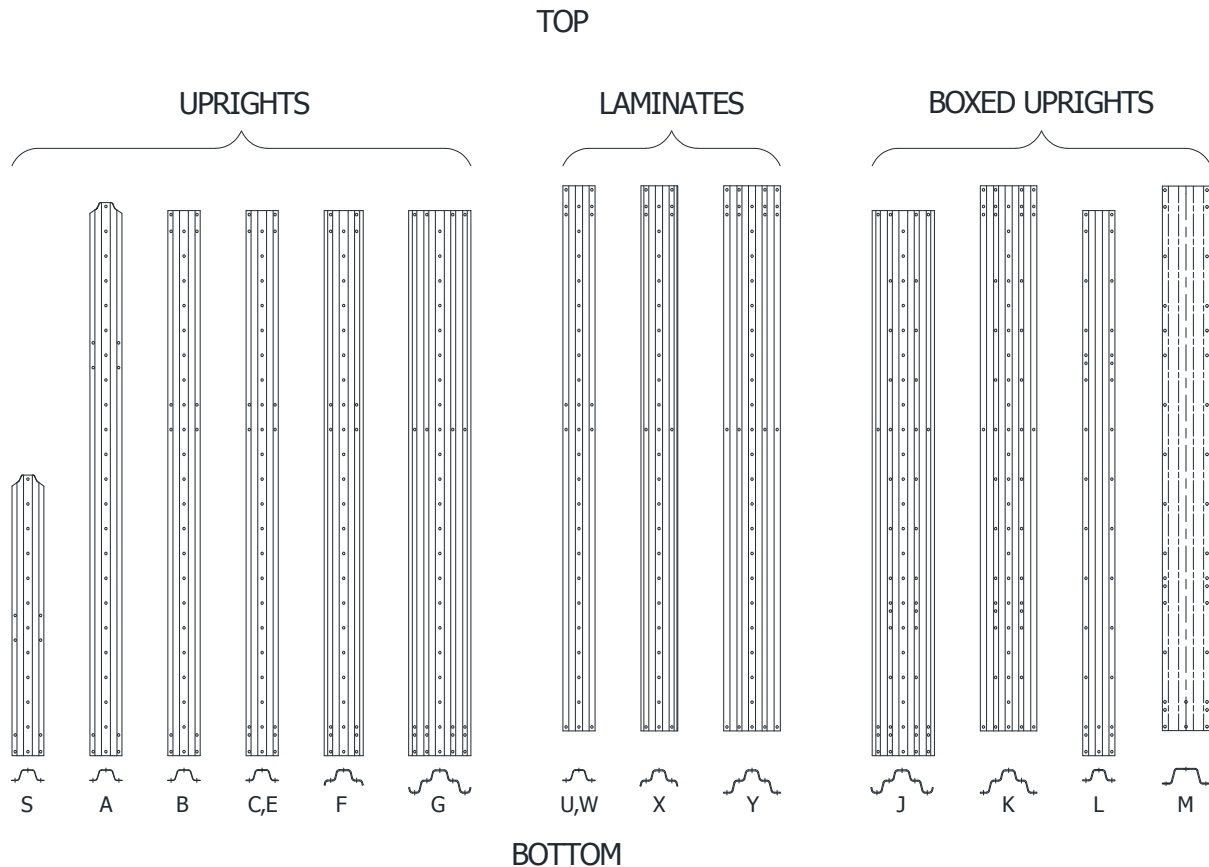
For proper upright orientation align the bottom of the first upright with the bottom edge of a wall sheet.

Figure 34. Upright Orientation Detail**Table 21. Upright/Laminate Identification Table**

	Identifying Letter	Part Number	Description	Gauge	Label Colour	Width of Section (in)
Uprights	S	232700S	Upright S .076" Short	14	Light green	5.5"
	A	232701A	Upright A .076" x 5.5"	14	Yellow	5.5"
	B	232702B	Upright B .076" x 5.5"	14	Light green	5.5"
	C	232703C	Upright C .116" x 5.5"	12	Blue	5.5"
	E	232705E	Upright E .168" x 5.5"	8	Brown	5.5"
	F	232706F	Upright F .168" x 6.5"	8	Silver	6.5"
	G	232707G	Upright G .168" x 10"	8	Gold	10"
Laminates	U	232713U	Laminate U .116" x 5.5"	12	Blue striped	5.5"
	W	232715W	Laminate W .168" x 5.5"	8	Brown striped	5.5"
	X	232716X	Laminate X .168" x 6.5"	8	Silver Striped	6.5"
	Y	232717Y	Laminate Y .168" x 10"	8	Gold striped	10"
Boxed	J	232709J	Upright Boxed J .168" x 10"	8	Red	10"
	K	232710K	Laminate Boxed K .168" x 10"	8	Red Striped	10"
	L	232711L	Laminate Boxed L .168" x 5.5"	8	Red Striped	5.5"

Note

Not all sections are used on all bins.

Figure 35. Upright and Laminate Components**Catwalk Support Uprights**

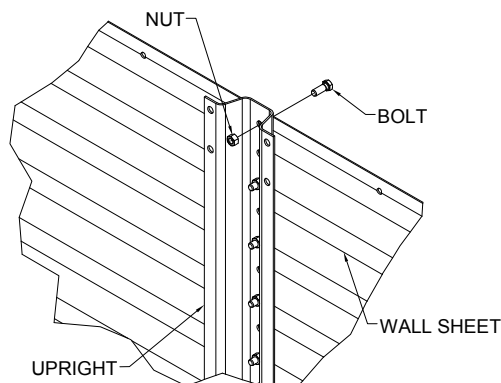
The upright/laminate requirements under catwalk support locations are likely different from the normal upright/laminate order. Consult your Westeel representative for specifications.

Bolt/Nut Orientation

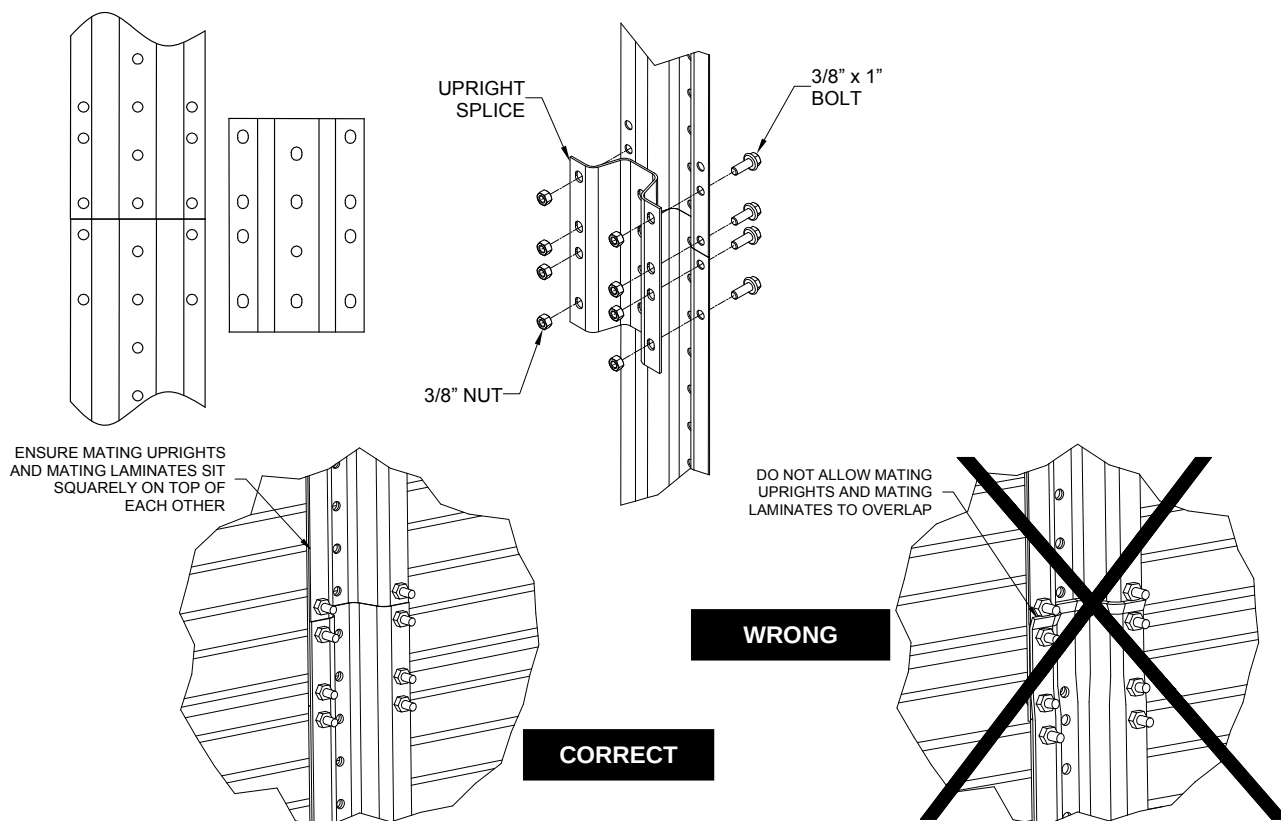
To allow for a good seal install the bolts from the inside of the bin as shown for externally stiffened bins.

Upright/Splice Pre-Assemblies

At the top of a bin, laminates are not utilized and a splice is required to make the connection between mating uprights. The splice nests inside the upright similar to a laminate. When pre-assembling uprights to splices, insure that the splice goes on the top end of the upright, such that the label on the bottom of the upright remains visible. Keeping the label visible will help prevent subsequent errors. This practice will also prevent ground interference when adjusting jack locations.

Figure 36. Upright / Wall Sheet Bolt and Nut Orientation**Tip**

It may be advantageous to conduct pre-assemblies during the installation process. This can be a real time saver.

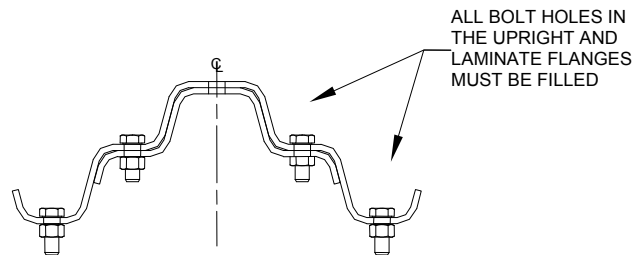
Figure 37. Pre-Assembling Uprights/Splices**Upright and Laminate Assembly**

Uprights and laminates are designed to transfer vertical loads through an end to end, butt connection. Ensure that mating uprights sit squarely on top of each other and do not overlap. Ensure that mating laminates sit squarely on top of each other and do not overlap. Secure the joints with the nuts and bolts provided. Failure to do so can result in structural failure.

Upright/Laminate Pre-assemblies

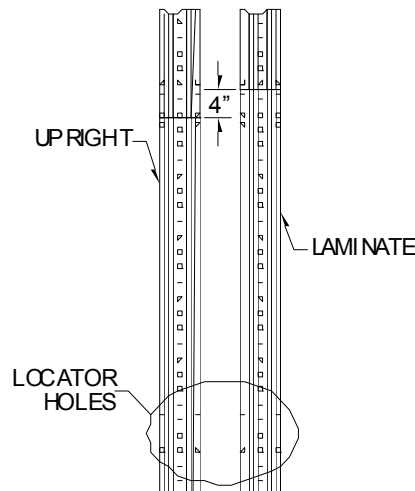
Laminates nest inside of uprights and are offset 4" above the uprights.

Figure 38. Nesting Laminates and Uprights



For proper orientation, ensure that the labels on the upright and laminate are both towards the bottom and that the locator holes in the middle portion of the upright and laminate line up. Bolts can be placed in these locator holes to create a pre-assembly. All locator holes, located in the middle of the upright and laminate flanges, need to be filled with bolts. These include holes in flanges that may only have one thickness of material.

Figure 39. Upright/Laminate Orientation



When properly assembled, both the upright label and the laminate label will remain visible during the pre-assembly phase. Once assembled on the bin the upright label will be covered.

In the assembly layouts, the combination of an upright and a laminate is called an assembly. For example, the combination of a "C" upright and a "U" laminate would be called a "CU Assembly". Both the "C" on the upright label and the "U" on the laminate label, would remain visible, and would therefore remain distinguishable from other pre-assemblies.

Tip

When creating pre-assemblies, ensure that the labels on both the upright and the laminate are on the same end, and that the locator holes align with each other in the middle of the parts. Once pre-assembled, both labels should remain visible.

Note

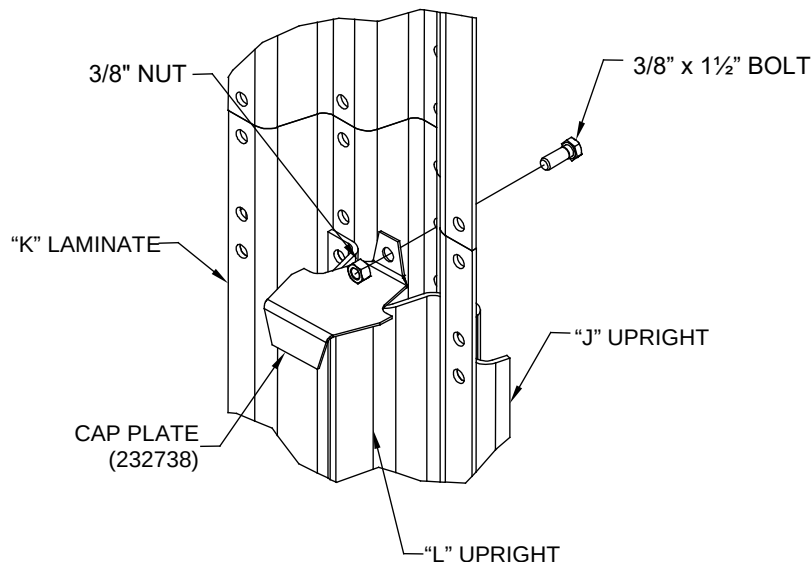
Once completely assembled onto the bin, all visible holes in the upright and laminate flanges must be filled. All mating wall sheet/upright holes must be filled.

Boxed Upright Cap Plates

Important

To prevent water from damming in the boxed section, install the cap plate (232738) as shown.

Figure 40. Installing Cap Plates on Boxed Uprights



5.19.1 Connection of the Upright Connectors to the Grain Bin for the Goal Posts

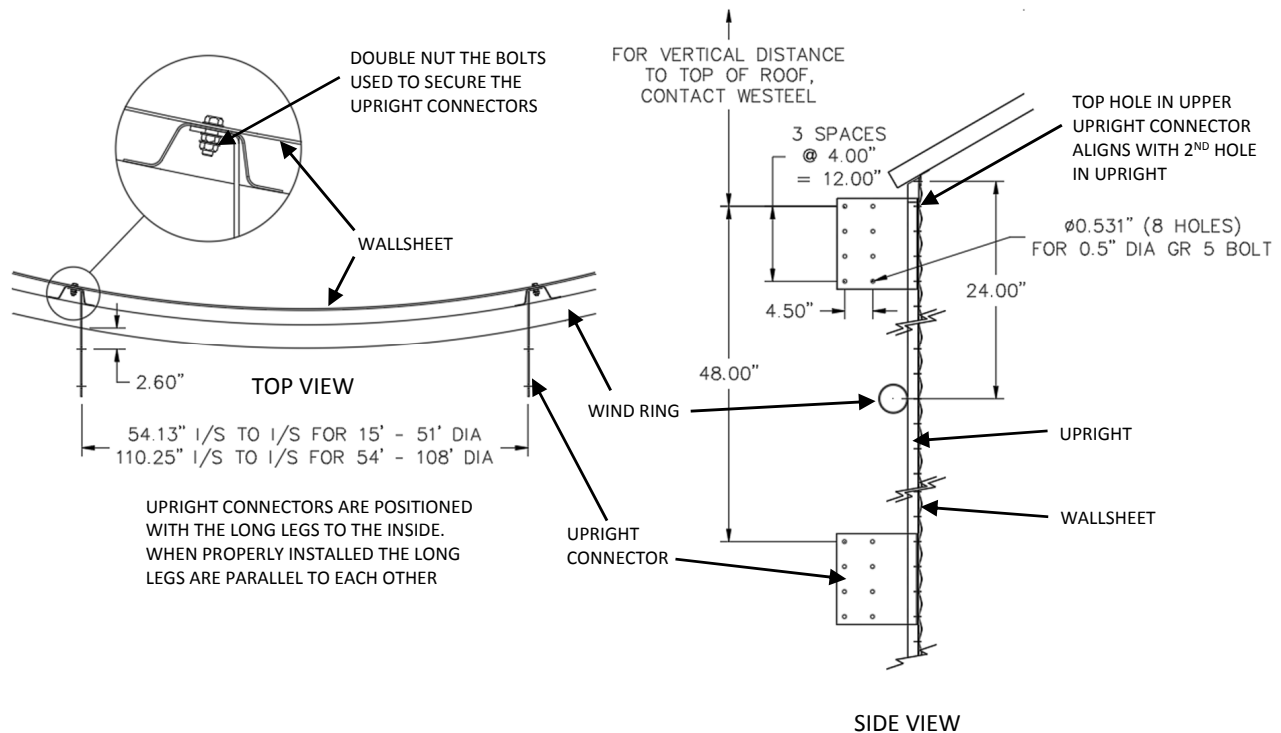
Goal posts connect into the grain bin uprights using upright connectors which bolt into the center section of the upright hat sections utilizing the same hardware that connects the uprights to the grain bin walls (3/8" x 1 1/2" hex bolts). For this reason, **the best time to attach the upright connectors is at the same time the uprights are bolted to the bin.**

Note

If retrofitting an existing bin, the appropriate connection points will need to be disassembled to facilitate the addition of the upright connectors. Ensure the bin structure is capable of supporting the loads.

Upright connectors are angled to match the curvature of the bin so there are unique connectors for every bin diameter. Make sure the connector part numbers match the bin diameter in question.

The timing and location of upright connectors is also important. Since the upright connectors are potentially being installed well in advance of any subsequent catwalk considerations, consult the site layouts to ensure that later repositioning of the connectors is averted.

Figure 41. Placement of upright connectors**Important**

Goal Posts need to be fully braced laterally between the left and right upright connectors. This is typically done with horizontal and diagonal bracing. Westeel goal posts are fully braced. If non-Westeel goal posts are used, bracing needs to be incorporated into the design.

For 44.5 kN (10,000 lb) capacity connections there are two upright connectors per upright (See [Figure 41 on page 68](#)). The top hole in the upper upright connector aligns with the 2nd bolted hole in the mating upright as illustrated. The next upright connector is then positioned 48" lower (measured from top of connector to top of connector) on the upright and bolted in.

Note

All bolts used to make the upright connector connections should be double nutted such that the outer nut serves as a jam nut.

5.19.2 Base Assembly 233013 (Laminate Sections)

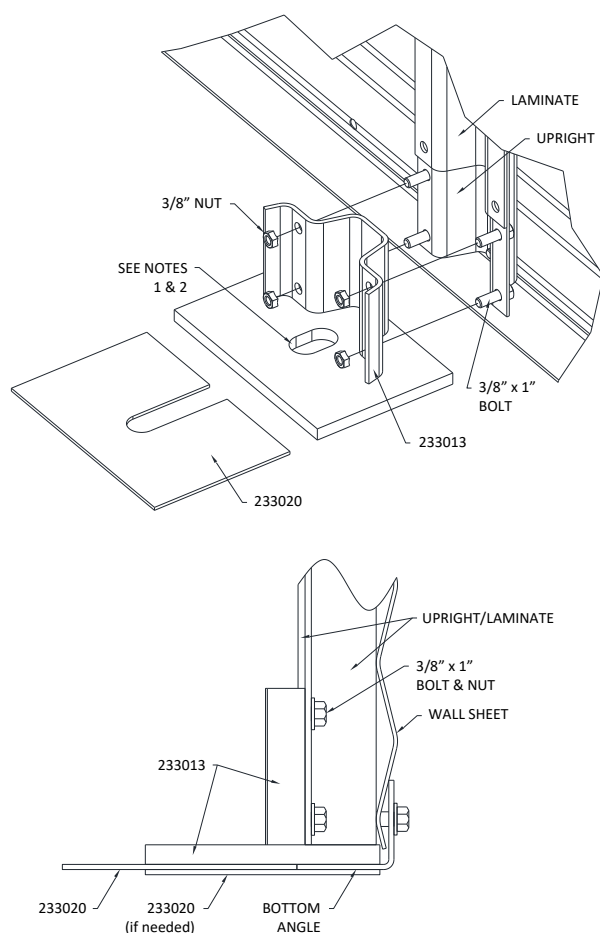
233013 — Base Assembly for Uprights with Laminate Sections

At the bottom of an assembled bin that has laminates, there will be a 4" gap between the bottom laminate and the base plate. It is imperative that this area is filled with the 4" laminate section that protrudes from the base assembly. Use the combination of base assemblies and shims to secure the bin to the foundation, as shown below.

Tip

Depending on the assembly procedure, it may be convenient to bolt on the base assemblies when creating the upright/laminate pre-assemblies.

Figure 42. Base Assembly 233013 for Uprights with Laminate Sections



Important

- 1) When positioning anchor bolts, locate them as far forward (away from the bin) within the slot as possible.
- 2) The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.20" x 1.90" anchoring slot provided in the base plate. The base plate should not be able to pull up over the nut.

5.19.3 Base Assembly 233012 (Laminate Sections)

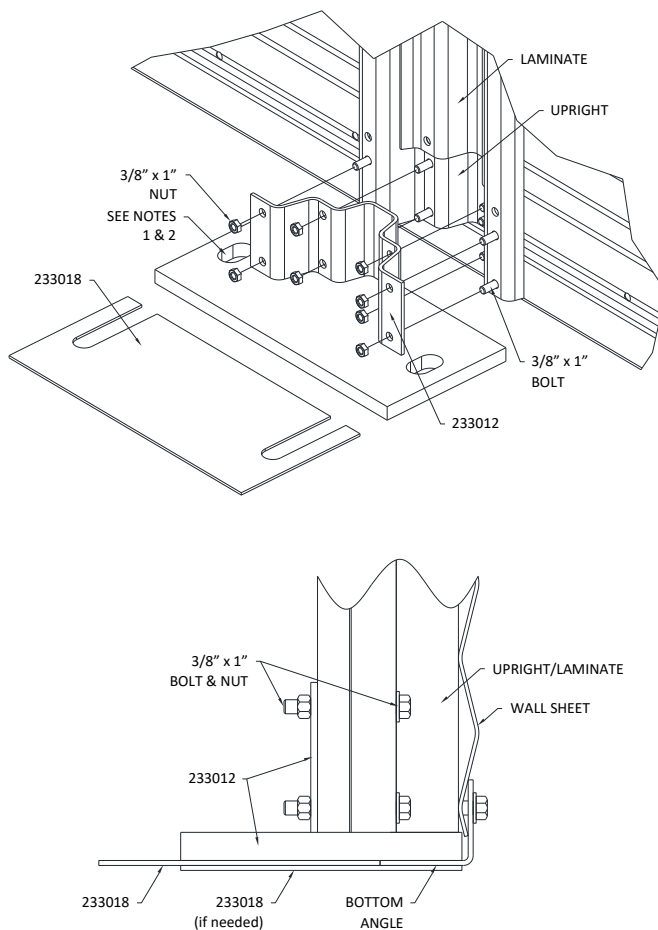
233012 – Base Assembly for Uprights (G or larger) with Laminate Sections

At the bottom of an assembled bin that has laminates, there will be a 4" gap between the bottom laminate and the base plate. It is imperative that this area is filled with the 4" laminate section that protrudes from the base assembly. Use the combination of base assemblies and shims to secure the bin to the foundation, as shown below.

Tip

Depending on the assembly procedure, it may be convenient to bolt on the base assemblies when creating the upright/laminate pre-assemblies.

Figure 43. Base Assembly 233012 for uprights with laminate sections



Important

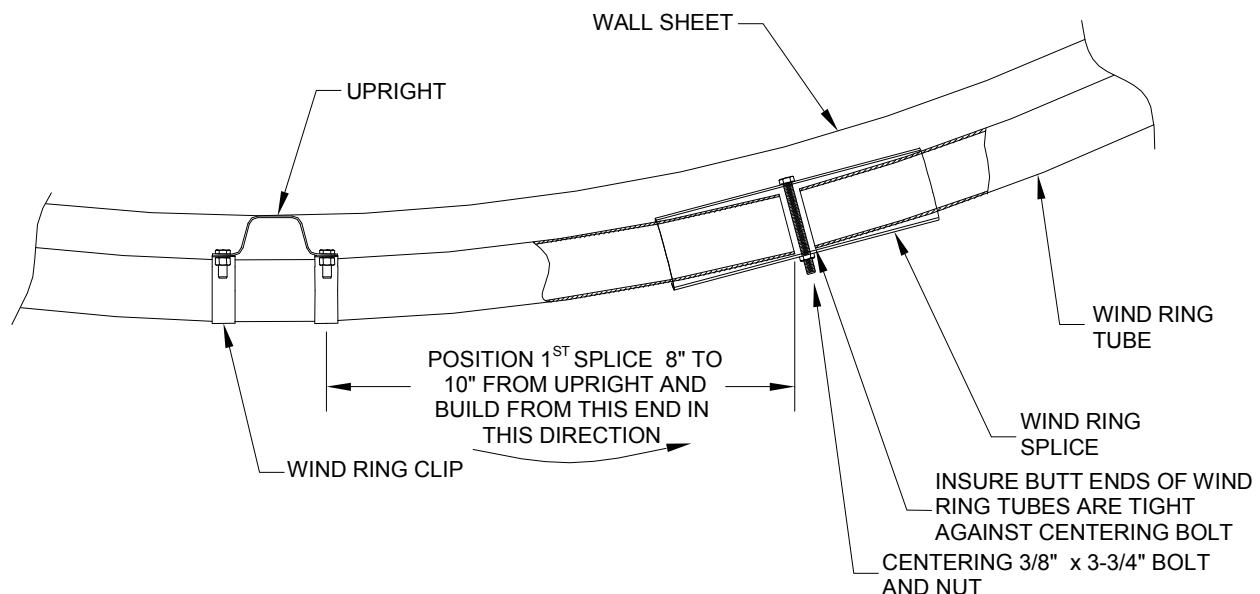
- 1) When positioning anchor bolts, locate them as far forward (away from the bin) as possible within the slots.
- 2) The anchor bolts at the upright locations should be chosen so that they can handle the loads imposed on them and that the head, either alone or with large diameter heavy washers, is large enough to sufficiently cover the 1.20" x 1.90" anchoring slots provided in the base plate. The base plate should not be able to pull up over the nuts.

5.20. Wind Ring Assembly

Wind rings fulfill their function when the bin is empty or partially filled. In high winds, the wind rings provide extra stiffness and help keep the bin round. Not all bins require wind rings. Bin diameter and height determine the location and the quantity of wind rings required.

Wind ring locations are identified by an O placed beside the relevant uprights within the wall sheet and upright layouts for the bin in question. At these locations wind ring tubes are secured to the upright flanges with a series of clips that bolt into the upright locator holes that are located in the flanges of the 5.5" wide upright and upright/laminate combinations. Adjacent tubes are aligned and secured to each other with wind ring splices. A 3/8" x 3-3/4" bolt through the splice keeps it centered on the connection.

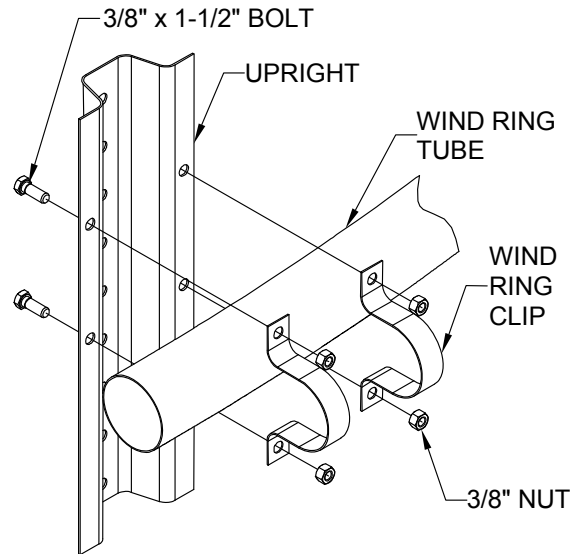
Figure 44. Wind Ring Assembly



Externally Stiffened Bins

Once the uprights have been secured to the bin walls, position the first wind ring tube and secure it to the upright using the wind ring clips provided. Two clips are required per upright, one on each flange. Position the wind ring such that a wind ring splice (with bolt inserted) can be slipped onto the end of the tube without interfering with the upright or the wind ring clips. The splice should be orientated such that the bolt is horizontal.

Insert the end of the next wind ring tube into the open end of the wind ring splice. Insure that the ends of both tubes are tight against the centering bolt. Secure the wind ring tube to the uprights with the wind ring clips. Continue around the bin.

Figure 45. Wind Ring Mounting Detail (externally stiffened bins)

All wind ring splice connections should be made in the space between uprights, and should not encroach into the area where the wind ring clips are securing the wind ring tube to the uprights. To avoid interference with uprights and the need to make multiple cuts, position an end of the first tube relatively close to an upright, such that the space between the end of the tube and the next upright is maximized, and build from that end. Insure that both ends of the tube are far enough away from the closest uprights to avoid interference with the splice. When progressing around the bin, this space between the end of the tube and the next uprights may shrink with each additional tube that is installed. On large diameter bins, if this space shrinks to the point where the wind ring splice interferes with the upright, then the tube will need to be cut. Make the cut such that the space that is created between the end of the tube and the next upright is similar to the identical space on the first tube that was installed. In this manner, there will not be a shortage of tube.

Note

Assembly Tip: When putting the first wind ring tube in place, locate one end close to an upright with a 8" to 10" overhang, and continue building from that end. This will reduce the need for multiple cuts.

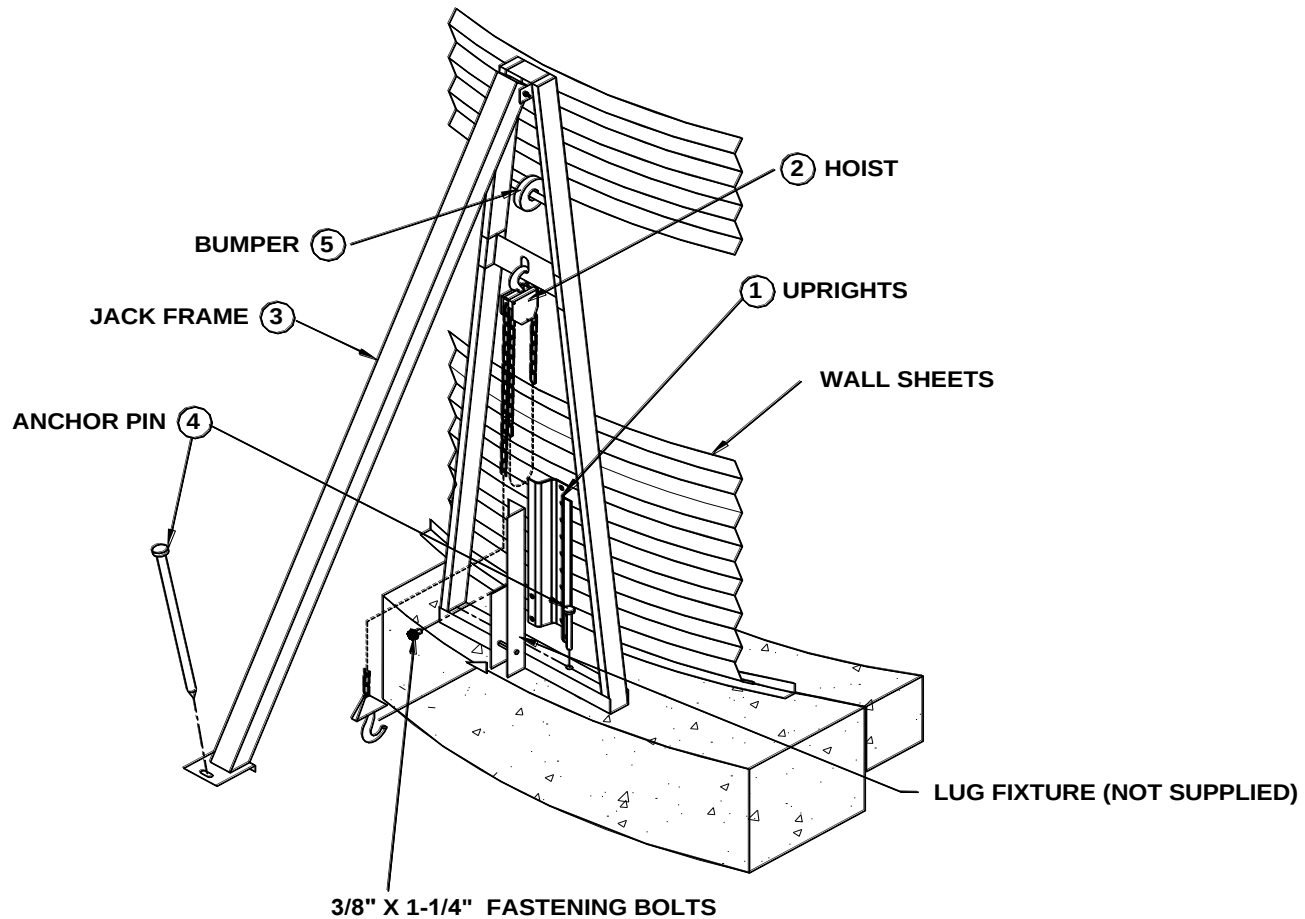
The final wind ring tube in a circle will need to be cut to length. Secure one end of the last tube in the previously installed wind ring splice as described above. Hold the tube in place and mark the cut-line relative to the previously installed tube at the other end. Insure that allowance is made for the 3/8" diameter bolt. Once the tube has been cut, install one end of the tube as described above. On the other end slide the wind ring splice completely onto the free end. Position this end relative to the previously installed tube, and slide the splice onto the second tube until it is centered. Insert the centering bolt. Install all wind ring clips. Tighten all bolts.

Note

Assembly Tip: When tightening wind ring clips, always tighten in sequence starting at the spliced end of the tube, which has already been secured, and work towards the free, and as yet unspliced, end.

5.21. Bin Jack Techniques

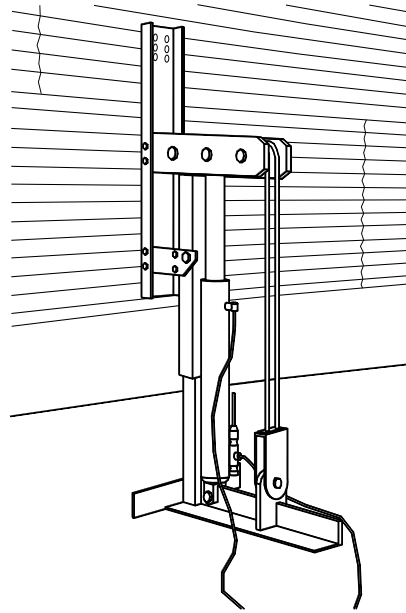
Figure 46. External Bin Jack Detail



1. Choose a hoist with a suitable capacity for the expected load. Make sure the rated capacity of the hoist is not exceeded.
2. Fabricate custom lifting lugs to suit the assembly equipment. Each lug should have a capacity 5 times the expected load.
3. Have jacks evenly spaced around the bin. Use one jack per wall sheet. Each jack should have a capacity 5 times the expected load.
4. Anchor the jack securely. Use a guy wire if necessary to ensure stability.
5. Use a minimum of four 3/8" x 1-1/4" bolts (Grade 8), not supplied, to fasten lug, if one jack per wall sheet is used

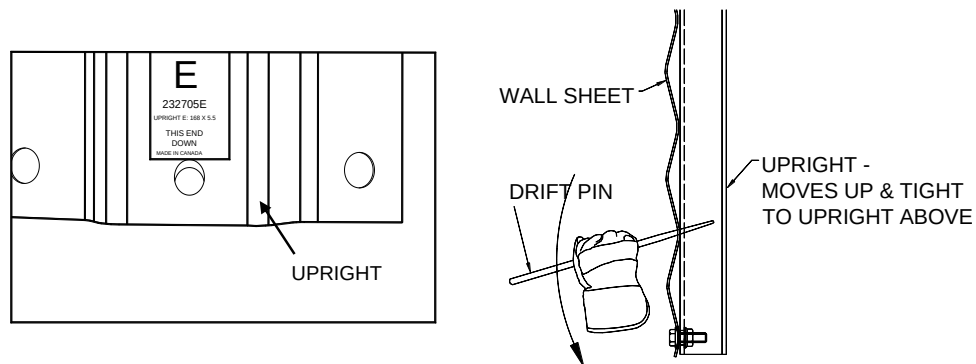
Note

The bin may also be erected using an internal jacking system (**not available from Westeel**).

Figure 47. Internal Bin Jack Detail

We recommend:

- Ensure jacks are sized for the job. Do not exceed rated capacity.
- Layout jacks in a proper radius. Setting the jacks in the round makes the bin round.
- Use a minimum of one jack per wall sheet. Anchor them securely to the concrete.
- Jack on uprights. Hang wall sheets for two tiers until uprights are installed. Monitor hole alignment and make adjustments with a drift pin as you go, before tightening. Lifting the uprights towards one another during construction makes for easier realignment should misalignment creep in.

Figure 48. Hole Alignment

- Protect electrical cords.
- A drift pin is your friend. Use them for carrying wall sheets, aligning holes and forcing uprights up.

6. Specifications

6.1. Centurion Commercial Grain Bin Specifications

Table 22. Centurion Commercial Grain Bin Specifications

MODEL	BIN DIAMETER	CAPACITY			HEIGHT					
					EAVES			OVERALL		
		bu	m ³	Tonnes	ft	ft - in	m	ft	ft - in	m
1513	14'11"	7310	243	198	47.8	47'10"	14.58	51.8	51'10"	15.80
1514	4.55 m	7860	262	213	51.5	51'6"	15.70	55.5	55'6"	16.92
1813	17'11"	10580	352	287	47.8	47'10"	14.58	52.7	52'8"	16.06
1814	5.46 m	11370	379	308	51.5	51'6"	15.70	56.4	56'4"	17.18
2113	20'11"	14480	482	393	47.8	47'10"	14.58	53.6	53'7"	16.33
2114	6.37 m	15550	518	422	51.5	51'6"	15.70	57.2	57'3"	17.44
2413	23'10" 7.28 m	19010	633	516	47.8	47'10"	14.58	54.4	54'5"	16.59
2414		20410	680	553	51.5	51'6"	15.70	58.1	58'1"	17.71
2415		21800	726	591	55.2	55'2"	16.82	61.8	61'9"	18.82
2416		23200	773	629	58.8	58'10"	17.93	65.4	65'5"	19.94
2713	26'10" 8.19 m	24180	806	656	47.8	47'10"	14.58	55.3	55'3"	16.85
2714		25950	864	704	51.5	51'6"	15.70	59.0	58'11"	17.97
2715		27720	923	752	55.2	55'2"	16.82	62.6	62'7"	19.09
2716		29490	982	800	58.8	58'10"	17.93	66.3	66'3"	20.20
3013	29'10" 9.10 m	30000	1000	814	47.8	47'10"	14.58	55.7	55'8"	16.98
3014		32180	1072	873	51.5	51'6"	15.70	59.4	59'4"	18.09
3015		34370	1145	932	55.2	55'2"	16.82	63.0	63'0"	19.21
3016		36560	1218	992	58.9	58'10"	17.94	66.7	66'9"	20.33
3313	32'10" 10.01 m	36470	1216	989	47.8	47'10"	14.58	56.6	56'7"	17.24
3314		39120	1304	1061	51.5	51'6"	15.70	60.2	60'3"	18.36
3315		41760	1392	1133	55.2	55'2"	16.82	63.9	63'11"	19.47
3316		44410	1480	1204	58.8	58'10"	17.93	67.6	67'7"	20.59
3613	35'10" 10.91 m	43620	1455	1183	47.8	47'10"	14.58	57.4	57'5"	17.50
3614		46770	1559	1269	51.5	51'6"	15.70	61.1	61'1"	18.62
3615		49910	1664	1354	55.2	55'2"	16.82	64.7	64'9"	19.74
3616		53060	1768	1439	58.8	58'10"	17.93	68.4	68'5"	20.85
3624		78450	2613	2128	88.2	88'2"	26.87	97.8	97'9"	29.79
3913	38'10" 11.82 m	51450	1716	1395	47.8	47'10"	14.58	58.3	58'3"	17.76
3914		55140	1839	1496	51.5	51'6"	15.70	61.9	61'11"	18.88
3915		58830	1962	1596	55.2	55'2"	16.82	65.6	65'7"	20.00
3916		62520	2084	1696	58.8	58'10"	17.93	69.3	69'3"	21.12
4213	41'9" 12.73 m	59960	2001	1626	47.8	47'10"	14.58	59.1	59'2"	18.03
4214		64240	2143	1742	51.5	51'6"	15.70	62.8	62'10"	19.14
4215		68520	2285	1859	55.2	55'2"	16.82	66.5	66'6"	20.26
4216		72810	2428	1975	58.8	58'10"	17.93	70.1	70'2"	21.38
4513	44'9" 13.64 m	69160	2308	1876	47.8	47'10"	14.58	60.0	60'0"	18.29
4514		74080	2472	2009	51.5	51'6"	15.70	63.7	63'8"	19.41
4515		79000	2635	2143	55.2	55'2"	16.82	67.3	67'4"	20.52
4516		83910	2799	2276	58.8	58'10"	17.93	71.0	71'0"	21.64
4813	47'9" 14.55 m	79070	2640	2145	47.8	47'10"	14.58	60.9	60'10"	18.55

Table 22 Centurion Commercial Grain Bin Specifications (continued)

MODEL	BIN DIAMETER	CAPACITY			HEIGHT					
					EAVES			OVERALL		
		bu	m ³	Tonnes	ft	ft - in	m	ft	ft - in	m
4814		84670	2826	2296	51.5	51'6"	15.70	64.5	64'6"	19.67
4815		90260	3012	2448	55.2	55'2"	16.82	68.2	68'2"	20.79
4816		95850	3198	2600	58.8	58'10"	17.93	71.9	71'10"	21.90
5111	50'9" 15.46 m	76510	2557	2075	40.5	40'6"	12.35	40.5	40'6"	12.35
5112		82830	2767	2247	44.2	44'2"	13.47	44.2	44'2"	13.47
5113		89690	2995	2433	47.8	47'10"	14.58	61.5	61'6"	18.75
5114		96010	3205	2604	51.5	51'6"	15.70	65.2	65'2"	19.87
5115		102320	3415	2775	55.2	55'2"	16.82	68.9	68'10"	20.99
5116		108640	3625	2947	58.8	58'10"	17.93	72.5	72'6"	22.11
5410	53'9" 16.37 m	79800	2669	2164	36.8	36'10"	11.23	51.4	51'5"	15.66
5411		86880	2904	2356	40.5	40'6"	12.35	55.1	55'1"	16.78
5412		93960	3140	2549	44.2	44'2"	13.46	58.7	58'9"	17.90
5413		101040	3375	2741	47.8	47'10"	14.58	62.4	62'5"	19.02
5414		108120	3610	2933	51.5	51'6"	15.70	66.1	66'1"	20.13
5415		115200	3846	3125	55.2	55'2"	16.82	69.7	69'9"	21.25
5416		122280	4081	3317	58.8	58'10"	17.93	73.4	73'5"	22.37
6008	59'8" 18.19 m	81760	2741	2218	29.5	29'6"	8.99	46.3	46'4"	14.13
6009		90500	3031	2455	33.2	33'2"	10.11	50.0	50'0"	15.24
6010		99240	3322	2692	36.8	36'10"	11.23	53.7	53'8"	16.36
6011		107980	3612	2929	40.5	40'6"	12.35	57.3	57'4"	17.48
6012		116720	3903	3166	44.2	44'2"	13.46	61.0	61'0"	18.60
6013		125460	4193	3403	47.8	47'10"	14.58	64.7	64'8"	19.71
6014		134200	4484	3640	51.5	51'6"	15.70	68.3	68'4"	20.83
6015		142940	4774	3877	55.2	55'2"	16.82	72.0	72'0"	21.95
6016		151730	5066	4116	58.9	58'10"	17.94	75.7	75'8"	23.07
Capacities shown include 28° roof cone.										
1. Based on 1.244 cu. ft. per bushel and 6% compaction in cylinder										
2. Based on 770 kg/m ³ and 6% compaction in cylinder										

6.2. Foundation Loads — Grain Bin Series CEN

Important

- Grain Bin Anchoring: Adequate anchoring is critical to a successful bin installation. The anchoring of the bin is dependent on local wind loading conditions and forms part of the site specific design requirements. The anchor bolt details must form part of the site specific foundation design.
- For stiffened bins the primary anchor bolt locations are through the base plates at each and every upright location. The anchor bolt design, plus connection details to the base plate, must accommodate the total shear and uplift loads that can occur due to wind loading at the site in question.
- In addition Westeel suggests the use of 1/2" x 3" embedment anchor bolts through the bottom ring angle to insure bin roundness, for sealing purposes, and for additional localized lateral stability.

Table 23. Foundation Loads — CEN Series 15' Bins (Imperial-Unfactored)

Model (CEN)		1513	1514
Vertical dead load	kips/upr	0.77	0.83
Vertical grain load	kips/upr	47.50	55.39
Vertical roof snow load *	kips/upr	0.79	0.79
Vertical roof peak load	kips/upr	0.40	0.40
Bin floor pressure	kips/ft ²	1.01	1.03
Number of anchor bolts for uprights		10	10
Number of anchor bolts for bottom angle ring		30	30
* Based on maximum snow load of 45 psf			

Table 24. Foundation Loads — CEN Series 18' Bins (Imperial-Unfactored)

Model (CEN)		1813	1814
Vertical dead load	kips/upr	0.76	0.82
Vertical grain load	kips/upr	46.76	54.55
Vertical roof snow load *	kips/upr	0.94	0.94
Vertical roof peak load	kips/upr	0.33	0.33
Bin floor pressure	kips/ft ²	1.16	1.18
Number of anchor bolts for uprights		12	12
Number of anchor bolts for bottom angle ring		36	36
* Based on maximum snow load of 45 psf			

Table 25. Foundation Loads — CEN Series 21' Bins (Imperial-Unfactored)

Model (CEN)		2113	2114
Vertical dead load	kips/upr	0.79	0.85
Vertical grain load	kips/upr	49.84	55.65
Vertical roof snow load *	kips/upr	1.13	1.13
Vertical roof peak load	kips/upr	0.29	0.29
Bin floor pressure	kips/ft ²	1.29	1.32
Number of anchor bolts for uprights		14	14
Number of anchor bolts for bottom angle ring		42	42
* Based on maximum snow load of 46 psf			

Table 26. Foundation Loads — CEN Series 24' Bins (Imperial-Unfactored)

Model (CEN)		2413	2414	2415	2416
Vertical dead load	kips/upr	0.83	0.92	1.00	1.09
Vertical grain load	kips/upr	53.19	59.51	66.00	72.64
Vertical roof snow load *	kips/upr	1.09	1.09	1.09	1.09
Vertical roof peak load	kips/upr	0.25	0.25	0.25	0.25
Bin floor pressure	kips/ft ²	1.40	1.44	1.48	1.51
Number of anchor bolts for uprights		16	16	16	16
Number of anchor bolts for bottom angle ring		48	48	48	48
* Based on maximum snow load of 39 psf					

Table 27. Foundation Loads — CEN Series 27' Bins (Imperial-Unfactored)

Model (CEN)		2713	2714	2715	2716
Vertical dead load	kips/upr	0.87	0.98	1.04	1.14
Vertical grain load	kips/upr	51.23	62.93	69.92	77.08
Vertical roof snow load *	kips/upr	1.23	1.23	1.23	1.23
Vertical roof peak load	kips/upr	0.28	0.28	0.28	0.28
Bin floor pressure	kips/ft ²	1.50	1.55	1.59	1.63
Number of anchor bolts for uprights		18	18	18	18
Number of anchor bolts for bottom angle ring		54	54	54	54
* Based on maximum snow load of 39 psf					

Table 28. Foundation Loads — CEN Series 30' Bins (Imperial-Unfactored)

Model (CEN)		3013	3014	3015	3016
Vertical dead load	kips/upr	0.91	1.00	1.13	1.28
Vertical grain load	kips/upr	46.19	57.62	70.55	81.08
Vertical roof snow load *	kips/upr	1.40	1.40	1.40	1.40
Vertical roof peak load	kips/upr	0.25	0.25	0.25	0.25
Bin floor pressure	kips/ft ²	1.59	1.64	1.69	1.74
Number of anchor bolts for uprights		20	20	20	40
Number of anchor bolts for bottom angle ring		60	60	60	60
* Based on maximum snow load of 40 psf					

Table 29. Foundation Loads — CEN Series 33' Bins (Imperial-Unfactored)

Model (CEN)		3313	3314	3315	3316
Vertical dead load	kips/upr	0.92	1.01	1.14	1.30
Vertical grain load	kips/upr	43.76	52.36	64.43	77.97
Vertical roof snow load *	kips/upr	1.69	1.69	1.69	1.69
Vertical roof peak load	kips/upr	0.23	0.23	0.23	0.23
Bin floor pressure	kips/ft ²	1.67	1.73	1.79	1.84
Number of anchor bolts for uprights		22	22	22	22
Number of anchor bolts for bottom angle ring		66	66	66	66
* Based on maximum snow load of 44 psf					

Table 30. Foundation Loads — CEN Series 36' Bins (Imperial-Unfactored)

Model (CEN)		3613	3614	3615	3616
Vertical dead load	kips/upr	0.97	1.05	1.15	1.28
Vertical grain load	kips/upr	45.37	51.01	58.95	71.64
Vertical roof snow load *	kips/upr	1.59	1.59	1.59	1.59
Vertical roof peak load	kips/upr	0.21	0.21	0.21	0.21
Bin floor pressure	kips/ft ²	1.74	1.81	1.87	1.93
Number of anchor bolts for uprights		24	24	24	24
Number of anchor bolts for bottom angle ring		72	72	72	72
* Based on maximum snow load of 38 psf					

Table 31. Foundation Loads — CEN Series 39' Bins (Imperial-Unfactored)

Model (CEN)		3913	3914	3915	3916
Vertical dead load	kips/upr	1.04	1.11	1.21	1.31
Vertical grain load	kips/upr	46.87	52.73	58.81	65.93
Vertical roof snow load *	kips/upr	1.64	1.64	1.64	1.64
Vertical roof peak load	kips/upr	0.19	0.19	0.19	0.19
Bin floor pressure	kips/ft ²	1.81	1.88	1.95	2.01
Number of anchor bolts for uprights		26	26	26	26
Number of anchor bolts for bottom angle ring		78	78	78	78
* Based on maximum snow load of 36 psf					

Table 32. Foundation Loads — CEN Series 42' Bins (Imperial-Unfactored)

Model (CEN)		4213	4214	4215	4216
Vertical dead load	kips/upr	1.08	1.16	1.26	1.36
Vertical grain load	kips/upr	48.28	54.34	60.64	67.16
Vertical roof snow load *	kips/upr	1.66	1.66	1.66	1.66
Vertical roof peak load	kips/upr	0.18	0.18	0.18	0.18
Bin floor pressure	kips/ft ²	1.87	1.94	2.02	2.09
Number of anchor bolts for uprights		28	28	28	28
Number of anchor bolts for bottom angle ring		84	84	84	84
* Based on maximum snow load of 34 psf					

Table 33. Foundation Loads — CEN Series 45' Bins (Imperial-Unfactored)

Model (CEN)		4513	4514	4515	4516
Vertical dead load	kips/upr	1.12	1.22	1.33	1.44
Vertical grain load	kips/upr	49.62	55.86	62.36	69.10
Vertical roof snow load *	kips/upr	1.68	1.68	1.68	1.68
Vertical roof peak load	kips/upr	0.17	0.17	0.17	0.17
Bin floor pressure	kips/ft ²	1.92	2.01	2.08	2.16
Number of anchor bolts for uprights		30	30	30	30
Number of anchor bolts for bottom angle ring		90	90	90	90
* Based on maximum snow load of 32 psf					

Table 34. Foundation Loads — CEN Series 48' Bins (Imperial-Unfactored)

Model (CEN)		4813	4814	4815	4816
Vertical dead load	kips/upr	1.26	1.38	1.47	1.62
Vertical grain load	kips/upr	50.89	57.30	63.99	70.93
Vertical roof snow load *	kips/upr	2.18	2.18	2.18	2.18
Vertical roof peak load	kips/upr	0.63	0.63	0.63	0.63
Bin floor pressure	kips/ft ²	1.97	2.06	2.14	2.22
Number of anchor bolts for uprights		32	32	32	32
Number of anchor bolts for bottom angle ring		96	96	96	96
* Based on maximum snow load of 39 psf					

Table 35. Foundation Loads — CEN Series 51' Bins (Imperial-Unfactored)

Model (CEN)		5111	5112	5113	5114	5115	5116
Vertical dead load	kips/upr	1.09	1.21	1.30	1.44	1.53	1.67
Vertical grain load	kips/upr	39.25	45.18	51.43	57.98	64.81	71.91
Vertical roof snow load *	kips/upr	2.32	2.32	2.32	2.32	2.32	2.32
Vertical roof peak load	kips/upr	0.59	0.59	0.59	0.59	0.59	0.59
Bin floor pressure	kips/ft ²	1.82	1.92	2.01	2.11	2.19	2.27
Number of anchor bolts for uprights		34	34	34	34	34	34
Number of anchor bolts for bottom angle ring		102	102	102	102	102	102
* Based on maximum snow load of 39 psf							

Table 36. Foundation Loads — CEN Series 54' Bins (Imperial-Unfactored)

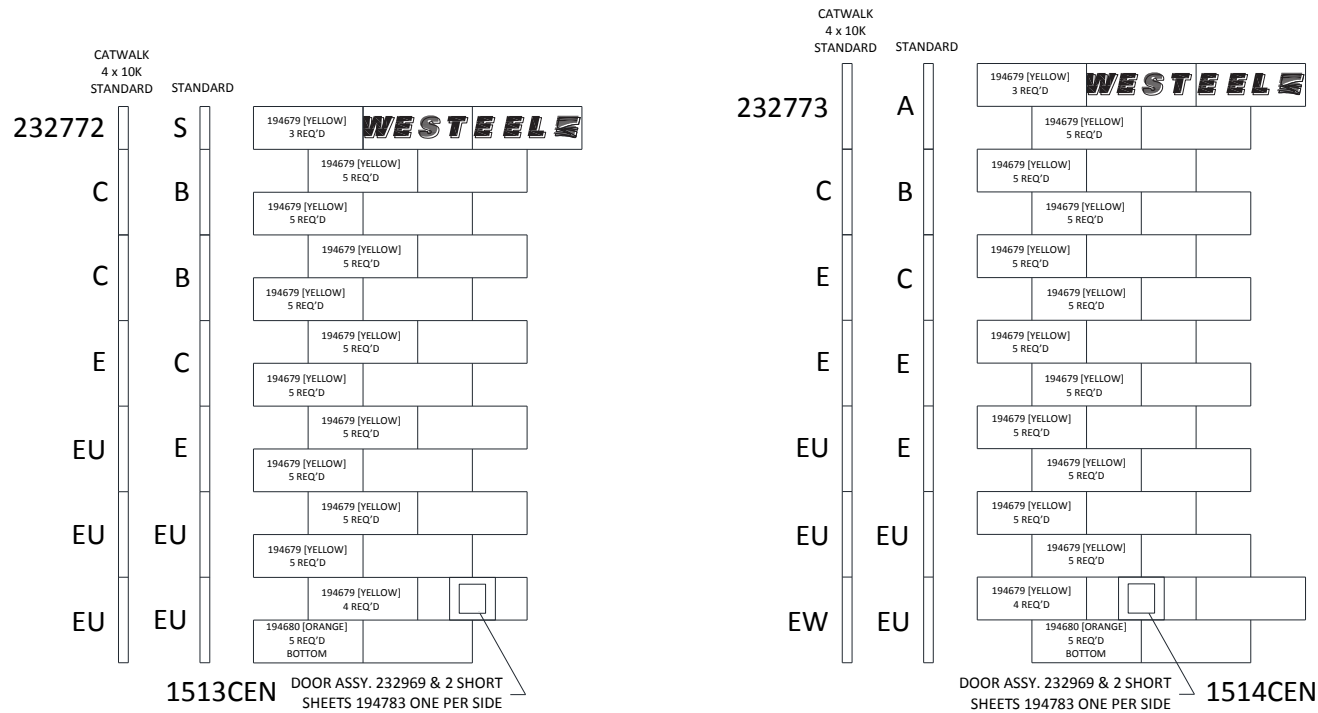
Model (CEN)		5410	5411	5412	5413	5414	5415	5416
Vertical dead load	kips/upr	1.38	1.47	1.58	1.67	1.81	1.93	2.06
Vertical grain load	kips/upr	34.39	40.10	46.16	52.55	59.24	66.23	73.51
Vertical roof snow load *	kips/upr	2.45	2.45	2.45	2.45	2.45	2.45	2.45
Vertical roof peak load	kips/upr	0.56	0.56	0.56	0.56	0.56	0.56	0.56
Bin floor pressure	kips/ft ²	1.74	1.85	1.96	2.06	2.15	2.24	2.33
Number of anchor bolts for uprights		36	36	36	36	36	36	36
Number of anchor bolts for bottom angle ring		108	108	108	108	108	108	108
* Based on maximum snow load of 39 psf								

Table 37. Foundation Loads — CEN Series 60' Bins (Imperial-Unfactored)

Model (CEN)		6008	6009	6010	6011	6012	6013	6014	6015	6016
Vertical dead load	kips/upr	1.29	1.37	1.48	1.59	1.71	1.84	1.96	2.09	2.24
Vertical grain load	kips/upr	25.12	30.27	35.82	41.74	48.03	54.67	61.64	68.92	76.51
Vertical roof snow load *	kips/upr	2.66	2.66	2.66	2.66	2.66	2.66	2.66	2.66	2.66
Vertical roof peak load	kips/upr	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Bin floor pressure	kips/ft ²	1.55	1.68	1.81	1.92	2.03	2.14	2.24	2.34	2.43
Number of anchor bolts for uprights		40	40	40	40	40	40	40	40	80
Number of anchor bolts for bottom angle ring		120	120	120	120	120	120	120	120	120
* Based on maximum snow load of 38										

6.3. Wall Sheet and Upright Layouts

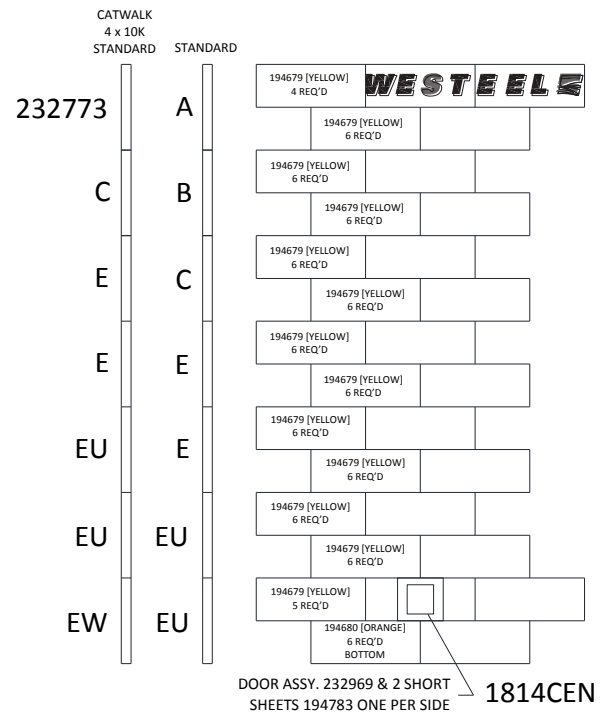
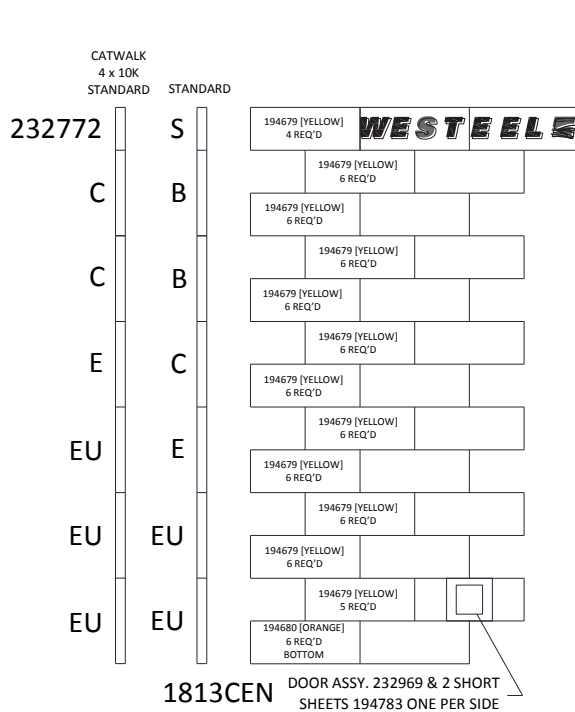
Figure 49. Model 1513 to 1514



Notes:

1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194654 and 194657 [YELLOW].
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232969 for all models.

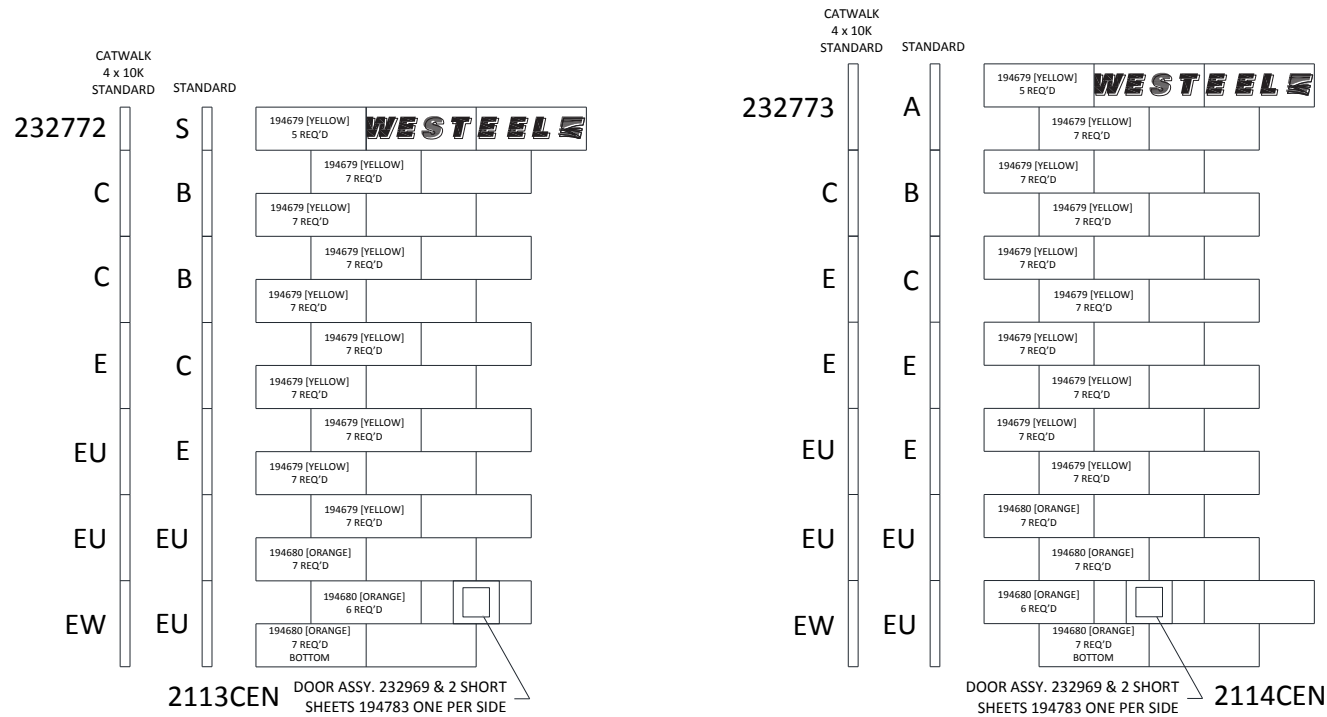
Figure 50. Model 1813 to 1814



Notes:

1. **Colors** match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194654 and 194657 [YELLOW].
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232969 for all models.

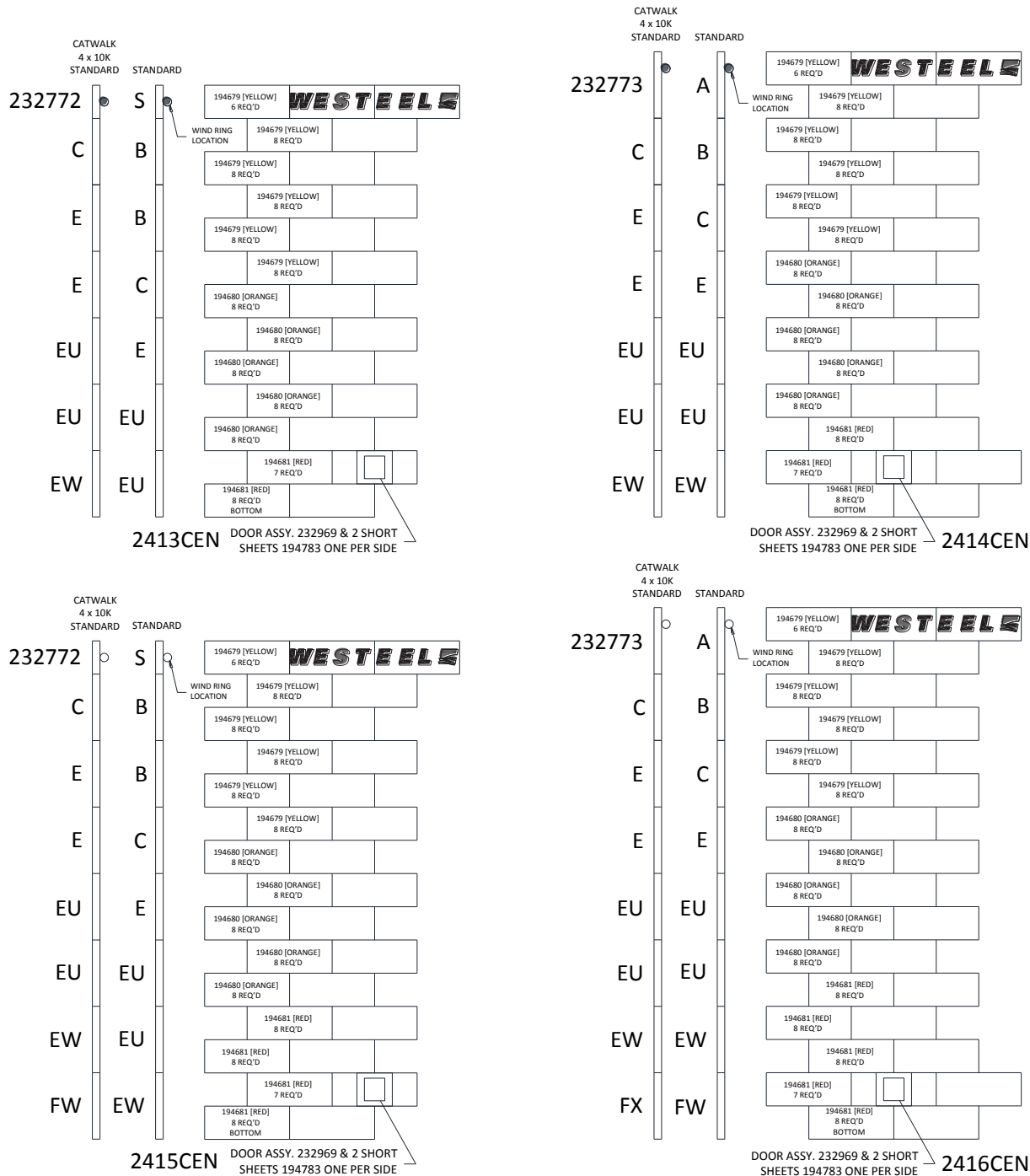
Figure 51. Model 2113 to 2114



Notes:

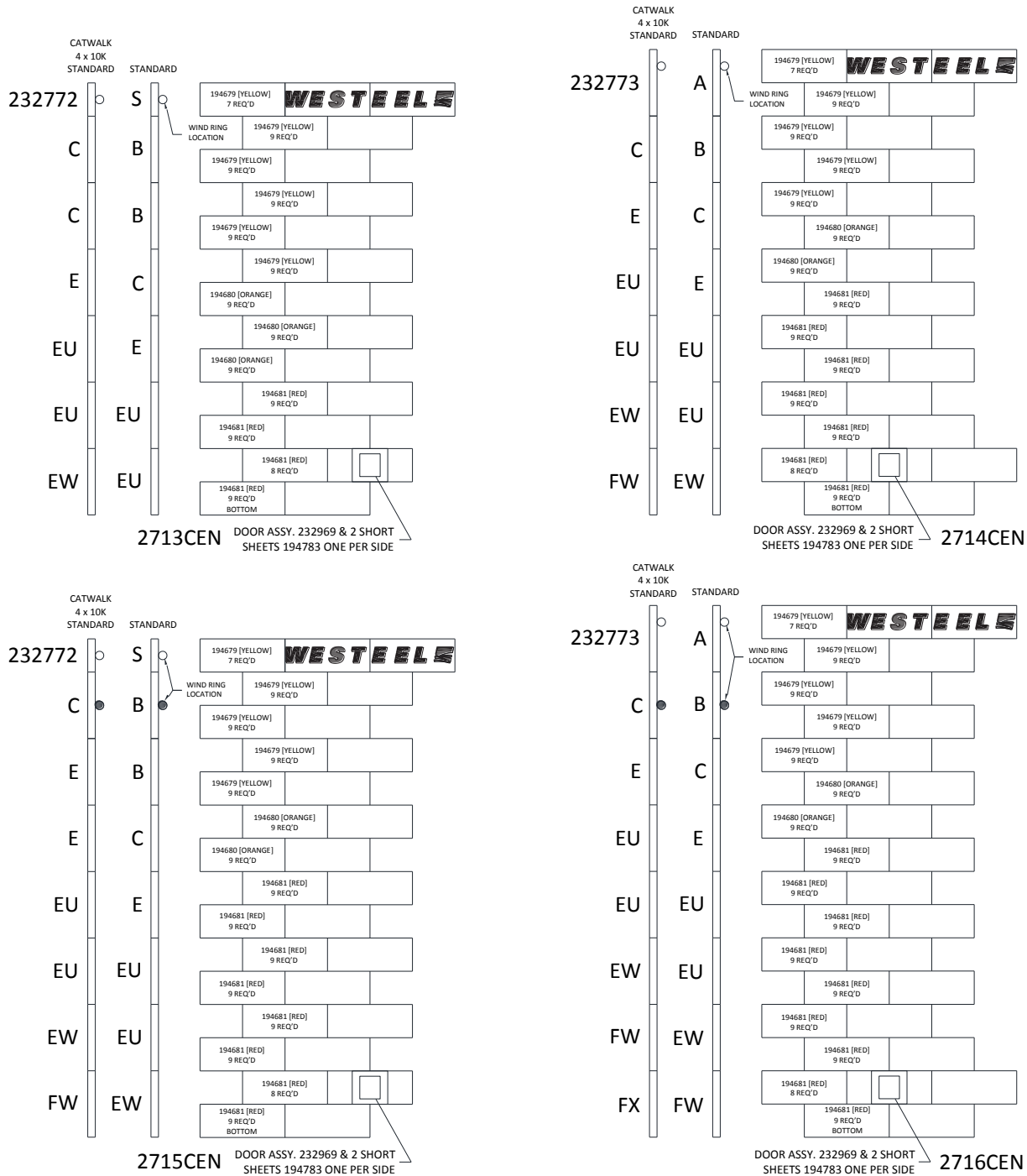
1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194654 and 194657 [YELLOW].
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232969 for all models.

Figure 52. Model 2413 to 2416

**Notes:**

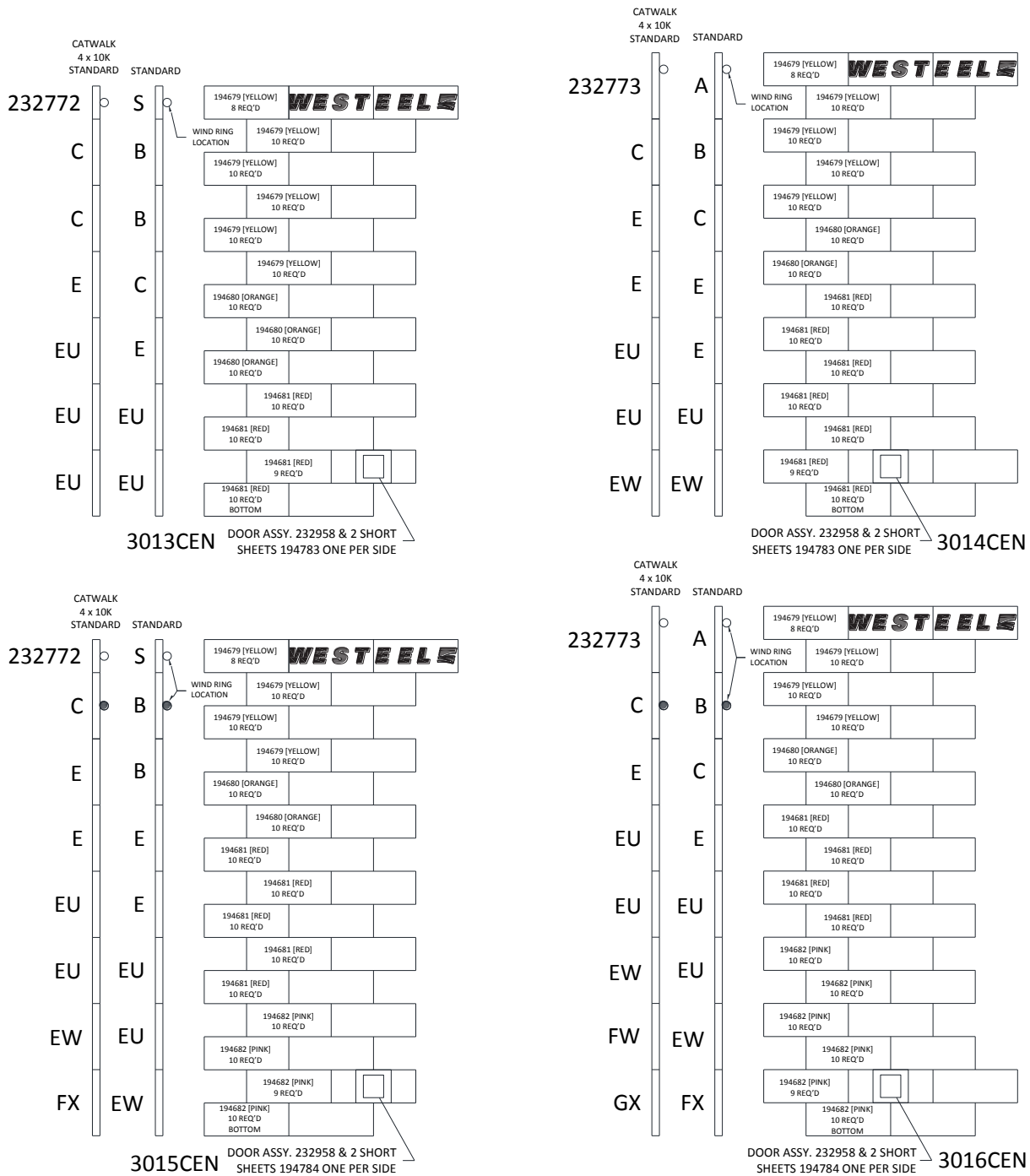
1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194654 and 194657 [YELLOW]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232969 for all models.
5. ○ — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 53. Model 2713 to 2716

**Notes:**

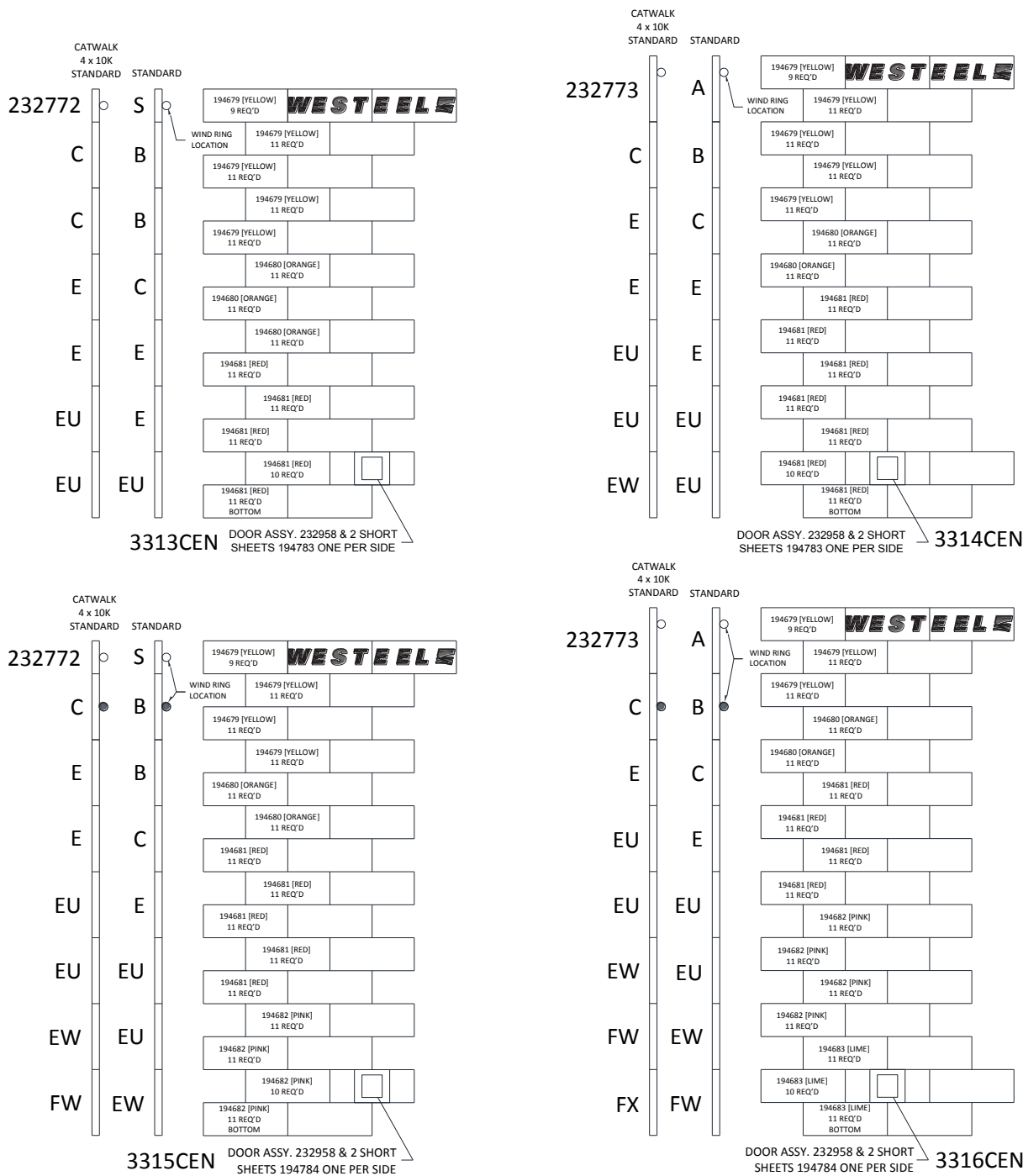
- Colors** match part number label and indicate wall sheet thickness.
- Stencil sheets are 194654 and 194657 [YELLOW]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
- All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
- One tier door 232969 for all models.
- — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 54. Model 3013 to 3016

**Notes:**

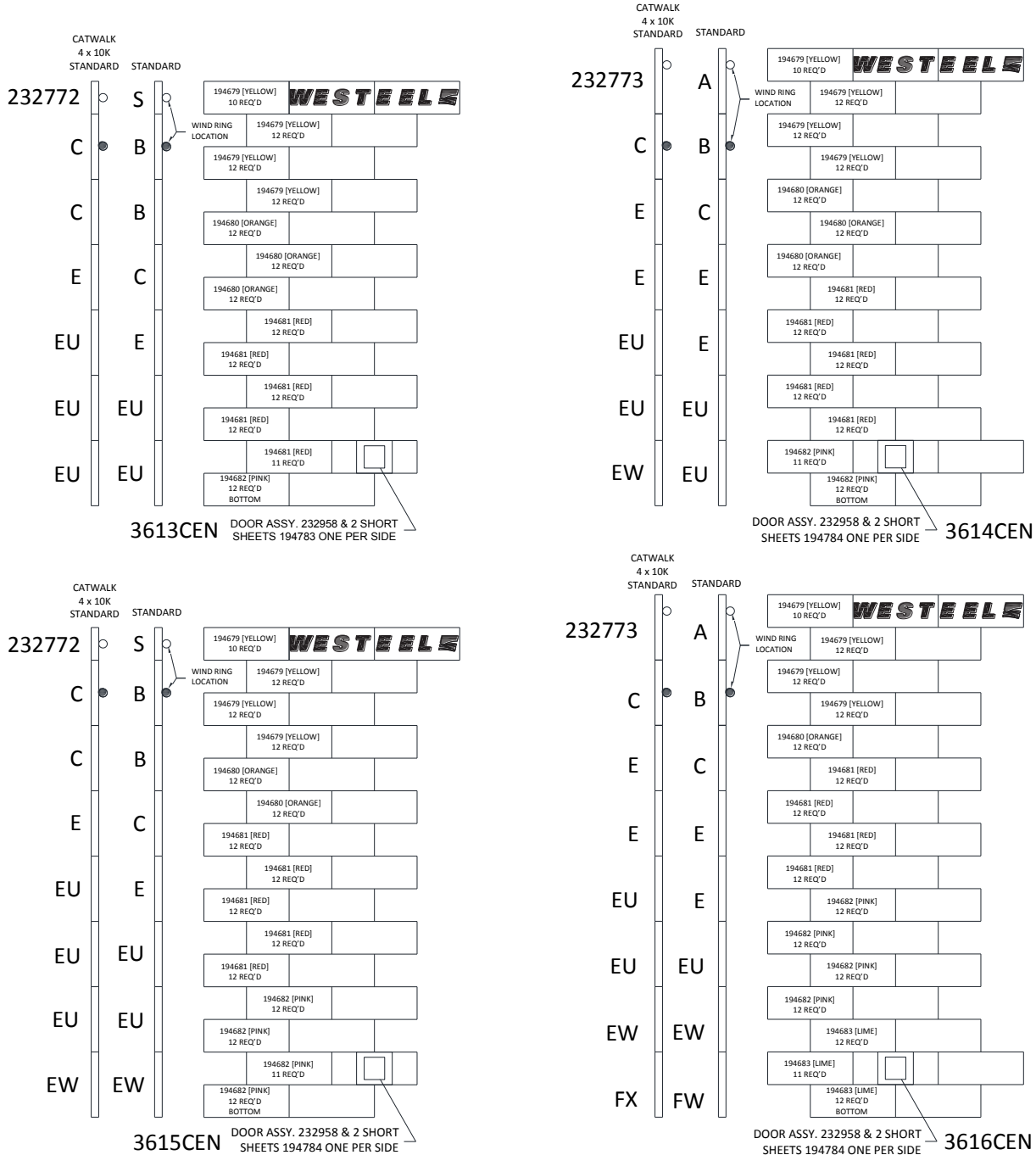
1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194654 and 194657 [YELLOW]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5.
 - — Indicates standard wind ring placement.
 - — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 55. Model 3313 to 3316

**Notes:**

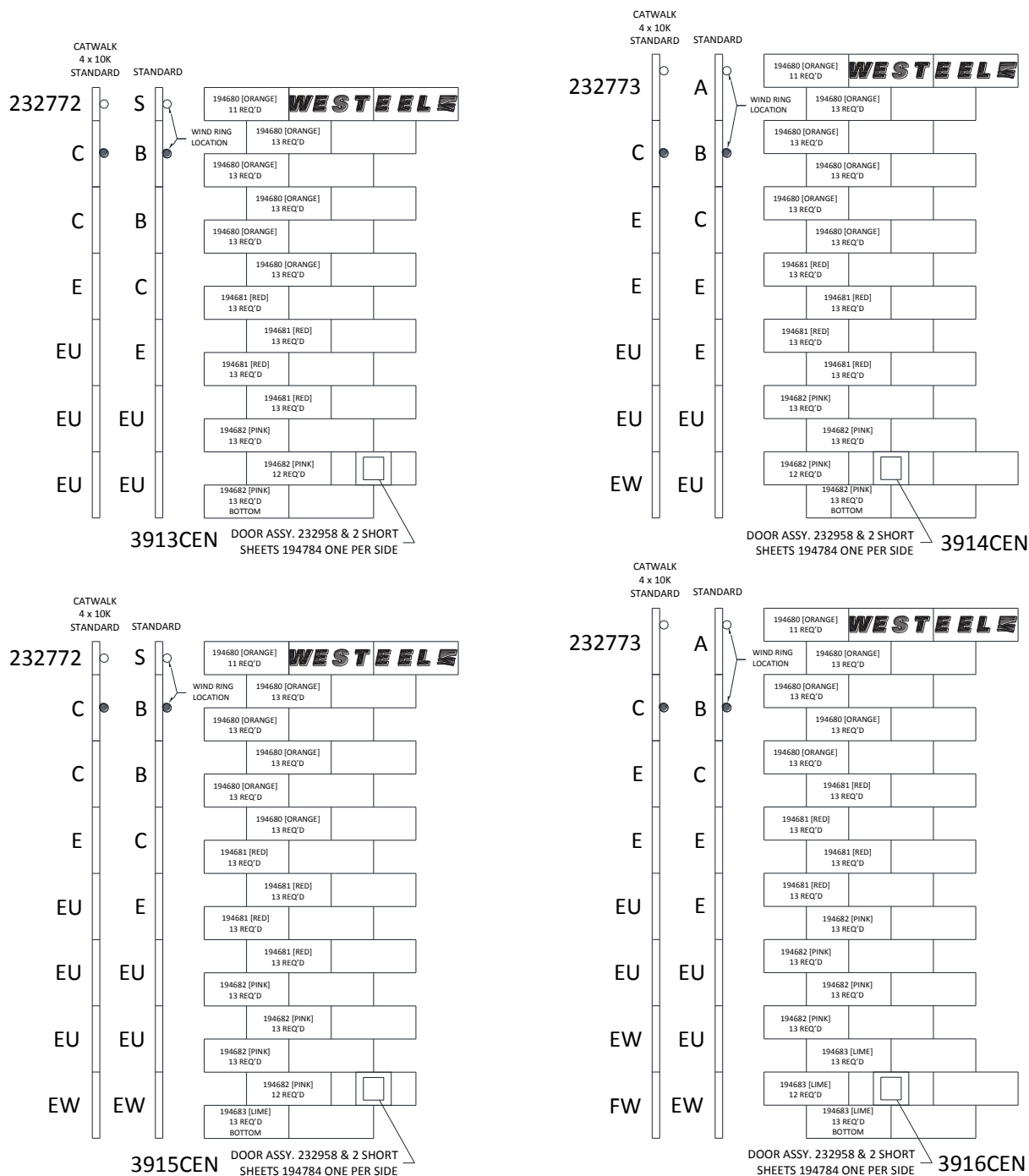
1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194654 and 194657 [YELLOW]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5.
 - — Indicates standard wind ring placement.
 - — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately)..

Figure 56. Model 3613 to 3616

**Notes:**

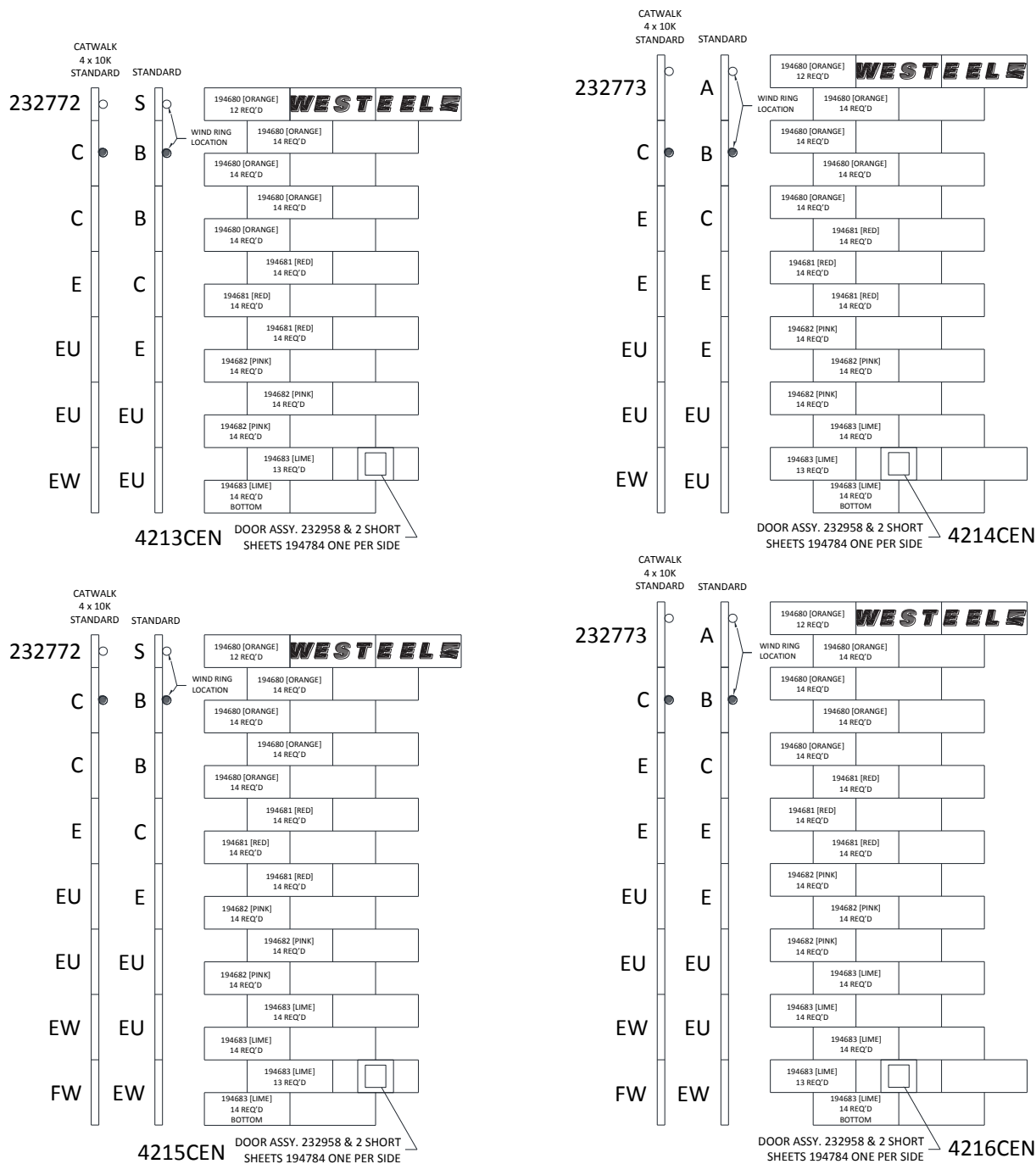
- Colors match part number label and indicate wall sheet thickness.
- Stencil sheets are 194654 and 194657 [YELLOW]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
- All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
- One tier door 232958 for all models.
- — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 57. Model 3913 to 3916

**Notes:**

- Colors** match part number label and indicate wall sheet thickness.
- Stencil sheets are 194655 and 194658 [ORANGE]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
- All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
- One tier door 232958 for all models.
- — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

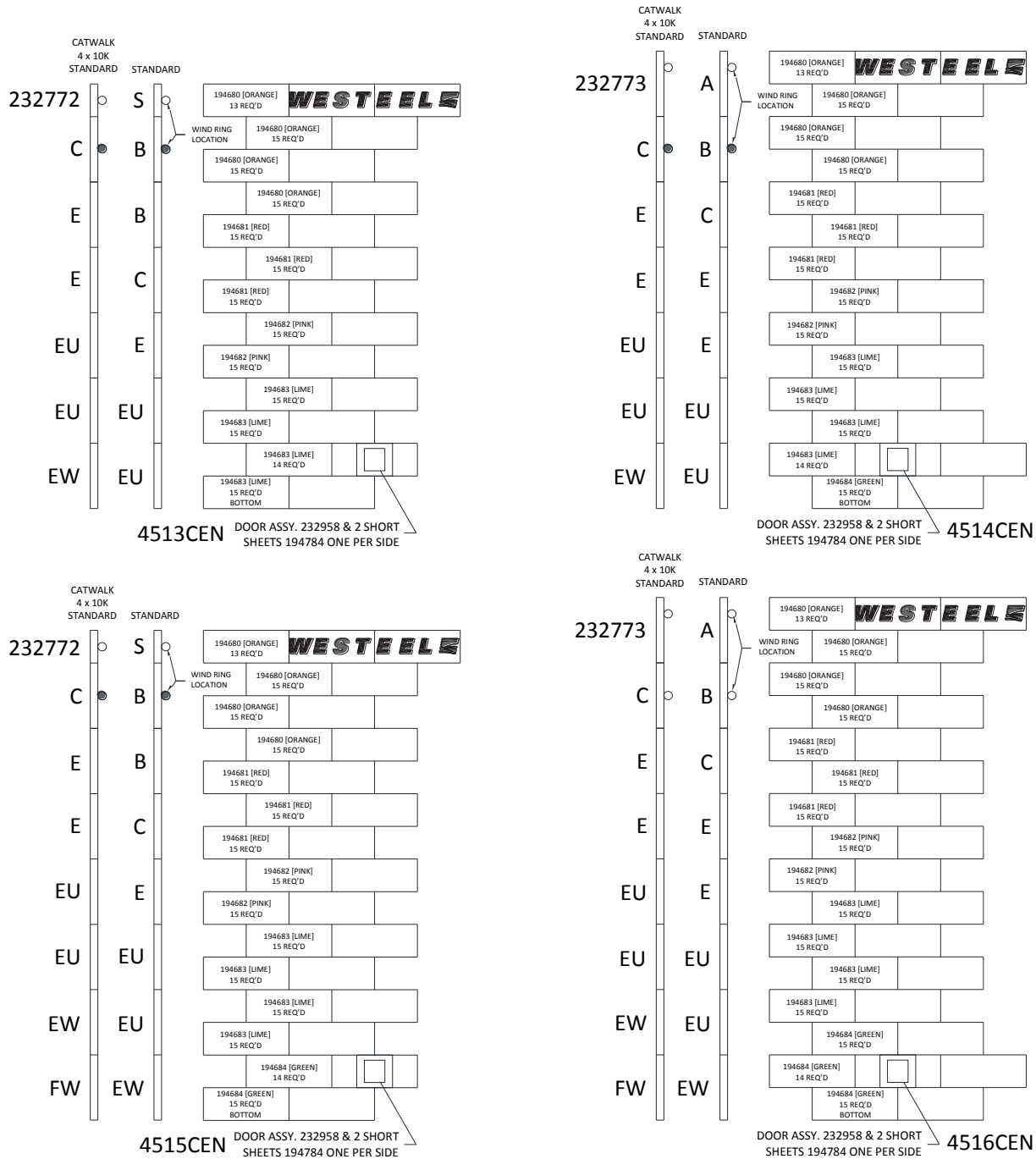
Figure 58. Model 4213 to 4216



Notes:

1. **Colors** match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194655 and 194658 [ORANGE]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5.
 - — Indicates standard wind ring placement.
 - — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

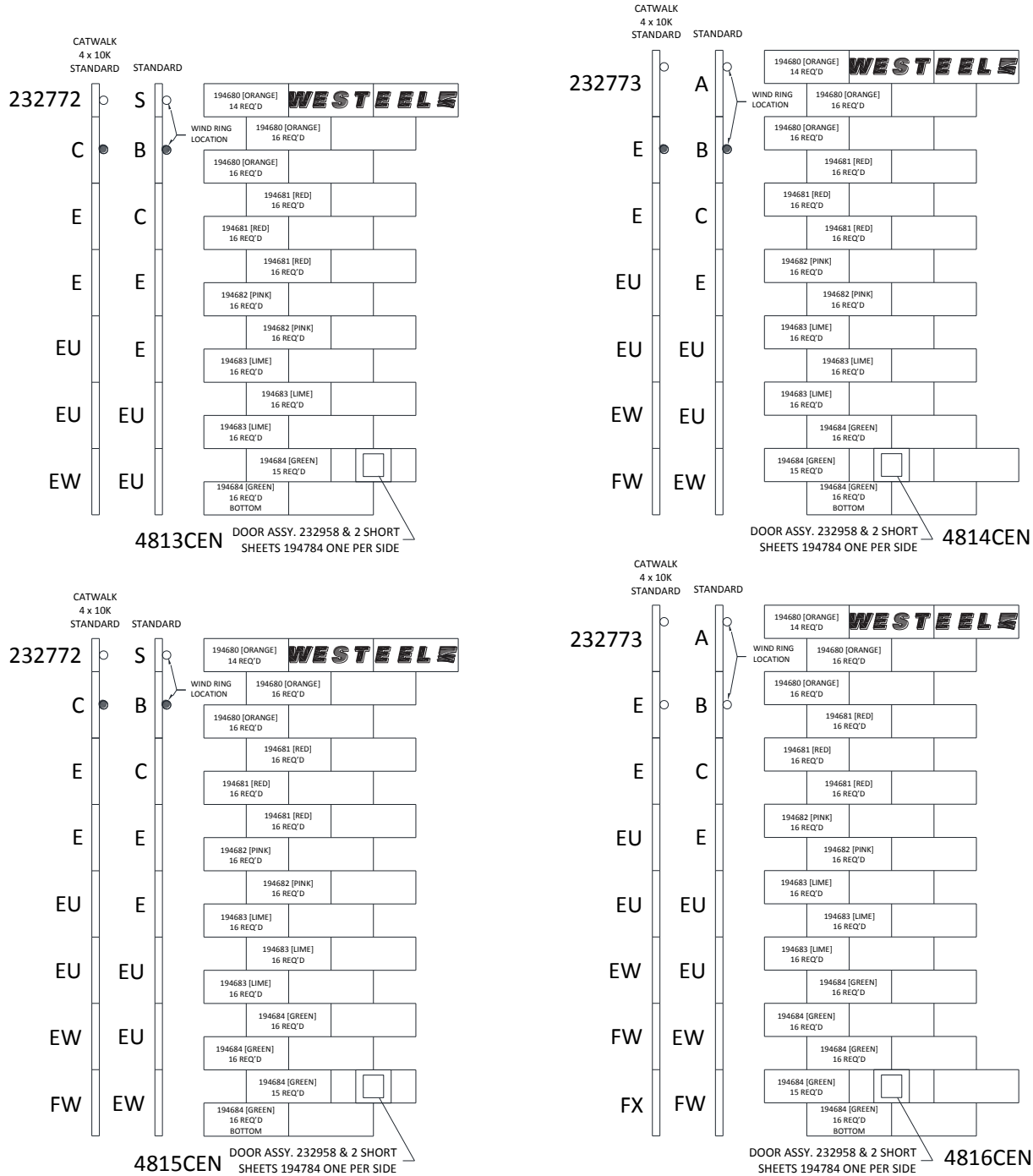
Figure 59. Model 4513 to 4516



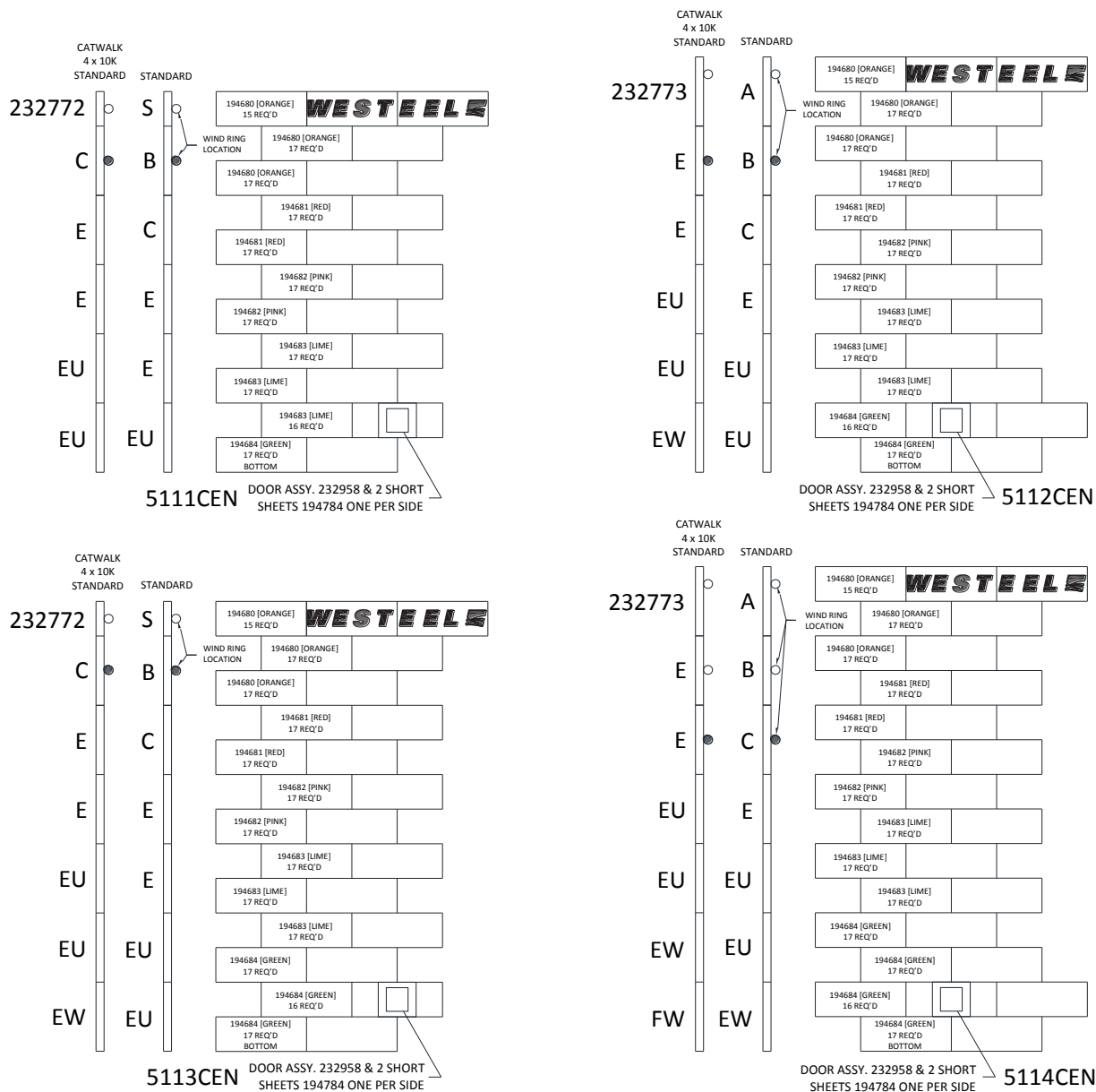
Notes:

1. **Colors** match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194655 and 194658 [ORANGE]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5.
 - — Indicates standard wind ring placement.
 - — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 60. Model 4813 to 4816

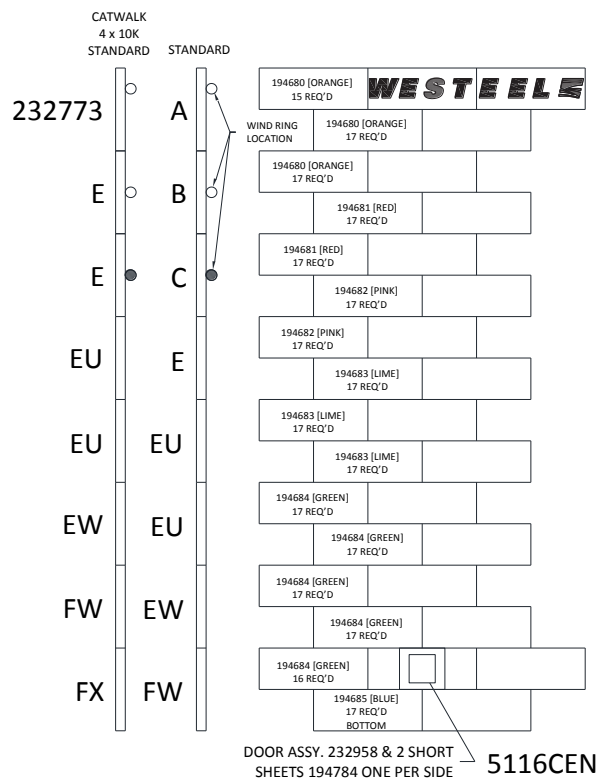
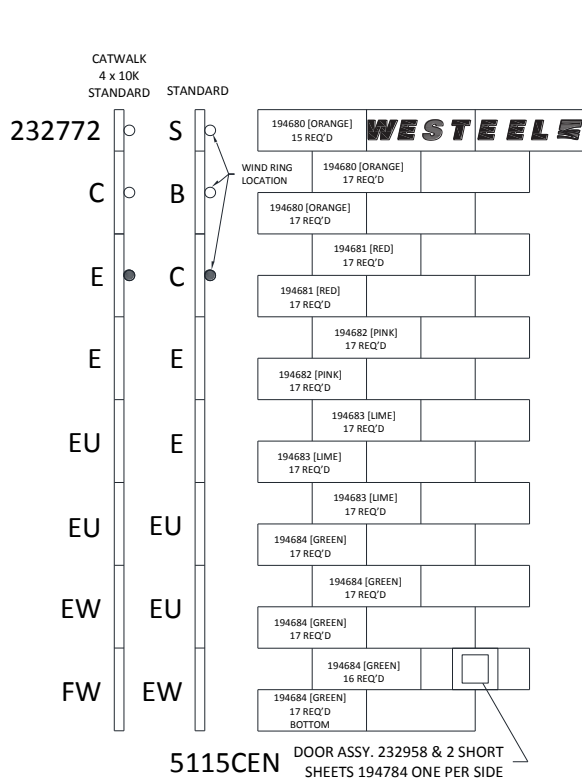
**Notes:**

1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194655 and 194658 [ORANGE]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5. ○ — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 61. Model 5111 to 5116**Notes:**

- Colors match part number label and indicate wall sheet thickness.
- Stencil sheets are 194655 and 194658 [ORANGE]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
- All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
- One tier door 232958 for all models.
- — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

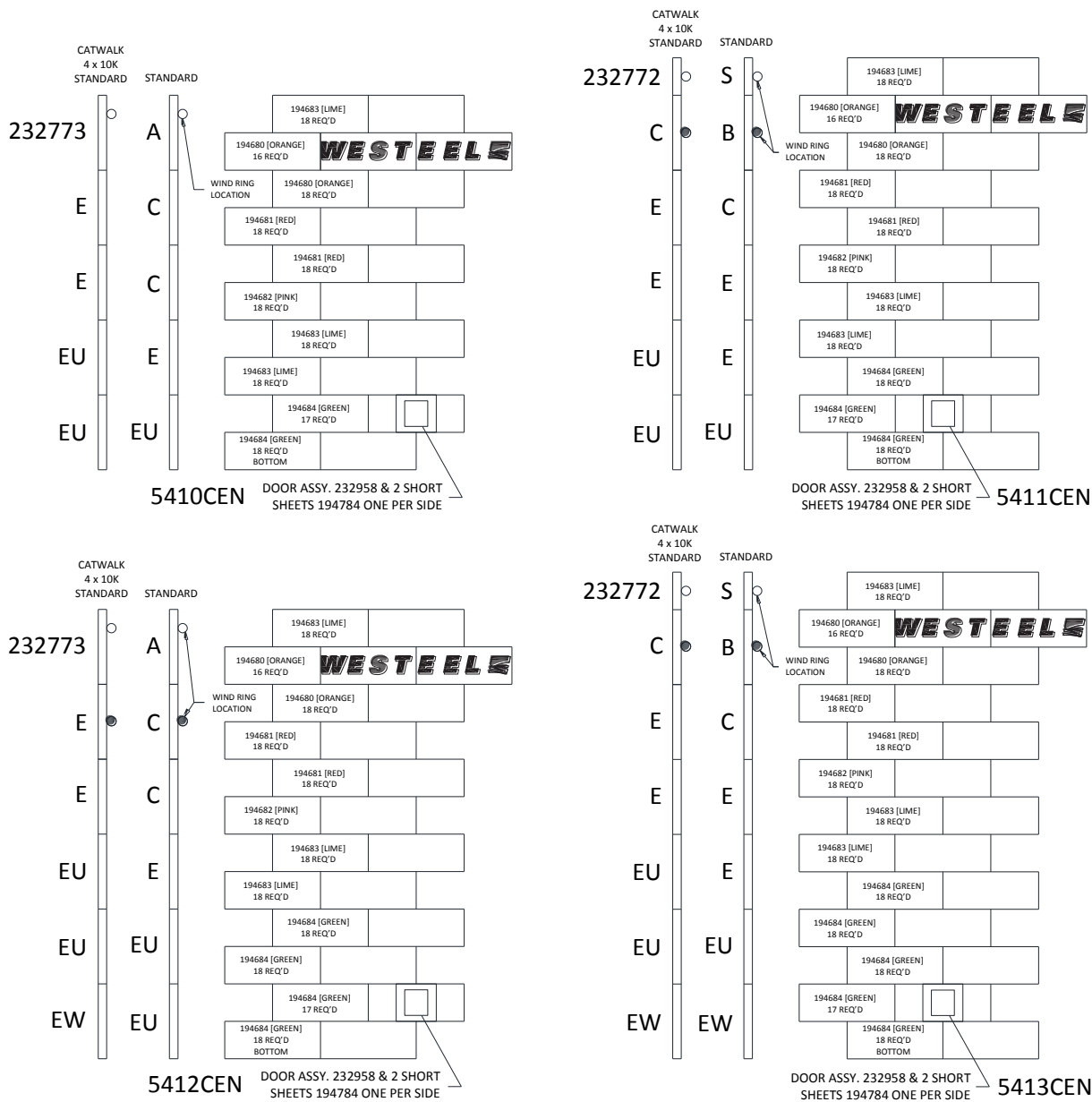
Figure 61 Model 5111 to 5116 (continued)



Notes:

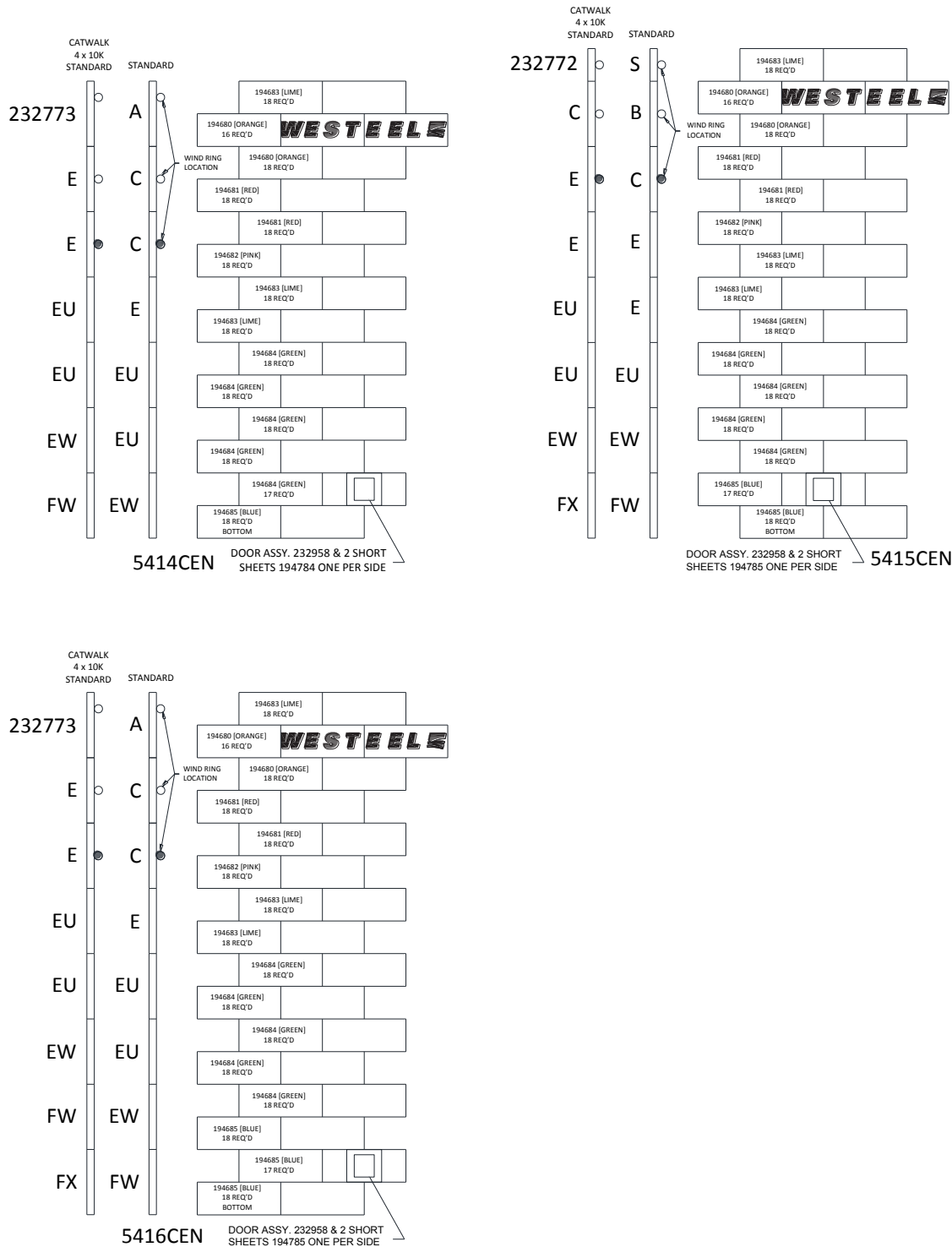
1. **Colors** match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194655 and 194658 [ORANGE]. They can be positioned in the 2nd tier from the top in order to keep them clear of the wind ring if desired.
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5.
 - — Indicates standard wind ring placement.
 - — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 62. Model 5410 to 5416

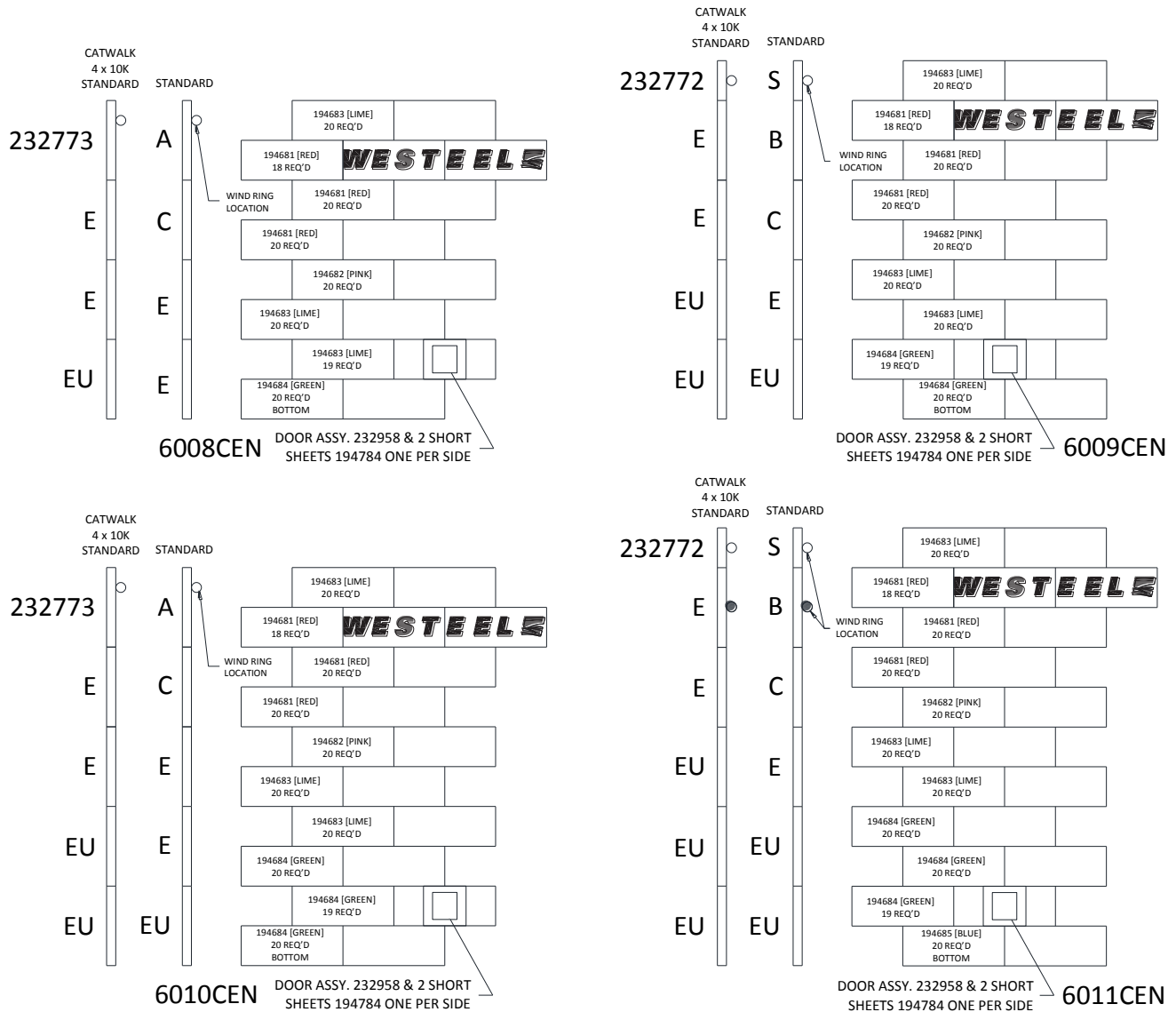
**Notes:**

- Colors match part number label and indicate wall sheet thickness.
- Stencil sheets are 194655 and 194658 [ORANGE].
- All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
- One tier door 232958 for all models.
- — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

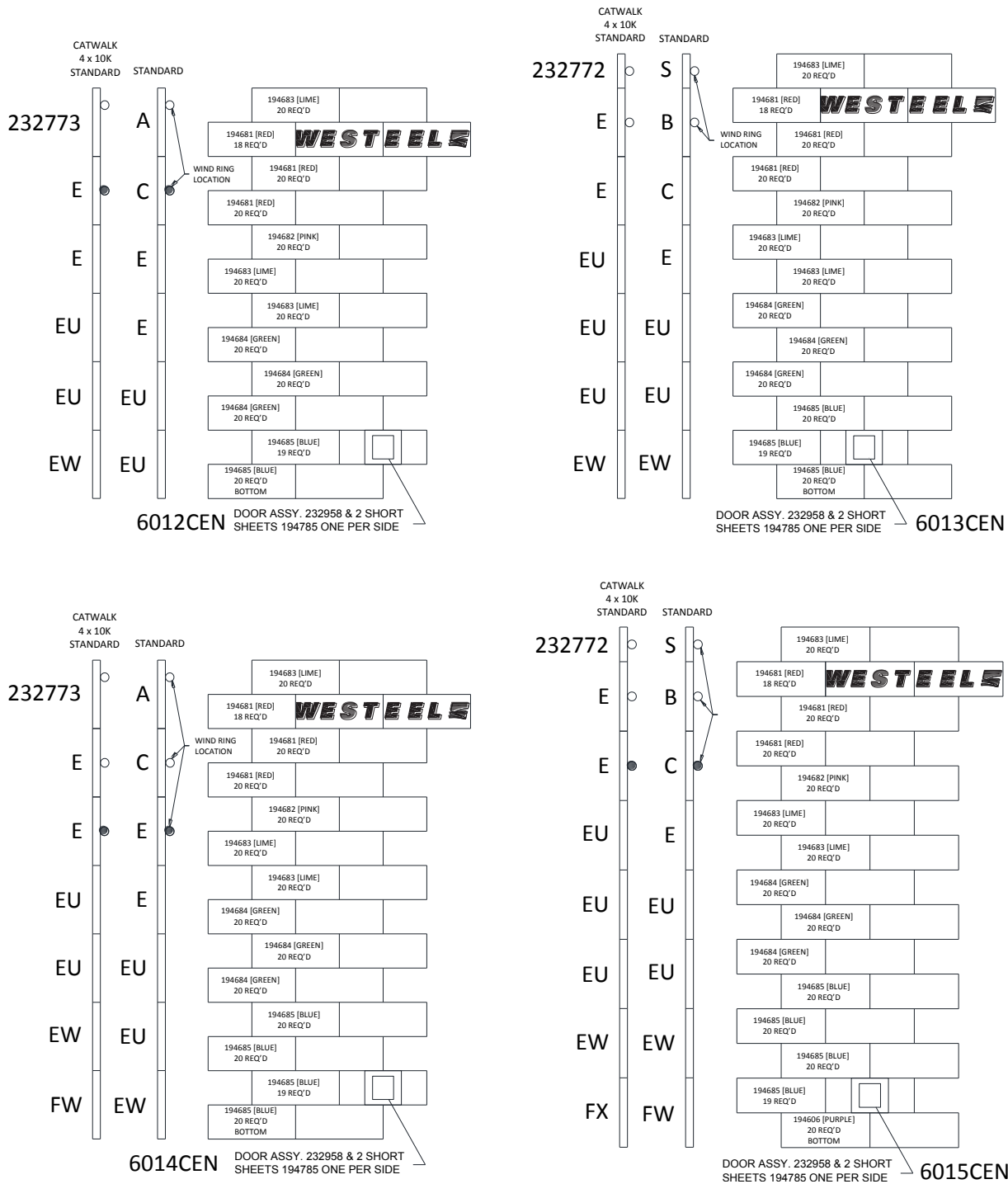
Figure 62 Model 5410 to 5416 (continued)

**Notes:**

- Colors match part number label and indicate wall sheet thickness.
- Stencil sheets are 194655 and 194658 [ORANGE].
- All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
One tier door 232958 for all models.
- — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

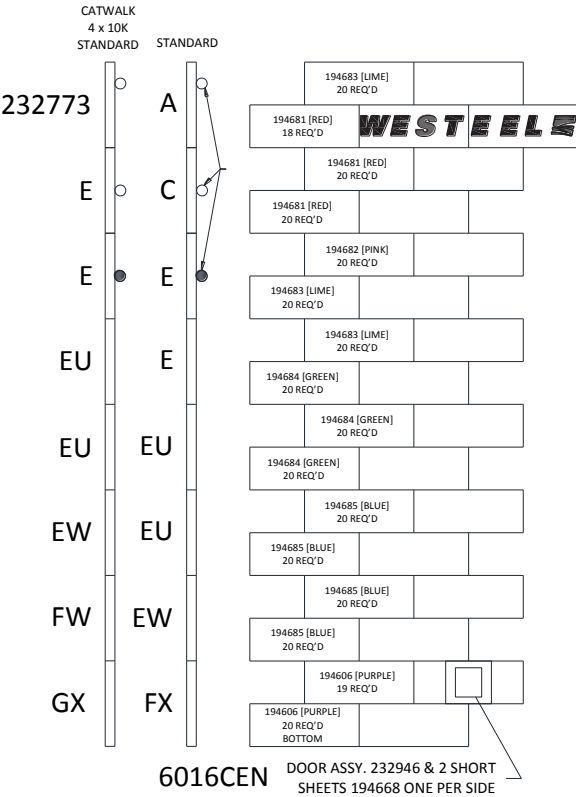
Figure 63. Model 6008 to 6016**Notes:**

1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194656 and 194659 [LIME].
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5.
 - — Indicates standard wind ring placement.
 - — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 63 Model 6008 to 6016 (continued)**Notes:**

1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194656 and 194659 [LIME].
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232958 for all models.
5.
 - — Indicates standard wind ring placement.
 - — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

Figure 63 Model 6008 to 6016 (continued)



Notes:

1. Colors match part number label and indicate wall sheet thickness.
2. Stencil sheets are 194656 and 194659 [LIME].
3. All uprights except for the top "S" and 232772 catwalk upright are 2 tiers long.
4. One tier door 232946.
5. ○ — Indicates standard wind ring placement.
● — Indicates additional wind ring placement if using Westeel side draw system (must be ordered separately).

7. Appendix

7.1. Roof Parts Box Listing

Table 38. Roof Parts Box Listing

Part Number	Description	Unit Weight (lbs)	15'	18'	21'	24'	27'	30'	33'	36'	39'	42'	45'	48'	51'	54'
			235760	235761	235762	235763	235764	235765	235766	235767	990170	235768	235778	235769	235777	235797
212201	PEAK RING 15	30.7	1	-	-	-	-	-	-	-	-	-	-	-	-	-
212203	PEAK RING 18	30.7	-	1	-	-	-	-	-	-	-	-	-	-	-	-
212204	PEAK RING 21	30.7	-	-	1	-	-	-	-	-	-	-	-	-	-	-
212205	PEAK RING 24	30.7	-	-	-	1	-	-	-	-	-	-	-	-	-	-
212206	PEAK RING 27	30.7	-	-	-	-	1	-	-	-	-	-	-	-	-	-
Varies	PEAK RING (30' - 54')	-	-	-	-	-	-									
212228	PEAK RING FOAM for 15-27, 51-54	0.4	1	1	1	1	1	-	-	-	-	-	-	-	2	2
212229	PEAK RING FOAM for 30-48	0.5	-	-	-	-	-	1	1	1	1	1	1	1	-	-
212230	BIRD STOP	0.127	15	18	21	24	27	30	33	36	39	42	45	48	51	54
212231	FOAM ROOF RIB CLOSURE (12)	0.06	2	2	2	2	3	3	3	3	4	4	4	4	5	5
170445	CAULKING - 300 ml TUBE	1	2	2	2	2	2	2	2	2	2	2	2	-	-	-
185010	CARTON 37x37x9 for BIN PARTS 15-27	9.4	1	1	1	1	1	-	-	-	-	-	-	-	-	-
185011	CARTON 53x27x7 for BIN PARTS 30-54	8.3	-	-	-	-	-	1	1	1	1	1	1	1	1	1
193062	LADDER RUNG 14.5 (6.0 CTR)	1	-	-	-	-	-	-	-	-	-	-	-	1	1	1
193063	LADDER RUNG 14.5 (8.0 CTR)	1	-	-	-	1	1	-	-	1	1	1	1	1	1	1
193064	LADDER RUNG 14.5 (10.0 CTR)	1	-	-	1	-	-	1	1	-	-	1	1	-	1	1
193065	LADDER RUNG 14.5 (12.0 CTR)	1	-	1	-	-	1	-	-	1	1	1	1	1	1	1
193066	LADDER RUNG 16.5 (14.0 CTR)	1.1	1	-	-	1	-	1	1	-	1	-	1	1	-	1
193067	LADDER RUNG 18.5 (16.0 CTR)	1.3	-	-	1	-	1	-	-	1	1	1	-	1	1	-
193068	LADDER RUNG 20.5 (18.0 CTR)	1.4	-	1	-	1	-	1	-	1	-	1	1	-	-	1
193069	LADDER RUNG 22.5 (20.0 CTR)	1.5	-	-	1	-	1	-	1	-	1	-	-	1	1	1
193070	LADDER RUNG 24.5 (22.0 CTR)	1.7	1	-	-	1	-	1	-	1	-	-	1	1	1	-
193071	LADDER RUNG 26.5 (24.0 CTR)	3.2	-	1	-	-	1	-	1	-	-	1	1	-	-	1
193072	LADDER RUNG 28.5 (26.0 CTR)	3.4	-	-	1	-	-	1	1	-	1	1	1	-	1	-
193073	LADDER RUNG 30.5 (28.0 CTR)	3.6	1	-	-	1	1	-	-	1	1	1	-	1	-	1
193074	LADDER RUNG 32.5 (30.0 CTR)	3.9	-	1	-	-	-	1	1	1	-	-	1	1	1	1
193075	LADDER RUNG 34.5 (32.0 CTR)	4.1	-	-	1	1	1	-	-	-	1	1	1	1	1	1
193076	LADDER RUNG 36.5 (34.0 CTR)	4.4	-	-	-	-	-	1	1	1	1	1	1	1	1	1
193077	LADDER RUNG 38.5 (36.0 CTR)	4.6	1	1	1	1	1	1	1	1	1	1	-	-	-	-
193078	LADDER RUNG 40.5 (38.0 CTR)	4.8	-	-	-	-	-	-	-	-	-	-	1	1	1	1
195063	STIFFENING RING BRACKET	0.31	-	-	-	-	-	-	33	36	39	42	45	96	153	162
195074	STIFFENING RING SPLICE 1.375	1.35	-	-	-	-	-	-	3	3	3	3	3	6	9	9
195080	STIFFENING RING GASKET - BAG 50	0.05	-	-	-	-	-	-	1	1	1	1	1	2	4	4
195085	STIFFENING RING EXPANDER 1.375	4.66	-	-	-	-	-	-	2	2	2	2	2	5	9	9
195149	PEAK RING BULB GASKET 105"	0.9	1	1	1	1	1	-	-	-	-	-	-	-	2	2
195150	PEAK RING BULB GASKET 168"	1.44	-	-	-	-	-	1	1	1	1	1	1	1	-	-
195695	DOOR TIE BACK CHAIN 17.75" LONG	0.52	1	1	1	1	1	1	1	1	1	1	1	-	-	-
212731	LOAD SPREADER TUBE 15	8.6	1	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 38 Roof Parts Box Listing (continued)

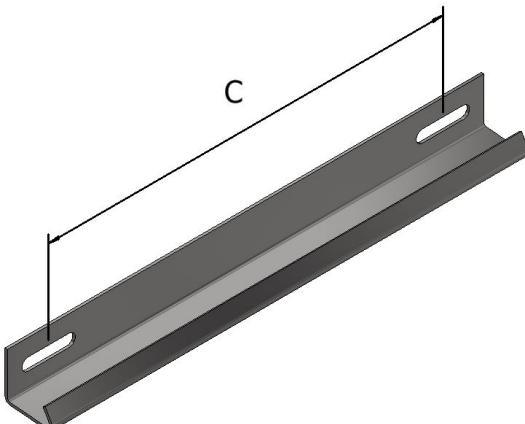
Part Number	Description	Unit Weight (lbs)	15'	18'	21'	24'	27'	30'	33'	36'	39'	42'	45'	48'	51'	54'
			235760	235761	235762	235763	235764	235765	235766	235767	990170	235768	235778	235769	235777	235797
212732	LOAD SPREADER TUBE 18	8.6	-	1	-	-	-	-	-	-	-	-	-	-	-	-
212733	LOAD SPREADER TUBE 21	8.6	-	-	1	-	-	-	-	-	-	-	-	-	-	-
212734	LOAD SPREADER TUBE 24	8.6	-	-	-	1	-	-	-	-	-	-	-	-	-	-
212735	LOAD SPREADER TUBE 27	8.6	-	-	-	-	1	-	-	-	-	-	-	-	-	-
212736	LOAD SPREADER TUBE 30	8.6	-	-	-	-	-	1	-	-	-	-	-	-	-	-
212737	LOAD SPREADER TUBE 33-36	8.6	-	-	-	-	-	-	1	1	-	-	-	-	-	-
212738	LOAD SPREADER TUBE 39-45	8.6	-	-	-	-	-	-	-	-	1	1	1	-	-	-
212739	LOAD SPREADER TUBE 48-54	8.6	-	-	-	-	-	-	-	-	-	-	-	1	1	1
212740	FALL RESTRAINT BRACKET	0.3	2	2	2	2	2	2	2	2	2	2	2	2	2	2
212789	RUBBER PAD	0.06	2	2	2	2	2	2	2	2	2	2	2	2	2	2
232720	UPRIGHT SPLICE	2.14	1	1	1	1	1	1	1	1	1	1	1	1	1	1
232767	WIND RING CLIP	0.44	2	2	2	2	2	2	2	2	2	2	2	2	2	2
232798	STIFFENING RING EXPANDER CLIP	0.125	-	-	-	-	-	-	2	2	2	2	2	5	9	9
234157	U-BOLT, ROUND .312 x 1.75W x 2.8L	0.12	-	-	-	-	-	-	33	36	39	42	45	96	153	162
235151	SELFDRILL SCREW .25 x 1.0 - BAG 7	0.13	-	-	-	-	-	-	1	1	1	1	1	2	3	3
235372	SEALING CLIP for BOTTOM ANGLE	0.12	5	6	7	8	9	10	11	12	13	14	15	-	-	-
235882	INSPECTION HATCH BULB GASKET 76"	0.5	1	1	1	1	1	1	1	1	1	1	1	1	1	1
235890	INSPECTION HATCH LID	7.48	1	1	1	1	1	1	1	1	1	1	1	1	1	1
235891	INSPECTION HATCH LATCH	0.81	1	1	1	1	1	1	1	1	1	1	1	1	1	1
235914	BOLT HFS .313 x 1.00 GR8.2 - BAG 250	8.5	1	1	1	2	2	2	3	3	3	4	5	5	6	7
235915	BOLT HFS .313 x 1.00 GR8.2 - BAG 50	1.7	-	1	3	-	-	2	-	2	4	2	-	3	2	-
235916	BOLT HFS .313 x 1.25 GR8.2 - BAG 80	3.04	1	-	-	1	1	2	2	1	2	2	3	3	3	2
235917	BOLT HFS .313 x 1.25 GR8.2 - BAG 50	1.9	-	2	2	1	1	-	-	2	1	1	-	-	-	2
235923	HEX FLANGE NUT .313 - BAG 250	3.5	1	2	2	2	3	3	4	4	5	6	6	7	9	9
235925	HEX FLANGE NUT .313 - BAG 50	0.7	3	-	1	3	-	1	-	3	2	-	3	3	-	3
235954	HEX FLANGE NUT .375 GR5 - Bag 300	5.7	-	-	-	-	-	-	-	-	1	1	1	-	-	-
235955	HEX FLANGE NUT .375 GR5 - Bag 50	0.95	2	2	2	2	3	3	3	3	-	-	-	-	-	-
235956	FLAT WASHER .375 - BAG 200	2.8	-	1	1	1	1	1	1	1	1	1	1	-	-	-
235957	FLAT WASHER .375 - BAG 75	1.05	2	-	-	-	-	1	1	1	1	1	2	-	-	-
235973	WSHR SEAL .313 STL/NEO - BAG 25	0.1	1	1	1	1	1	1	1	1	1	2	2	2	2	2
235974	WSHR SEAL .375 STL/NEO - BAG 25	0.2	1	1	1	1	1	1	1	1	1	1	1	1	1	1
212454	MANUAL – ROOF NON STRUCTURAL	0.3	-	-	-	-	-	-	-	-	-	-	-	-	-	1
Shaded items are not included in the parts box, rather they are shipped individually.																

Note

For structural roof parts box listing, please refer to **Structural Roof Manual (212453)**.

7.2. Roof Parts Box Part Identification

Table 39. Roof Parts Box Part Identification



P/N	C
193061	4
193062	6
193063	8
193064	10
193065	12
193066	14
193067	16
193068	18
193069	20
193070	22

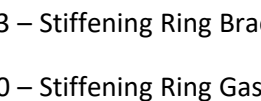
P/N	C
193071	24
193072	26
193073	28
193074	30
193075	32
193076	34
193077	36
193078	38

"C" Dimension is in inches

193061-78 – Roof Ladder Rung



195063 – Stiffening Ring Bracket



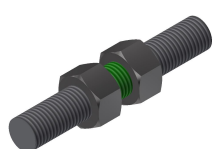
195080 – Stiffening Ring Gasket
(Bag of 50)



194120 – Grain Gauge



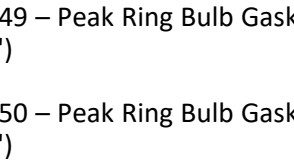
195074 – Stiffening Ring Splice
(1-3/8" Dia.)



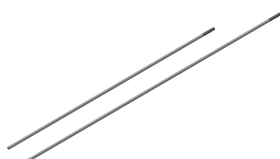
195085 – Stiffening Ring Expander
(1-3/8" Dia.)



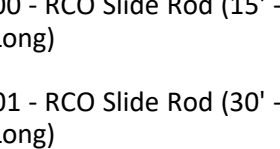
195149 – Peak Ring Bulb Gasket
(105")



195150 – Peak Ring Bulb Gasket
(168")



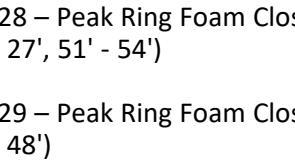
212400 - RCO Slide Rod (15' - 27')
(45" Long)



212401 - RCO Slide Rod (30' - 54')
(55" Long)



212228 – Peak Ring Foam Closure
(15' - 27', 51' - 54')



212229 – Peak Ring Foam Closure
(30' - 48')



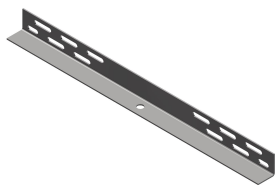
212230 – Bird Stop



212231 – Roof Rib Foam Closure



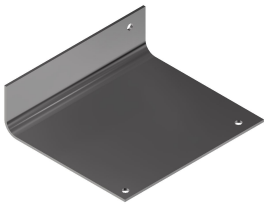
195695 – Door Tie Back Chain
(17-3/4" Long)



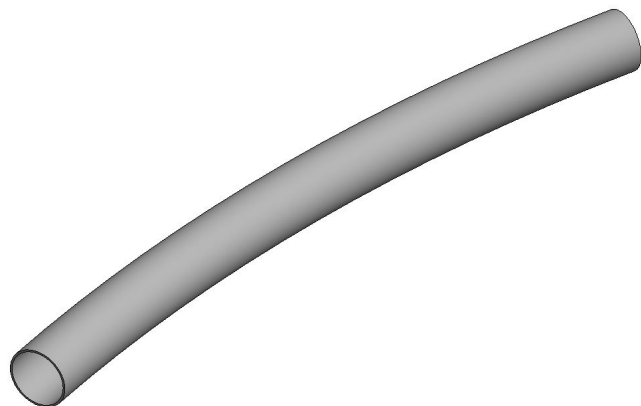
212402 – RCO Slide Rod Angle (15' - 48')



212403 – RCO Guide Rail (30' - 54') (51-7/8" Long)



212404 – RCO Rope Guide Plate (15' - 27')

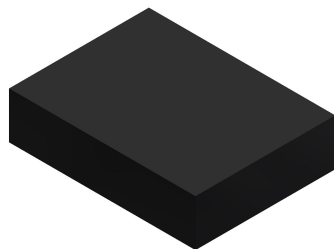


212731-39 — Load Spreader Tube

P/N	DIAMETER
212731	15
212732	18
212733	21
212734	24
212735	27
212736	30
212737	33 - 36
212738	39 - 45
212739	48 - 54



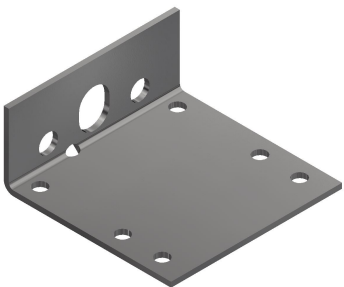
212740 – Fall Restraint Bracket



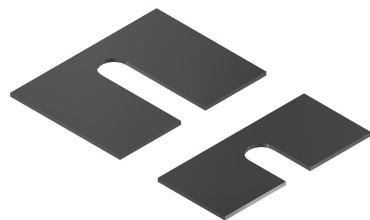
212789 - Rubber Pad



232720 – Upright Splice



232735 – Anchor Bracket



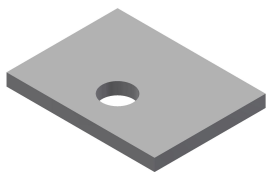
232736 – Shim (4-1/2" x 6")
232737 – Shim (2-3/4" x 6")



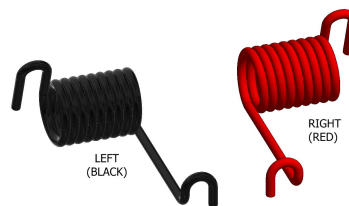
232767 – Wind Ring Clip



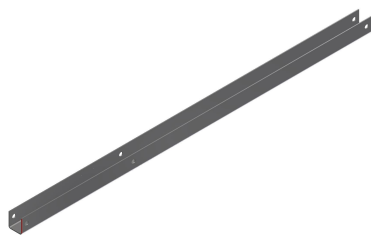
232798 - Stiffening Ring Expander Retaining Clip



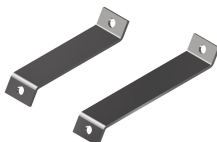
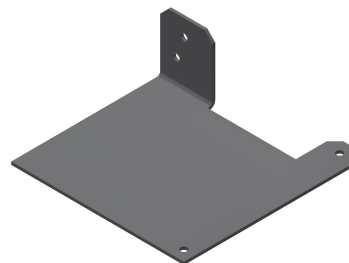
232836 – Base Plate

235009 – RCO Hardware Package
(15' - 27')

235012-13 – RCO Spring (15' - 27')



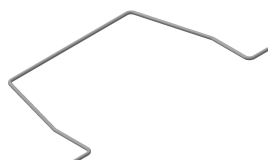
235218 – RCO Rope Arm (36")

235219 - RCO Rope Arm Support
(15' - 27')235279 – RCO Rope Guide Plate
(30' - 54')235332 – RCO Pivot Arm Bracket
(15' - 27')

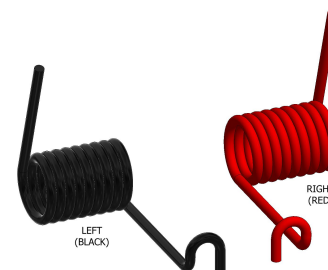
235333 – RCO Pivot Arm (15' - 27')



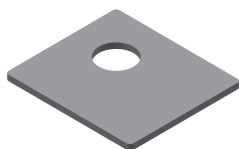
235335 – RCO Hardware Bag (30' - 48')

235337 - RCO Pivot Arm Bracket
(30' - 48')

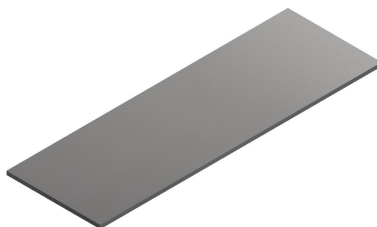
235338 – RCO Pivot Arm (30' - 54')



235341-42 – RCO Spring (30' - 54')



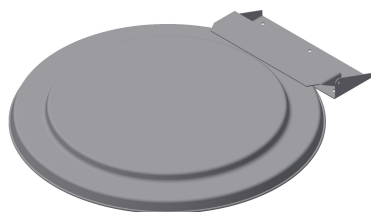
235372 – Bottom Angle Sealing Clip



236583 – SSK Shim (6" x 2")



235882 – Inspection Hatch Bulb Gasket



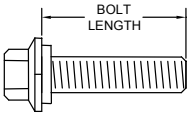
235890 – Inspection Hatch Lid Assembly



235891 – Inspection Hatch Latch Assembly

7.3. Hardware Usage

Table 40. Roof Hardware

	1/4" x 1" Self Drilling Screw	5/16" x 1" Flanged Hex Bolt (Washer)	5/16" x 1-1/4" Flanged Hex Bolt (Washer)	5/16" Flanged Lock Nut	5/16" STL/NEO Sealing Washer	3/8" x 1-1/2" Flanged Hex Bolt (Washer)	3/8" Hex Nut	3/8" STL/NEO Sealing Washer
	235151 (7)	235914 (250) 235915 (50)	235916 (80) 235917 (50)	235923 (250) 235925 (50)	235973 (25)	235946 (100) 235935 (55)	232850 (700) 235950 (300) 235951 (100)	235974 (25)
TOP RING ANGLE to WALL SHEET			•	•				
ROOF SHEET to PEAK RING		•		•				
ROOF SHEET to TOP RING ANGLE		•		•				
ROOF SHEET RIB to ROOF SHEET RIB		•		•				
LADDER RUNG to ROOF SHEET RIB			•	•	•			
STIFFENING RING BRACKET to ROOF SHEET RIB			•	•				
STIFFENING RING EXPANDER CLIP to STIFFENING RING	•							
ROOF VENT to ROOF SHEET VENT CUTOUT		•		•				
INSPECTION HATCH LID ASSEMBLY to ROOF SHEET HATCH CUTOUT		•		•				
INSPECTION HATCH LATCH to ROOF SHEET		•		•	•			
ROOF SHEET to PEAK RING to FALL RESTRAINT BRACKET		•		•				
WIND RING CLIP to WALL SHEET to UPRIGHT SPLICE						•	•	•
FALL RESTRAINT BRACKET to UPRIGHT SPLICE		•		•				
BIRD STOP to TOP RING ANGLE		•		•				

Note

For structural roof hardware usage, please refer to **Structural Roof Manual (212453)**.

Table 41. Bin Hardware

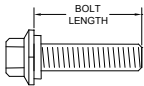
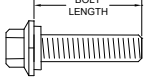
	3/8" x 1" Flanged Hex Bolt (Washer)	3/8" x 1" Round Head Bolt	3/8" x 1-1/2" Flanged Hex Bolt (Washer)	3/8" x 3-3/4" Hex Bolt	3/8" Flanged Lock Nut	3/8" Hex Nut	3/8" Wingnut	3/8" Flat Washer	3/8" STL/NEO Sealing Washer	7/16" x 1-1/2" Flanged Hex Bolt (Washer)	7/16" x 1-3/4" Flanged Hex Bolt (Washer)	7/16" Hex Nut	1/2" Flat Washer
	232850 (700) 235941 (325) 235943 (50)	150594	232852 (500) 235946 (100)	235949 (10)	235954 (300) 235955 (50)	232850 (700) 232852 (500) 235950 (300) 235951 (100)	154208	235956 (200) 235957 (75)	235975 (100)	232855 (400)	232856 (300)	232855 (400) 232856 (300)	154981
INSIDE ROOF CONNECTOR UPRIGHT to WALL SHEET to OUTSIDE UPRIGHT (DOUBLE NUT)			•		•	•			•				
WALL SHEETS 194679 to 194685, and 194606 to 194607 (0.040" to 0.139")	•					•		• ☀					
WALL SHEET 194608 (0.168")			•			•		• ☀					
UPRIGHT to WALL SHEETS 194679 to 194685 (0.040" to 0.116")	•					•							
UPRIGHT to WALL SHEETS 194606 to 194608, and 194604 to 194618 (0.126" to 0.168", AND 0.096" LAM to 0.139" LAM)			•			•							
WALL SHEETS 194604 to 194605, and 194616 to 194617 (0.096" LAM to 0.139" LAM)										•		•	• ♦
WALL SHEET 194618 (0.168" LAM)											•	•	• ♦
UPRIGHT to WALL SHEET AT HORIZONTAL SEAMS			•			•							
UPRIGHT to LAMINATE to CAP PLATE to WALL SHEET (FOR BINS WITH BOXED UPRIGHTS ONLY)			•			•							
UPRIGHT to UPRIGHT SPLICE	•					•							
UPRIGHT to LAMINATE	•					•							
UPRIGHT to LAMINATE to BOXED UPRIGHT			•			•							
WALL SHEET to UPRIGHT to LAMINATE to CAP PLATE (FOR BINS WITH BOXED UPRIGHTS)			•			•							

Table 41 Bin Hardware (continued)

	3/8" x 1" Flanged Hex Bolt (Washer)	3/8" x 1" Round Head Bolt	3/8" x 1-1/2" Flanged Hex Bolt (Washer)	3/8" x 3-3/4" Hex Bolt	3/8" Flanged Lock Nut	3/8" Hex Nut	3/8" Wingnut	3/8" Flat Washer	3/8" STL/NEO Sealing Washer	7/16" x 1-1/2" Flanged Hex Bolt (Washer)	7/16" x 1-3/4" Flanged Hex Bolt (Washer)	7/16" Hex Nut	1/2" Flat Washer
	232850 (700) 235941 (325) 235943 (50)	150594	232852 (500) 235946 (100)	235949 (10)	235954 (300) 235955 (50)	232850 (700) 232852 (500) 235950 (300) 235951 (100)	154208	235956 (200) 235957 (75)	235975 (100)	232855 (400)	232856 (300)	232855 (400) 232856 (300)	154981
WIND RING CLIP to UPRIGHT			•			•							
WIND RING SPLICE				•		•							
WALL SHEET to BOTTOM RING ANGLE	•				•	•		•					
DRYING FLOOR FLASHING HOLES in BOTTOM WALL SHEET	•					•							
WALL SHEET to DOOR			•			•		• ☀					
DOOR TIE-BACK to WALL SHEET	•					•							
AUGER CHUTE HOOD to AUGER DOOR BOARD		•				•							
AUGER CHUTE BLOCK-OFF PLATE to AUGER DOOR BOARD			•			•	•						
BIN WALL to HOPPER ASSEMBLY			•			•		•					

Note

☀ — Use washers only at wall sheet to bottom ring angle, non-laminated to laminated wall sheet horizontal seam and wall sheet vertical seams to door (non-laminated sheets only; 3/8" bolts).

◆ — Use washers only at wall sheet to bottom ring angle.

7.4. Recommended Bolt Assembly

When tightening bolts, tighten the nut on the bolt until a “snug-tightened condition” has been achieved. A “snug-tightened condition” is defined in *Specification for Structural Joints Using ASTM A325 or A490 Bolts* (Research Council on Structural Connections: June 2004), which states:

*“The snug-tightened condition is the tightness that is attained with a few impacts of an impact wrench or the full effort of an ironworker using an ordinary spud wrench to bring the connected plies into **firm contact**.”*

A properly tightened bolt will compress the sealing washer noticeably. All assembly crew members must be made aware of this requirement, and must know how to achieve a snug-tightened condition using common bin-building tools.

It is important that the bolts in the vertical wall sheet seams are tightened enough to squeeze the caulking and bring the wall sheet surfaces into firm contact with each other. This is especially important to monitor when installing bolts in temperatures approaching -10°C (14°F).

The following table shows the minimum impact gun torque capacity necessary to achieve a snug-tightened condition for bolts used in the assembly process.

Table 42. Recommended Impact Gun Torque Values Capacity to Achieve Snug-Tightened Bolts

Bolt Diameter	Bolt Grade	Grade Mark	Recommended Torque Capacity		
			in-lb	ft-lb	N-m
1/4"	Grade 8.2		75	6	8
5/16"	Grade 8.2		215	18	24
3/8"	Grade 8.2		370	31	42
7/16"	Grade 8.2		600	50	68
1/2"	Grade 8.2		960	80	108
5/8"	Grade 8.2		1800	150	203
3/4"	Grade 5		3230	269	365

For proper sealing, do not overtighten the wall seam connections. Sealing is not critical on upright splice connections; these connections should be tightened securely to prevent loosening.

Hold the bolt head securely when tightening the nut to prevent damage to the sealing washer.

Important

ALWAYS TIGHTEN THE NUT, NOT THE BOLT.

Avoid bin assembly at temperatures below -10°C (14°F) if possible. Erection in low temperatures does not ensure strong, well sealed connections. Do not substitute bolts in place of those supplied by Westeel.

8. Limited Warranty: Westeel Grain Bin Products

Westeel – Ag Growth International ("Westeel") warrants products that it has manufactured and/or that are branded with its name (the "goods") subject to the following terms and limitations, (the "warranty"):

Duration of Warranty

This warranty will run from the date of purchase from the dealer or distributor, authorized by Westeel. The duration of the warranty is limited as follows:

Galvanized Bins	5 years
EasyFlow2	24 months
Westeel Fans	36 months
Floors	12 months
Catwalk	12 months
Bulk Feed Tanks	24 months
SeedStor-K Cones	
Paint	12 months
Structural	30 months
Elite Cones	
Paint	30 months
Structural	10 years
WESTEEL cones	
Paint	No Warranty
Structural	12 months
Smooth Wall Bins	
Paint	60 months
Structural	10 years
Commercial Smooth Wall Bins	
Paint	12 months
Structural	10 years

Limitation of Remedies Replacement

Within the warranty period, Westeel will replace the goods and/or original manufactured components thereof which are found, to Westeel's satisfaction, to be defective. Westeel is not responsible for direct, indirect, special, consequential, or any other damages of any kind, including personal injury to any individual, howsoever caused, including caused by transportation of the goods for repair or replacement.

Procedure for Obtaining Service

In the event of a warranty claim, the purchaser must complete any and all information required by Westeel in order to properly assess or investigate the claim. Westeel will not be responsible for the removal of any of the goods found to be defective, or transportation charges to and from Westeel's authorized dealer or distributor, or for installation of any replacement goods and/or parts furnished under the warranty.

Limitations as to Scope of Warranty

The warranty does not extend to defects or damage caused, in whole or in part, by:

1. use of a kind and/or to a degree not reasonably expected to be made of the goods;
2. improper storage of the goods both prior to and after purchase;
3. damage caused by, or in the course of, installation or assembly;
4. any use of the goods which is not an intended use as specified in Westeel's published product literature, or otherwise specified by Westeel in writing;
5. any equipment attached to or used in conjunction with the goods;
6. any field modifications or substitutions to original bin components;
7. inadequate ventilation or any other circumstance not in keeping with proper maintenance and/or use of the goods;
8. Acts of God, accident, neglect or abuse of the goods by the purchaser and/or any other individual or entity; or
9. Any use or installation inconsistent with Westeel's Standard Disclaimers.

Limitations as to Manufacturer

The warranty does not cover products sold by Westeel that are not manufactured by Westeel. In those circumstances, the purchaser is referred to the manufacturer of those products.

Limitation of Implied Warranties and Other Remedies

To the extent allowed by law, neither Westeel nor its dealers, nor any company affiliated with Westeel makes any warranties, representations, or promises as to the quality, performance, or freedom from defect of any Product covered by this Warranty.

WESTEEL HEREBY DISCLAIMS, TO THE EXTENT APPLICABLE, ANY AND ALL IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. A PURCHASER'S ONLY REMEDIES IN CONNECTION WITH THIS WARRANTY ARE THOSE SET FORTH IN THIS WARRANTY. IN NO EVENT WILL WESTEEL, ITS DEALERS, OR ANY COMPANY AFFILIATED WITH WESTEEL BE LIABLE FOR INCIDENTAL, CONSEQUENTIAL OR PUNITIVE DAMAGES.

Some jurisdictions do not allow waivers of certain warranties, so the above waivers may not apply to you. In that event, any implied warranties are limited in duration to ninety (90) days from delivery of the products. You may also have other rights which vary from jurisdiction to jurisdiction.

Exclusive Warranty

This warranty is the only warranty provided by Westeel and all other warranties and/or commitments, whether express or implied and no matter by whom made, statutory or otherwise, are subsumed and replaced by it and are of no legal effect. If any provision of the warranty is held by a court of

competent jurisdiction to be void or unenforceable, in whole or in part, such provision shall be deemed severable and will not affect or impair the legal validity of any other provision of the warranty.

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Please include the part number listed on the cover page in your message.